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2774-550 Paço de Arcos - Portugal

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Table of Contents

RoR Record of Revisions.....	RoR-1
Comp Compliance Statement.....	Comp-1
0 General Information and Units Of Measurement.....	0-1
0.1 General Information.....	0-1
0.1.1 Introduction.....	0-1
0.1.2 Type Specific Definitions and Abbreviations.....	0-1
0.1.3 Supplementary Publications.....	0-1
0.1.4 Units of Measurements and Conversion Tables.....	0-2
0.1.5 Aeroplane Dimensions.....	0-2
0.1.6 Conventions.....	0-2
1 Limitations.....	1-1
1.1 General.....	1-1
1.2 Certification Status.....	1-2
1.3 Types of Operation.....	1-3
1.4 Crew Composition.....	1-4
1.5 Operational Limitations.....	1-5
1.5.1 Mass and Centre of Gravity.....	1-5
1.5.2 Speed Limitations.....	1-5
1.5.3 Flight Envelope.....	1-5
1.5.4 Wind Limitations.....	1-5
1.5.5 Performance Limitations.....	1-5
1.5.6 Runway Slope.....	1-5
1.5.7 Limitations on Wet or Contaminated Runways.....	1-5
1.5.8 Airframe Contamination.....	1-5
1.5.9 System Limitations.....	1-5
1.5.10 Passenger Seating Configuration.....	1-5
1.5.11 Animal Seating Configuration.....	1-6
2 Normal Procedures.....	2-1
2.1 General Operating Procedures.....	2-1
2.1.1 Automation.....	2-1
2.1.2 Crew Alerting System.....	2-7
2.1.3 Runway Situation Awareness System.....	2-7
2.1.4 Data Link Communications.....	2-8
2.2 Pre-Flight.....	2-12
2.2.1 Task Allocation.....	2-12
2.2.2 Fuelling Procedures.....	2-13
2.2.3 Cabin Preparation.....	2-14
2.2.4 Exterior Inspection.....	2-15
2.3 Pre-Departure.....	2-17
2.3.1 Documentation and Forms.....	2-17
2.3.2 Cockpit Preparation.....	2-17
2.3.3 Departure Briefing.....	2-28
2.4 Engine Start.....	2-29
2.4.1 Normal Start.....	2-29
2.4.2 Alternative Start Procedures.....	2-38
2.5 Taxi.....	2-39
2.5.1 Taxi Procedure.....	2-39
2.5.2 Taxi Flow.....	2-40
2.6 Line-Up.....	2-42
2.6.1 Line-Up Flow.....	2-42
2.6.2 Line-Up Checklist.....	2-44
2.7 Take-Off.....	2-45
2.7.1 Normal Take-Off.....	2-45

Uncontrolled Copy when Printed

2.7.2 Crosswind Take-Off.....	2-47
2.7.3 Take-Off on Wet Runway.....	2-47
2.7.4 Noise Abatement.....	2-47
2.7.5 Departure with level-off at or below 3000 ft AAL.....	2-47
2.7.6 Wake Turbulence Separation - Departure.....	2-48
2.7.7 After Take-Off / Go-Around.....	2-49
2.8 Climb.....	2-52
2.8.1 Climb Flow.....	2-52
2.8.2 Climb Checklist.....	2-53
2.9 Cruise.....	2-54
2.9.1 Cruise Flow.....	2-54
2.10 Arrival Preparation and Briefing.....	2-56
2.11 Descent.....	2-57
2.11.1 Descent Flow.....	2-57
2.11.2 Descent Checklist.....	2-58
2.11.3 Holding.....	2-58
2.12 Approach General.....	2-59
2.12.1 Approach Category.....	2-59
2.12.2 Approach Manoeuvring Speeds.....	2-59
2.12.3 Configurations Gates.....	2-59
2.12.4 Final Approach Target Speed and Gust Corrections.....	2-60
2.12.5 Approach Automation Modes and Settings.....	2-61
2.12.6 Approach Configuration Callouts.....	2-62
2.12.7 Wake Turbulence Separation - Final Approach.....	2-63
2.12.8 Flaps and Gear Extension.....	2-64
2.12.9 Navigation Aids.....	2-64
2.12.10 Warning Flags.....	2-64
2.12.11 Approach Plate.....	2-65
2.12.12 Approach Flow.....	2-65
2.12.13 Approach Checklist.....	2-66
2.12.14 Stabilised Approach Concept.....	2-67
2.13 IFR Approach.....	2-68
2.13.1 Precision Approach - PA.....	2-68
2.13.2 Approach with Vertical Guidance - APV.....	2-72
2.13.3 Non-Precision Approach - NPA.....	2-74
2.13.4 Circling Approach.....	2-78
2.14 VFR Approach.....	2-81
2.15 Missed Approach / Balked Landing.....	2-82
2.15.1 Missed Approach.....	2-82
2.15.2 Balked Landing.....	2-84
2.16 Landing.....	2-85
2.16.1 Normal Landing.....	2-85
2.16.2 Crosswind Landing.....	2-86
2.16.3 Reduced Flap Landing.....	2-87
2.16.4 Landing on Wet / Contaminated Runway.....	2-87
2.17 After Landing.....	2-88
2.17.1 After Landing Flow.....	2-88
2.17.2 After Landing Checklist.....	2-89
2.18 Shutdown.....	2-90
2.18.1 Before Shutdown.....	2-90
2.18.2 Shutdown Flow.....	2-90
2.18.3 Shutdown Checklist.....	2-92
2.18.4 Releasing the Brakes.....	2-92
2.19 Post-Flight.....	2-93
2.20 Turnaround.....	2-94
2.21 Securing Aeroplane.....	2-95
2.21.1 Aeroplane Condition.....	2-95
2.21.2 Expanded Securing Flow.....	2-95
2.21.3 Securing Checklist.....	2-96

2.21.4	Battery Removal.....	2-96
2.21.5	Garmin and Navigation Database Update.....	2-96
2.21.6	Oil Servicing.....	2-96
2.22	Cold Weather Operations.....	2-97
2.22.1	Aeroplane Preparation.....	2-97
2.22.2	De-icing and Anti-icing.....	2-98
2.22.3	Operations on Contaminated Surfaces.....	2-106
2.22.4	Climb.....	2-107
2.22.5	Descent.....	2-107
2.22.6	Approach.....	2-107
2.22.7	Landing.....	2-108
2.22.8	Taxi In.....	2-108
2.22.9	Parking and Post-Flight.....	2-108
2.23	Supplementary Normal Procedures.....	2-109
2.23.1	Pushback and Towing.....	2-109
2.23.2	QFE Operations.....	2-109
2.23.3	Low Visibility Operations.....	2-109
2.23.4	Take-Off with Alternative Bleed Configuration.....	2-109
2.23.5	High Elevation Airport.....	2-109
2.23.6	Metric Altimetry.....	2-110
2.23.7	Steep Approach.....	2-110
2.23.8	Non-ETOPS 180 Operations.....	2-112
2.23.9	High Latitude and Polar Operations.....	2-112
2.23.10	Hot Weather Operations.....	2-112
2.23.11	Operations in Volcanic Ash.....	2-112
2.24	Normal Checklist.....	2-113
2.24.1	Checklist Source.....	2-113
2.24.2	Normal Procedures Checklists.....	2-114
2.24.3	Checklist Task Allocation.....	2-120
2.25	Expanded Flows and Checklists.....	2-121
2.26	Flight Profiles.....	2-122
2.26.1	Normal Take-Off.....	2-122
2.26.2	Alternative Take-Off.....	2-122
2.26.3	ILS – to CAT I minimum.....	2-122
2.26.4	RNP – to LPV minimum.....	2-123
2.26.5	PAR.....	2-123
2.26.6	RNP – to LNAV/VNAV minimum.....	2-123
2.26.7	RNP AR.....	2-123
2.26.8	RNP – to LNAV minimum.....	2-123
2.26.9	FMS Overlay.....	2-124
2.26.10	Non-FMS Overlay.....	2-124
2.26.11	IGS.....	2-124
2.26.12	Circling Approach.....	2-125
2.26.13	Visual Approach.....	2-125
2.26.14	Steep Approach.....	2-126
2.26.15	All Engine Missed Approach.....	2-127
3	Abnormal and Emergency Procedures.....	3-1
3.1	Abnormal / Emergency Procedures Management.....	3-1
3.1.1	Communication.....	3-1
3.1.2	Departure Contingency Procedures.....	3-1
3.1.3	Diversion in Case of Serious Technical Failure.....	3-2
3.2	Abnormal / Emergency Checklist Procedures.....	3-3
3.2.1	Checklist Methodology.....	3-3
3.2.2	Emergency Evacuation.....	3-3
3.2.3	Memory Items.....	3-3
3.2.4	Bomb Threat In Flight.....	3-5
3.2.5	Reserved.....	3-6
3.3	System Failures.....	3-7

3.3.1	Engine Failure On the Ground.....	3-7
3.3.2	Engine Failure In Flight.....	3-7
3.3.3	Rejected Take-off.....	3-7
3.3.4	Cabin Pressurisation.....	3-8
3.3.5	Power-up Anomalies.....	3-8
3.3.6	Loss of Position Information.....	3-9
3.4	Fire and Smoke.....	3-12
3.4.1	Engine Fire On Ground.....	3-12
3.4.2	Hot Start / Tail Pipe Fire.....	3-12
3.4.3	Engine Fire In Flight.....	3-13
3.4.4	APU Fire.....	3-13
3.4.5	Aeroplane Interior Smoke and Fire.....	3-14
3.5	Crew Incapacitation.....	3-15
3.6	Emergency Landing and Ditching.....	3-16
3.6.1	Emergency Landing.....	3-16
3.6.2	Ditching.....	3-16
3.7	Exceeding Structural Limits.....	3-17
3.7.1	Overweight Landing.....	3-17
3.7.2	Hard Landing.....	3-18
3.7.3	Exceeding Speed Limitations.....	3-19
3.8	TAWS / Ground Proximity Warning.....	3-20
3.8.1	Caution Alerts.....	3-20
3.8.2	Warning Alerts.....	3-20
3.9	Windshear.....	3-22
3.9.1	Windshear Recognition.....	3-22
3.9.2	Standard Response / Recovery Technique.....	3-23
3.10	TCAS	3-25
3.10.1	Traffic Advisory (TA).....	3-25
3.10.2	Resolution Advisory (RA).....	3-25
3.10.3	Clear of Conflict.....	3-25
3.11	Lightning Strikes.....	3-26
3.12	Exceeding Cosmic Radiation Limits.....	3-27
3.13	Profile Diagrams.....	3-28
3.13.1	Rejected Take-off.....	3-28
3.13.2	Engine Failure After V1.....	3-29
3.13.3	Engine Failure During the Cruise.....	3-29
3.13.4	One Engine Inoperative Approach and Landing.....	3-30
3.13.5	One Engine Inoperative Go-Around.....	3-30
3.13.6	Emergency Descent.....	3-31
4	Performance.....	4-1
4.1	Performance Data.....	4-1
4.1.1	Supplementary Performance Data.....	4-1
4.1.2	Reserved.....	4-1
4.2	Additional Performance Data.....	4-2
5	Flight Planning.....	5-1
5.1	Pre-Flight / In-Flight Planning.....	5-1
5.2	In-Flight Fuel Monitoring.....	5-2
5.3	ETOPS Flight Planning.....	5-3
6	Mass and Balance.....	6-1
6.1	General.....	6-1
6.2	Pre-Flight Procedure.....	6-2
7	Loading.....	7-1
8	Configuration Deviation List.....	8-1
8.1	Publication.....	8-1
8.2	Location.....	8-2

9	Minimum Equipment List.....	9-1
9.1	Publication.....	9-1
9.2	Location.....	9-2
10	Survival and Emergency Equipment Including Oxygen.....	10-1
10.1	Survival and Emergency Equipment.....	10-1
10.2	Oxygen Requirements.....	10-2
11	Emergency Evacuation Procedures.....	11-1
11.1	Preparation for Emergency Evacuation.....	11-1
11.2	Emergency Evacuation.....	11-2
12	Aeroplane Systems.....	12-1
12.1	Garmin Navigational Database Update.....	12-1

Uncontrolled Copy when Printed

List of Effective Pages

Uncontrolled Copy when Printed

Page Number	Revision Date	Revision Number	Page Number	Revision Date	Revision Number
Cover Page			1-4	29 MAY 2023	10.0.0
APPR-1	29 JUN 2026	12.0.0	1-5	29 MAY 2023	10.0.0
APPR-2	29 JUN 2026	12.0.0	1-6	16 FEB 2026	11.0.0
Table of Contents			Chapter 2 - Normal Procedures		
TOC-1	29 JUN 2026	12.0.0	2-1	16 FEB 2026	11.0.0
TOC-2	29 JUN 2026	12.0.0	2-2	16 FEB 2026	11.0.0
TOC-3	29 JUN 2026	12.0.0	2-3	16 FEB 2026	11.0.0
TOC-4	29 JUN 2026	12.0.0	2-4	16 FEB 2026	11.0.0
TOC-5	29 JUN 2026	12.0.0	2-5	16 FEB 2026	11.0.0
List of Effective Pages			2-6	16 FEB 2026	11.0.0
LEP-1	29 JUN 2026	12.0.0	2-7	16 FEB 2026	11.0.0
LEP-2	29 JUN 2026	12.0.0	2-8	16 FEB 2026	11.0.0
LEP-3	29 JUN 2026	12.0.0	2-9	16 FEB 2026	11.0.0
Revision Highlights			2-10	16 FEB 2026	11.0.0
RH-1	29 JUN 2026	12.0.0	2-11	16 FEB 2026	11.0.0
Abbreviations			2-12	16 FEB 2026	11.0.0
ABBR-1	29 JUN 2026	12.0.0	2-13	16 FEB 2026	11.0.0
ABBR-2	29 JUN 2026	12.0.0	2-14	16 FEB 2026	11.0.0
ABBR-3	29 JUN 2026	12.0.0	2-15	16 FEB 2026	11.0.0
Chapter RoR - Record of Revisions			2-16	16 FEB 2026	11.0.0
ROR-1	29 JUN 2026	12.0.0	2-17	16 FEB 2026	11.0.0
ROR-2	29 JUN 2026	12.0.0	2-18	16 FEB 2026	11.0.0
Chapter Comp - Compliance Statement			2-19	16 FEB 2026	11.0.0
COMP-1	16 FEB 2026	11.0.0	2-20	16 FEB 2026	11.0.0
COMP-2	16 FEB 2026	11.0.0	2-21	16 FEB 2026	11.0.0
Chapter 0 - General Information and Units Of Measurement			2-22	16 FEB 2026	11.0.0
0-1	29 MAY 2023	10.0.0	2-23	16 FEB 2026	11.0.0
0-2	16 FEB 2026	11.0.0	2-24	16 FEB 2026	11.0.0
Chapter 1 - Limitations			2-25	16 FEB 2026	11.0.0
1-1	29 MAY 2023	10.0.0	2-26	16 FEB 2026	11.0.0
1-2	29 MAY 2023	10.0.0	2-27	16 FEB 2026	11.0.0
1-3	29 MAY 2023	10.0.0	2-28	16 FEB 2026	11.0.0
			2-29	16 FEB 2026	11.0.0
			2-30	16 FEB 2026	11.0.0
			2-31	16 FEB 2026	11.0.0
			2-32	16 FEB 2026	11.0.0
			2-33	16 FEB 2026	11.0.0
			2-34	16 FEB 2026	11.0.0
			2-35	16 FEB 2026	11.0.0
			2-36	16 FEB 2026	11.0.0
			2-37	16 FEB 2026	11.0.0
			2-38	16 FEB 2026	11.0.0

Page Number	Revision Date	Revision Number	Page Number	Revision Date	Revision Number
2-39	16 FEB 2026	11.0.0	2-84	16 FEB 2026	11.0.0
2-40	16 FEB 2026	11.0.0	2-85	16 FEB 2026	11.0.0
2-41	16 FEB 2026	11.0.0	2-86	16 FEB 2026	11.0.0
2-42	16 FEB 2026	11.0.0	2-87	16 FEB 2026	11.0.0
2-43	16 FEB 2026	11.0.0	2-88	16 FEB 2026	11.0.0
2-44	16 FEB 2026	11.0.0	2-89	16 FEB 2026	11.0.0
2-45	16 FEB 2026	11.0.0	2-90	16 FEB 2026	11.0.0
2-46	16 FEB 2026	11.0.0	2-91	16 FEB 2026	11.0.0
2-47	16 FEB 2026	11.0.0	2-92	16 FEB 2026	11.0.0
2-48	16 FEB 2026	11.0.0	2-93	16 FEB 2026	11.0.0
2-49	16 FEB 2026	11.0.0	2-94	16 FEB 2026	11.0.0
2-50	16 FEB 2026	11.0.0	2-95	16 FEB 2026	11.0.0
2-51	16 FEB 2026	11.0.0	2-96	16 FEB 2026	11.0.0
2-52	16 FEB 2026	11.0.0	2-97	16 FEB 2026	11.0.0
2-53	16 FEB 2026	11.0.0	2-98	16 FEB 2026	11.0.0
2-54	16 FEB 2026	11.0.0	2-99	16 FEB 2026	11.0.0
2-55	16 FEB 2026	11.0.0	2-100	16 FEB 2026	11.0.0
2-56	16 FEB 2026	11.0.0	2-101	16 FEB 2026	11.0.0
2-57	16 FEB 2026	11.0.0	2-102	16 FEB 2026	11.0.0
2-58	16 FEB 2026	11.0.0	2-103	16 FEB 2026	11.0.0
2-59	16 FEB 2026	11.0.0	2-104	16 FEB 2026	11.0.0
2-60	16 FEB 2026	11.0.0	2-105	16 FEB 2026	11.0.0
2-61	16 FEB 2026	11.0.0	2-106	16 FEB 2026	11.0.0
2-62	16 FEB 2026	11.0.0	2-107	16 FEB 2026	11.0.0
2-63	16 FEB 2026	11.0.0	2-108	16 FEB 2026	11.0.0
2-64	16 FEB 2026	11.0.0	2-109	16 FEB 2026	11.0.0
2-65	16 FEB 2026	11.0.0	2-110	16 FEB 2026	11.0.0
2-66	16 FEB 2026	11.0.0	2-111	16 FEB 2026	11.0.0
2-67	16 FEB 2026	11.0.0	2-112	16 FEB 2026	11.0.0
2-68	16 FEB 2026	11.0.0	2-113	16 FEB 2026	11.0.0
2-69	16 FEB 2026	11.0.0	2-114	29 JUN 2026	12.0.0
2-70	16 FEB 2026	11.0.0	2-115	29 JUN 2026	12.0.0
2-71	16 FEB 2026	11.0.0	2-116	29 JUN 2026	12.0.0
2-72	16 FEB 2026	11.0.0	2-117	29 JUN 2026	12.0.0
2-73	16 FEB 2026	11.0.0	2-118	29 JUN 2026	12.0.0
2-74	16 FEB 2026	11.0.0	2-119	29 JUN 2026	12.0.0
2-75	16 FEB 2026	11.0.0	2-120	16 FEB 2026	11.0.0
2-76	16 FEB 2026	11.0.0	2-121	16 FEB 2026	11.0.0
2-77	16 FEB 2026	11.0.0	2-122	16 FEB 2026	11.0.0
2-78	16 FEB 2026	11.0.0	2-123	16 FEB 2026	11.0.0
2-79	16 FEB 2026	11.0.0	2-124	16 FEB 2026	11.0.0
2-80	16 FEB 2026	11.0.0	2-125	16 FEB 2026	11.0.0
2-81	16 FEB 2026	11.0.0	2-126	16 FEB 2026	11.0.0
2-82	16 FEB 2026	11.0.0	2-127	16 FEB 2026	11.0.0
2-83	16 FEB 2026	11.0.0	2-128	16 FEB 2026	11.0.0

Page Number	Revision Date	Revision Number	Page Number	Revision Date	Revision Number
Chapter 3 - Abnormal and Emergency Procedures			5-2	29 MAY 2023	10.0.0
3-1	29 MAY 2023	10.0.0	5-3	29 MAY 2023	10.0.0
3-2	29 MAY 2023	10.0.0	5-4	29 MAY 2023	10.0.0
3-3	29 MAY 2023	10.0.0	Chapter 6 - Mass and Balance		
3-4	29 MAY 2023	10.0.0	6-1	29 MAY 2023	10.0.0
3-5	29 MAY 2023	10.0.0	6-2	29 MAY 2023	10.0.0
3-6	29 MAY 2023	10.0.0	6-3	29 MAY 2023	10.0.0
3-7	16 FEB 2026	11.0.0	6-4	29 MAY 2023	10.0.0
3-8	29 MAY 2023	10.0.0	Chapter 7 - Loading		
3-9	29 MAY 2023	10.0.0	7-1	29 MAY 2023	10.0.0
3-10	16 FEB 2026	11.0.0	7-2	29 MAY 2023	10.0.0
3-11	29 MAY 2023	10.0.0	Chapter 8 - Configuration Deviation List		
3-12	29 MAY 2023	10.0.0	8-1	29 MAY 2023	10.0.0
3-13	29 MAY 2023	10.0.0	8-2	29 MAY 2023	10.0.0
3-14	29 MAY 2023	10.0.0	Chapter 9 - Minimum Equipment List		
3-15	29 MAY 2023	10.0.0	9-1	29 MAY 2023	10.0.0
3-16	29 MAY 2023	10.0.0	9-2	29 MAY 2023	10.0.0
3-17	29 MAY 2023	10.0.0	Chapter 10 - Survival and Emergency Equipment Including Oxygen		
3-18	29 MAY 2023	10.0.0	10-1	29 MAY 2023	10.0.0
3-19	29 MAY 2023	10.0.0	10-2	29 MAY 2023	10.0.0
3-20	29 MAY 2023	10.0.0	Chapter 11 - Emergency Evacuation Procedures		
3-21	29 MAY 2023	10.0.0	11-1	29 MAY 2023	10.0.0
3-22	29 MAY 2023	10.0.0	11-2	29 MAY 2023	10.0.0
3-23	29 MAY 2023	10.0.0	Chapter 12 - Aeroplane Systems		
3-24	29 MAY 2023	10.0.0	12-1	29 MAY 2023	10.0.0
3-25	29 MAY 2023	10.0.0	12-2	29 MAY 2023	10.0.0
3-26	29 MAY 2023	10.0.0			
3-27	29 MAY 2023	10.0.0			
3-28	16 FEB 2026	11.0.0			
3-29	16 FEB 2026	11.0.0			
3-30	16 FEB 2026	11.0.0			
3-31	16 FEB 2026	11.0.0			
3-32	16 FEB 2026	11.0.0			
Chapter 4 - Performance					
4-1	16 FEB 2026	11.0.0			
4-2	29 MAY 2023	10.0.0			
Chapter 5 - Flight Planning					
5-1	29 MAY 2023	10.0.0			

Uncontrolled Copy when Printed

Highlights[RoR Record of Revisions](#)[RoR](#)

New revision.

[2.24.2 Normal Procedures Checklists](#)[2.24.2](#)

Updated Normal Procedures Checklists.

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Abbreviations

ABBREVIATION	DESCRIPTION
AAD	Altitude Alerting Device
AAL	Above Aerodrome Level
AFCS	Automatic Flight Control System
AFM	Aeroplane Flight Manual
AGL	Above Ground Level
AIR OPS	Commission Regulation (EU) NO 965/2012 and Amendments Thereof Pursuant to the Regulation (EC) NO 216/2008 of the European Parliament and of the Council
AMCC	Aircraft Maintenance Control Centre
ANAC	Autoridade Nacional de Aviação Civil
AOC	Air Operator Certificate
AP	Autopilot
APU	Auxiliary Power Unit
ATC	Air Traffic Control
CAS	Crew-Alerting System
CAT I	Category 1 (One)
CDFA	Continuous Descent Final Approach
CDL	Configuration Deviation List
CG	Centre of Gravity
CPDLC	Controller Pilot Data Link Communications
DA	Decision Altitude
DME	Distance-Measuring Equipment
DR	Dead Reckoning
EFB	Electronic Flight Bag
EICAS	Engine Indication Crew Alert System
ETOPS	Extended Range Two-Engined Operations
FAF	Final Approach Fix
FBO	Fixed-Base Operator
FCC	Flight Crew Compartment
FCM	Flight Crew Member
FD	Flight Director
FL	Flight Level
FMS	Flight Management System
GP	Glidepath
GPS	Global Positioning System
GPU	Ground Power Unit
GS	Glide Slope
GTC	Garmin Touchscreen Controller
IAS	Indicated Airspeed
ICAO	International Civil Aviation Organisation
IFR	Instrument Flight Rules
ILS	Instrument Landing System
IMC	Instrument Meteorological Conditions
KIAS	Knots Indicated Air Speed
LH	Left Hand

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ABBREVIATION	DESCRIPTION
LHS	Left Hand Seat
LNAV	Lateral Navigation
LOC	Localiser
LPV	Localizer Performance with Vertical guidance
LVTO	Low Visibility Take-Off
MDA	Minimum Descent Altitude
MEL	Minimum Equipment List
MOCA	Minimum Obstacle Clearance Altitude
MORA	Minimum Off-Route Altitude
MSA	Minimum Sector Altitude
NADP	Noise Abatement Departure Procedure
NAV	Navigation Equipment
NAVAID	Navigation Aid
NDB	Non-Directional Beacon / Radio Beacon
NJE	NetJets Europe
NJMP	NetJets Maintenance Procedures
NM	Nautical Miles
NOTAM	Notice To Airmen
NPA	Non-Precision Approach
NTA	NetJets Transportes Aéreos, S.A.
OAT	Outside Air Temperature
OEI	One Engine Inoperative
OEM	Original Equipment Manufacturer
OFP	Operational Flight Plan
OM-A	Operations Manual Part A
OM-B	Operations Manual Part B
OM-C	Operations Manual Part C
PAR	Precision Approach Radar
PAX	Passenger
PBE	Protective Breathing Equipment
PDCS	Performance Data Computation System
PF	Pilot Flying
PFD	Primary Flight Display
PIC	Pilot-In-Command
PM	Pilot Monitoring
POH	Pilot Operating Handbook
QFE	Height above Airport Elevation
QNH	Atmospheric Pressure Adjusted to Mean Sea Level
QRH	Quick Reference Handbook
RA	Resolution Advisory (ACAS)
RAIM	Receiver Autonomous Integrity Monitoring
RH	Right Hand
RHS	Right Hand Seat
RNAV	Area Navigation
RTO	Rejected Take-Off
RVR	Runway Visual Range

ABBREVIATION	DESCRIPTION
RVSM	Reduced Vertical Separation Minimum
RWY	Runway
SAT	Static Air Temperature
SBAS	Satellite Based Augmentation System
SEP	Safety and Emergency Procedures
SIC	Second In Command
SID	Standard Instrument Departure
SRA	Surveillance Radar Approach
TAT	Total Air Temperature
TAWS	Terrain Awareness and Warning System
TBS	Time Based Separation
TCAS	Traffic Collision Avoidance System
TOD	Top Of Descent
VFR	Visual Flight Rule
VHF	Very High Frequency
VNAV	Vertical Navigation
YD	Yaw Damper

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Record of Revisions

Revision Number	Issue Date
Original	20 Jun 14
1	26 Sep 14
2	14 Oct 14
TR01	29 Dec 14 (incorporated into Rev.3)
TR02	01 Oct 15 (incorporated into Rev.3)
3	23 Nov 15
4	12 Dec 16
5	06 Feb 18
6	07 Feb 19
7	30 Jan 20
TR03	05 May 20
8	12 Mar 21
9	13 Apr 22
10.0.0	29 May 2023
11.0.0	16 Feb 2026
12.0.0	29 Jun 2026

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Compliance Statement

The nominated person hereunder certifies that this list of effective pages and the contents of this volume of the operations manual satisfy the requirements of AIR OPS and are identical to the copy of this volume submitted to the ANAC.

On behalf of the operator by the nominated person:

Name	Richard Weeks
Title	Accountable Manager
Signature	

DocuSigned by:

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Chapter 0 - General Information and Units Of Measurement

0.1 General Information

0.1.1 Introduction

This Operations Manual Part B is the NetJets Transportes Aéreos S.A. OM-B for the EMB-505 and is approved by the Autoridade Nacional de Aviação Civil (ANAC).

For details relating to the Administration and Control of this Operations Manual, refer to OM-A, Administration and Control of Operations Manual.

0.1.2 Type Specific Definitions and Abbreviations

Abbreviations in this OM-B will be introduced using standard convention, e.g. General Aviation (GA). For abbreviations used in Air Navigation and Aeroplane Operations, see Jeppesen Route Manual part GEN, NTA OM-A, or OEM documentation.

0.1.3 Supplementary Publications

The Operations Manual Part B is supplemented by the following manuals and documents:

- Pilots Operating Handbook (POH), including supplements and appendices;
- Aircraft Flight Manual (AFM), including supplements and appendices;
- Quick Reference Handbook (QRH), to be used for take-off and landing performance calculations, and as primary reference during emergency / abnormal situations;
- NTA Normal Procedures Checklists;
- NTA Icing Procedures Checklist;
- NTA EMB-505 MEL;
- Flight Operations Letters (FOLs) and Operational Bulletins;
- Other NTA Operating Manuals (Part A and Part C);
- Various OEM manuals for installed equipment.

Pilots shall be familiar with the information in the above mentioned publications, all of which can be found on Comply365.

CAUTION

The EMB-505 cannot be operated with exclusive reference to this manual. The procedures and checklists contained herein are optimised and designed for standard operating conditions. Pilots must refer to the approved manual when operating outside these conditions, and each time they need to review the expanded procedures. References to the appropriate manuals are presented when necessary.

0.1.4 Units of Measurements and Conversion Tables

Refer to Jeppesen Airway manual, volume 1 Tables and codes, and POH, Vol 1 Chapter 1.

0.1.5 Aeroplane Dimensions

Refer to AFM and POH.

0.1.6 Conventions

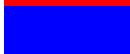
Conventions adopted within the OM-B are as follows:

Colour coding of Flow Diagrams

LHS



RHS



PF



PM



Any item marked with + is only applicable for the first flight of the day (FFD). If a challenge item on a checklist is marked this way, it is only required to be offered as a challenge on the FFD. Similarly, any response item marked this way only needs to be verbalised on the FFD.

Any item marked with # is to be checked if fitted to the aircraft – this is to allow for differences between the Phenom 300 and Phenom 300E.

Any checklist response item of 'A / R' requires the respondent to announce the actual condition of the item being checked. Whenever a Checklist item requires a check of the CAS messages, the response shall be 'CLEAR' only if no CAS messages are displayed. Any visible CAS messages should be summarised verbally.

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Chapter 1 - Limitations

1.1 General

Limitations are in accordance with AFM and NTA Operating Manuals whichever is more restrictive.

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1.2 Certification Status

The certification status is in accordance with:

- the AOC; and
- the AFM.

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1.3 Types of Operation

Types of operations are in accordance with the MOST restrictive of:

- the AOC; and
- the AFM.

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1.4 Crew Composition

Refer to OM-A, [Crew Composition](#).

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1.5 Operational Limitations

1.5.1 Mass and Centre of Gravity

Refer to AFM, Section 6.

1.5.2 Speed Limitations

Refer to AFM, Section 2.

1.5.3 Flight Envelope

Refer to AFM, Section 6.

1.5.4 Wind Limitations

Refer to POH, Section 3 for OEM Limitations. Refer to OM-A for NTA Limitations.

1.5.5 Performance Limitations

Refer to AFM, Section 2.

1.5.6 Runway Slope

Refer to AFM, Section 2.

1.5.7 Limitations on Wet or Contaminated Runways

Refer to POH, Section 2.

1.5.8 Airframe Contamination

Refer to POH, Section 3.

1.5.9 System Limitations

Refer to the AFM, POH and Pilot Training Manuals for appropriate system descriptions.

1.5.9.1 Use of VNAV

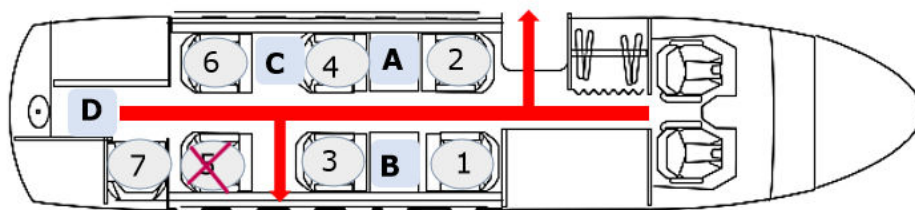
May be used for all stages of flight in accordance with the Garmin Manual or POH.

1.5.10 Passenger Seating Configuration

Please refer to the *Safety Equipment and Procedures Manual*.

1.5.11 Animal Seating Configuration

1.5.11.1 Approved Locations for Un-caged Dogs Phenom 300



Refer to OM-A 8.2.2.5.3 regarding weights and emergency exits restriction.

Area A

- 1 Small dog.

Area B

- 1 Small dog.

Area C

- 1 Very Small dog, provided seat 5 is not occupied by a passenger.

Area D

- 1 Medium dog, provided seat 7 is not occupied, or 1 small dog if seat 7 is occupied.

Phenom 300 - Un-caged dog seating capacity and locations					
No of Pax	All on unoccupied seats	Floor seated			Lap Seated
	L Dog (or smaller)	Medium Dog (or smaller)	Small Dog (or smaller)	Very Small Dog	Very Small Dog
7	0	0	3	0	6
6	1	1	2	0	6
5	2	1	2	1	5
4	3	1	2	1	4
3 ₁	4	1	2	1	3
2 ¹	5	1	2	1	2
1 ₁	6	1	2	1	1

¹ The total number of dogs that can be carried may be less than depicted here due to the OM-A requirement not to exceed 3 un-caged dogs per passenger (except when approved by the Chief Pilot).

Chapter 2 - Normal Procedures

2.1 General Operating Procedures

2.1.1 Automation

2.1.1.1 Autopilot

The point at which the autopilot is engaged is at the discretion of the PF and depends on the flight conditions / workload, etc. Use of the autopilot should be included in the departure briefing.

SITUATION	PF	PM
AP to be selected ON. PF: Request AP on PM: Selects the Autopilot ON and verifies AP Green annunciated on the PFD PF: Verifies AP ON	'AUTOPILOT ON'	'AP GREEN'
AP selected OFF PF: Select AP OFF PM: Verifies AP OFF	'AUTOPILOT OFF'	

2.1.1.2 Autothrottle

Reserved.

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2.1.1.3 FD modes selection (AP on and off)**Autopilot 'ON' Scenarios**

SITUATION	PF	PM
Automatic change or selection of FD mode PF: Selects mode or observes automatic change and calls out PFD annunciation PM: Verifies selected mode	'[XX] GREEN / WHITE'	

Autopilot 'OFF' Scenarios

SITUATION	PF	PM
Selection of FD mode		
PF: Requests mode.	'SELECT [XX]'	
PM: Selects mode and calls PFD annunciation.		'[XX] GREEN'
PF: Verifies selected mode.		

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2.1.1.4 FD mode values (AP on and off)**Autopilot 'ON' Scenarios**

SITUATION	PF	PM
Setting of FD mode value as required by ATC or a procedure. PM: Acknowledges ATC instruction. PF: Sets value (e.g. Heading 300 degrees or direct [XXX] in FMS). PM: Verifies correct value is set.		

Autopilot 'OFF' Scenarios

SITUATION	PF	PM
Setting of FD mode value as required by ATC or a procedure PM: Acknowledges ATC instruction. Sets value (e.g. Heading 300 degrees or direct [XXX] in FMS). PF: Verifies correct value is set.		
Setting of FD mode value on request of PF PF: Requests FD value change. PM: Selects new FD value and calls new value. PF: Verifies correct value set.	'SET [XXXX] FEET PER MIN' 'SET HEADING [XXX]' 'SET SPEED [XXX]'	'[XXXX] FEET PER MIN' 'HEADING [XXX]' 'SPEED [XXX]'

2.1.1.5 Altitude Alert Selection

SITUATION	PF	PM
Setting of new altitude or flight level (AP ON) PM: Acknowledges ATC instruction. PF: Sets new value. PM: Verifies correct value is set.	'FL [XXX] SET'	'CHECKED'
Setting of new altitude or flight level (AP OFF) PM: Acknowledges ATC instruction and sets new value. PF: Verifies correct value is set.	'CHECKED'	'FL [XXX] SET'
Setting of MAA (AP ON or OFF) PM: Sets new value. PF: Verifies correct value is set.	'CHECKED'	'[XXX] SET'

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2.1.1.6 FMS Modification and Changes

SITUATION	PF	PM
Setting of new waypoint (e.g. after ATC instruction (AP ON)) PM: Acknowledges ATC instruction. PF: Selects new waypoint and 'Direct To' (D□). PM: Verifies correct waypoint is selected. PF: Selects 'Activate Direct To' the waypoint.	'DIRECT TO [XXX]'	'CONFIRMED'

SITUATION	PF	PM
Setting of new waypoint (e.g. after ATC instruction (AP OFF)) PM: Acknowledges ATC instruction and selects new waypoint and 'Direct To' (D□). PF: Verifies correct waypoint is selected. PM: Selects 'Activate Direct To' the waypoint.	'CONFIRMED'	'DIRECT TO [XXX]'

NOTE

After any change to either Navigation or altitude data the PM must ensure that the correct FD mode is selected. This will be done by exception.

E.g.: When changing to a HDG following an ATC instruction to deviate from the standard route the PM must ensure that HDG mode is correctly engaged.

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2.1.2 Crew Alerting System

Any unexpected CAS messages (see note) shall be announced by the first crew member to notice them. They shall be announced stating the Warning, Caution or Advisory alert and the message as it is displayed. The other crew member shall confirm they have understood the event by repeating the message or, where applicable, initiating an immediate action (such as RTO).

Example: Initial call: 'Warning ENG 1 FIRE'. Reply: 'Warning ENG 1 FIRE confirmed'. PM cancels the Warning. PF: 'Memory items for ENG 1 FIRE'. Memory items are actioned by PM. PM: 'Memory items ENG 1 FIRE complete'.

This is just an example of the initial call outs. Following this, the nature and type of the situation will dictate the next response or actions.

At a convenient time, considering the CAS message, level of urgency and cockpit workload, the PF shall call for the associated checklist, or initiate the appropriate actions or discussion. Crew can elect to use either the paper QRH or electronic checklist for emergency handling.

NOTE

Should a CAS message be generated as an expected event e.g. during the start, a test procedure, or as a normal response to a particular switch selection, it shall be silently acknowledged by the crew member concerned. Any unexpected or additional CAS messages shall be brought to the other crew member's attention using the methodology outlined above.

Whenever a Checklist item requires a check of the CAS messages, the response shall be 'CLEAR' only if no CAS messages are displayed. Any visible CAS messages should be summarised verbally.

2.1.3 Runway Situation Awareness System

Reserved.

2.1.4 Data Link Communications

2.1.4.1 Controller Pilot Data Link Communications

The Phenom is equipped with a VHF Data Link Transceiver (using VHF 3) that provides the Controller Pilot Data Link Communications system (CPDLC). The VHF Data Link system supports CPDLC through the Aeronautical Telecommunications Network (ATN-CPDLC) or through the Aircraft Communications Addressing and Reporting System (ACARS) network known as FANS-CPDLC (Non-EuroControl CPDLC).

2.1.4.2 CPDLC Operating Principles

FCM should verify the availability of CPDLC on the intended route before flight.

When flying in airspace where CPDLC is available crew should try to logon with ATC and use data link message exchange as much as possible. When operating outside data link airspace crew should initiate a logon 10-25 minutes prior to entry into airspace where data link services are provided.

Data link communications coexist with VHF voice radio communications.

When using CPDLC for ATC communications, crew shall monitor the appropriate ATC frequency and revert to voice when required.

Voice communications and voice instructions always have precedence over data link communications. The use of ATN-CPDLC is restricted to supplementary means of communication for non-time-critical messages.

To minimise pilot head down time and potential distractions during critical stages of flight, the FCM should use voice communications when operating below 10,000 ft.

The flight crew or the air traffic controller involved has discretion over whether to discontinue or not the use of data link services.

Messages received via voice should be replied to via voice and messages received via data link should be replied to via data link. If a conflicting CPDLC clearance or instruction is received the crew should ask for clarification via voice.

Data Link Clearances and instructions should be dealt with efficiently whilst maintaining normal crew co-ordination.

The POH and AFM contain detailed descriptions of the system and its operation.

2.1.4.3 CPDLC Logon

A flight plan must be filed and should be inserted in the normal way via the GTC into Garmin 3000 before attempting to log on to the CPDLC system.

The information entered in the CPDLC Log-On display must match the filed flight plan.

If the flight plan has been inserted into the Garmin, the required information should automatically be seen on the CPDLC setup page.

- From the Home Screen touch CPDLC.
- The Status Tab displays the Logon Setup.
- The Facility Button displays a list of available ATC facilities.

After performing the Logon Setup check that the Link Status display indicates that a link with a ground station is available with a solid green line between the aircraft symbol and the ground station antenna symbol.

The Logon Button connects the aircraft to the ATC facility. The Link Status display indicates 'Connecting' with an animated dashed green line between the ground station antenna symbol and the ATC facility symbol.

The Cancel Button terminates the log-on process.

When connection is complete the display indicates 'Connected' with a solid green line between the ground station antenna symbol and the ATC facility symbol.

After a successful logon has been completed the system will go into a CPDLC waiting state ready for an ATC facility to initiate a CPDLC session.

2.1.4.4 CPDLC Messages and Icons

An incoming CPDLC message (Uplink Message) will be displayed on the PFDs and GTCs and is accompanied by an aural alert. Crew will have 100 seconds to respond to ATC before the message expires; this time window allows for the crew sending a CPDLC response and initiating the required action.

The CPDLC Message Tab displays the number of pending messages and also the status of each message.

Downlink CPDLC Status Icons

Icon	Text	Description
	Sent	Message sent to ATC.
	Standby	'Standby' message response sent to ATC. The exclamation point notifies the flight crew that an action is needed to close the message.
	Send Failed / Expired	Message failed to send or the message timed out.
	Closed	Message sent to ATC and message thread closed.

Uplink CPDLC Status Icons

Icon	Text	Description
	Standby (message not opened)	Unopened 'Standby' message response sent by ATC.
	Standby (message opened)	Opened 'Standby' message response sent by ATC.
	Need Response (message not opened)	Unopened message that requires a response to ATC.
	Need Response (message opened)	Opened message that requires a response to ATC.
	Expired (message not opened)	Unopened expired message sent by ATC.
	Expired (message opened)	Opened expired message sent by ATC.
	Closed (message not opened)	Unopened message that closed the message thread.
	Closed (message opened)	Opened message that closed the message thread.

2.1.4.5 CPDLC Crew Coordination

Operational experience has already shown that there are fundamental differences between using voice communications and CPDLC. These differences need to be considered when considering crew coordination while using CPDLC.

With voice communications there is a natural ability for each FCM to understand incoming and outgoing transmissions for themselves and this provides a certain level of situational awareness between the crew.

Each FCM should individually review each CPDLC uplink message prior to responding and / or executing any clearance. Similarly each FCM should individually review each CPDLC downlink message before it is transmitted. Reading a message individually is a key element to ensuring that the message is clearly understood by each FCM. Reading messages out loud may bias the other FCM and could lead to misunderstanding.

The procedures below are for the normal state of two pilots in the FCC. Should there be only one pilot, CPDLC can still be used and as per OM-A, 8.3.1.10, the returning pilot is briefed on all relevant changes if any.

2.1.4.5.1 Uplink Message (From ATC)

When a CPDLC uplink message is received, both FCM should maintain the same level of situational awareness as normal voice communications.

Once a new CPDLC message is received, the PM will access the CPDLC screen and check any new messages that have not yet been viewed.

Both FCM will individually read and agree the content of the received message. The ATC requirement shall be executed using established SOPs (frequency change, altitude change, direct to, etc.), following which the PM will send the appropriate reply and both FCM shall verify the message has been sent.

The communication process shall be monitored by PF and any discrepancies shall be called by exception.

If multiple clearances are received in a single message, the crew should only respond with WILCO if all clearances in the entire message can be complied with.

If the crew cannot comply with any part of the multi element message, then they should respond to the entire message with UNABLE.

2.1.4.5.2 Downlink Message (From Aircraft)

When a CPDLC downlink message is required to be sent, the PM should access the CREATE MESSAGE menu and select the applicable message in consultation with the PF.

PM shall send the message and both FCM shall verify it has been sent.

The timer then allows ATC 270 seconds to respond before the message expires.

2.1.4.6 CDA (Current Data Authority) and NDA (Next Data Authority)

Under normal conditions the transfer of CPDLC connections between CDA and the NDA is controlled by ATC and is transparent to the pilots. If the automatic transfer is unsuccessful, ATC will provide data link instructions to the pilot to manually disconnect from CDA and logon to the NDA.

2.1.4.7 CPDLC Logoff

All CPDLC connections will automatically be terminated at the end of the flight.

To manually logoff from a CPDLC session press the Logoff Button on the CPDLC Status Tab.

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2.2 Pre-Flight

2.2.1 Task Allocation

FIRST FLIGHT OF DAY (FFOD)	
PF + PM	
• Check-in; • Collect and organise flight documents; • Review day schedule; • Check general weather conditions; • Check if there is any special requirement (aerodrome qualification, operating limitations and passenger's request); • Commander allocates functions and gives any special instructions (Initial Briefing).	
PF	PM
• Perform Cockpit Emergency Equipment Check; • Perform 'Cockpit Preparation' and 'Power-up'; • Check validity of databases.	• Perform Pre-flight Exterior Inspection; • Perform 'Cabin Preparation'.
PIC	
• Check aeroplane documents and forms as per OM-A, Operating Procedures - First Flight Tour (FFT); • Review Tech / Appearance Logs (FFT).	

BEFORE EVERY FLIGHT	
PF and PM	
• Check all relevant weather, NOTAMs and Performance; • Review limitations and special requirements; • Compute fuel load; • Prepare, secure, and nominate primary Portable EFB.	
PF	PM
• Compute flight data and enter on OFP; • Check fuel quantity and oxygen pressure; • Check the load sheet and amend if necessary; • Complete Aircraft Flight Log iaw NJMP 1.01 Part 1; • Give Departure Briefing; • Load flight plan; • Perform the 'Duplicate Exterior Inspection'.	• Perform 'Pre-Flight Exterior Inspection' or 'Post-Flight / Turn-Around Inspection' as appropriate; • Monitor refuelling; • Load and set up catering; • Ensure cabin is prepared; • Meet passengers; • Give required 'Passenger Briefing' iaw OM-A; • Communicate any delays to Dispatch and PF; • Reply to release messages in consultation with PF.
Commander	
• Sign all required documents. • Ensure Security Check is completed and acknowledged in the Waypoint App.	

NOTE

This table is to be used as a guideline. Crews should exercise discretion to share and optimise the workload as conditions on the day dictate, except that the ^{*}'Duplicate Exterior Inspection' shall always be performed by the person who did not perform the 'Exterior Pre-Flight Inspection' or 'Turn-Around Inspection'.

2.2.2 Fuelling Procedures**Refuelling**

Refer to POH Fuel System Servicing, section 5.

If refuelling via over-wing fillers, it is forbidden for passengers to be on board whilst refuelling is being carried out. Passengers may remain on board during pressure refuelling provided the regulations in OM-A, Operating Procedures, are adhered to and the SIGNS / OUTLET switch is selected to OFF / ON, and the main cabin door is OPEN.

2.2.3 Cabin Preparation

The Pre-flight Cabin / Interior inspection shall be executed diligently, ensuring that the cabin presentation is up to required NJE Standards.

This includes the following items:

2.2.3.1 Cabin Emergency Equipment Check

Emergency Equipment and Security Check

- Infant life jacket and seat belts shall be checked.
- Check adult life-jacket in toilet cupboard for security and integrity.
- Check MedAire FAK sealed and in date. If the FAK has been opened, check the Tech Log for MEL expiry date.
- Check AED.
- Check if the fire extinguisher is secured (date of inspection controlled by AMCC).
- Emergency Exit cover in place and pin is removed (pin shall be placed in LHS pilot's stowage pocket).
- Briefing Cards shall be placed at the sides of the seats. Any cards in poor condition shall be replaced from a stock location when available.
- Check all life jackets for security and integrity.
- Extension Seat Belts in the small cabinet shall be checked to be in place.

2.2.3.2 Cabin Setup

Toilet Area

- Check that the toilet roll is in the holder and the cupboard is stocked with the standard items (deficits shall be noted in the passtown and requests made to the corresponding cleaning / restock company as well as Lis-LockerRep@ntasa.pt).
- Check the toilet is clean and odour-free and the flush functions correctly (external power is required).

Cabin - Rear to Front

- Ashtrays clean.
- Seats shall be clean, with seat belt buckles and metallic parts of the seats clean and free from fingerprints.
- Stock locations shall be checked, replenished if able from onboard stock, and any discrepancies noted using the Stock App.
- Newspapers shall be presented on the work surface in front of the main door along with any passenger requested magazines. Standard stock magazines shall be presented in the magazine rack which should be placed on seat 6. If 6 passengers are present, then the magazine rack can be stored in the roller cabinet.
- Tables, carpets, and windows are to be clean and free of fingerprints.
- iPads should be checked, sufficiently charged, and free of fingerprint marks on the screen.

- Cabin lighting shall be selected as appropriate for the time of day, GPU availability, and pax needs.
- If the cabin temperature is excessive and a GPU is available, then this shall be connected and the air conditioning switched ON.
- The galley and galley entrance door panelling shall be clear of fingerprints and stains.
- The steps should be free of dirt and stones.

This list is not exhaustive and crew should use their discretion where appropriate to enhance the Owner's comfort.

Please refer to the NJE Service Standards Manual for cabin preparation standards and presentational suggestions.

2.2.4 Exterior Inspection

The pre-flight exterior inspection shall be completed independently by one FCM (FCM 1), ensuring all gear pins and protective covers are removed and stowed, and if interrupted it shall be commenced from the beginning. FCM 1 shall sign for the pre-flight inspection in the AFL.

2.2.4.1 Pre-Flight Exterior Inspection

The Pre-flight Exterior Inspection shall be executed as per the relevant checklists contained in the 'Normal Procedures' section of the AFM and POH.

The pilot carrying out the pre-flight inspection shall ensure that the other FCM is shown the 3 ADS probe covers and 3 gear pins prior to stowing them.

2.2.4.2 Duplicated Exterior Inspection

When all pre-flight servicing is complete (fuel, toilet service, catering loaded, etc), the other FCM (FCM 2) shall conduct an independent external inspection, ensuring all servicing equipment has been removed, and all panels are secure; no panels should be opened during this inspection.

The FCM loading the baggage (normally FCM 2 – see OM-A 8.3.19.9) shall, having secured the baggage access door, ensure that no hazards are present that may affect the safety of the aircraft during taxi. If there is no baggage to load, this verification shall be conducted during the Duplicated Exterior inspection.

The pilot who did not perform the [Pre-Flight Exterior Inspection](#) will perform the 'Duplicate Exterior Inspection' checking as a minimum the following items:

- ADS Probe Covers and Gear Pins removed;
- All doors and panels secure and locked;
- Engine covers removed;
- Fuel Caps fitted correctly and latched;
- Fuel Panel closed and locked;
- Torque Link correctly connected, unless aeroplane is to be pushed back prior to engine start;

- Chocks clear unless it is agreed with ground handling that they will be removed once engines are running.

NOTE

Ensure parking brake is set and hydraulic emergency brake accumulator pressure is available prior to removing chocks.

2.2.4.3 Final Exterior Inspection

Immediately before closing the main passenger door, FCM 1 shall conduct a final visual check of the aeroplane, ensuring all panels (including baggage access door) are secure, the nose wheel torque-link pin is correctly configured for taxi / pushback as applicable, and the intended taxi route is clear of any obstructions.

2.2.4.4 Post-Flight / Turnaround Exterior Inspection

The Post-Flight / Turn-Around Inspection is accomplished on completion of the Shutdown Checklist and passenger disembarkation.

As a minimum, the following items should be visually inspected:

- General Condition:
 - Fuselage and surfaces;
 - Static Wicks.
- Engine oil:
 - Check quantity within 10 to 20 minutes of engine shutdown.
- Fluid leaks;
- Tyres and brakes;
- Hydraulic Differential Pressure Indicator (DPI) pop-ups;
- Cabin General Condition:
 - Cleanliness;
 - Stock;
 - If a toilet service has been carried out, a visual check of the servicing bay and functional check of the toilet (if external power available) should be carried out. If possible, a crew member should monitor the toilet service closely and ensure the servicing bay is correct afterwards.

A turn-around inspection should be followed by a [Duplicated Exterior Inspection](#) and a [Final Exterior Inspection](#).

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2.3 Pre-Departure

2.3.1 Documentation and Forms

Refer to OM-A: Operating Procedures.

2.3.2 Cockpit Preparation

Cockpit set-up incorporates the Safety Equipment Check, Cockpit Set-up and Power-up Checklist, and all other items to be configured before taxi.

Portable EFB - The proper usage of the portable EFB shall be accomplished in accordance with the OM-A.

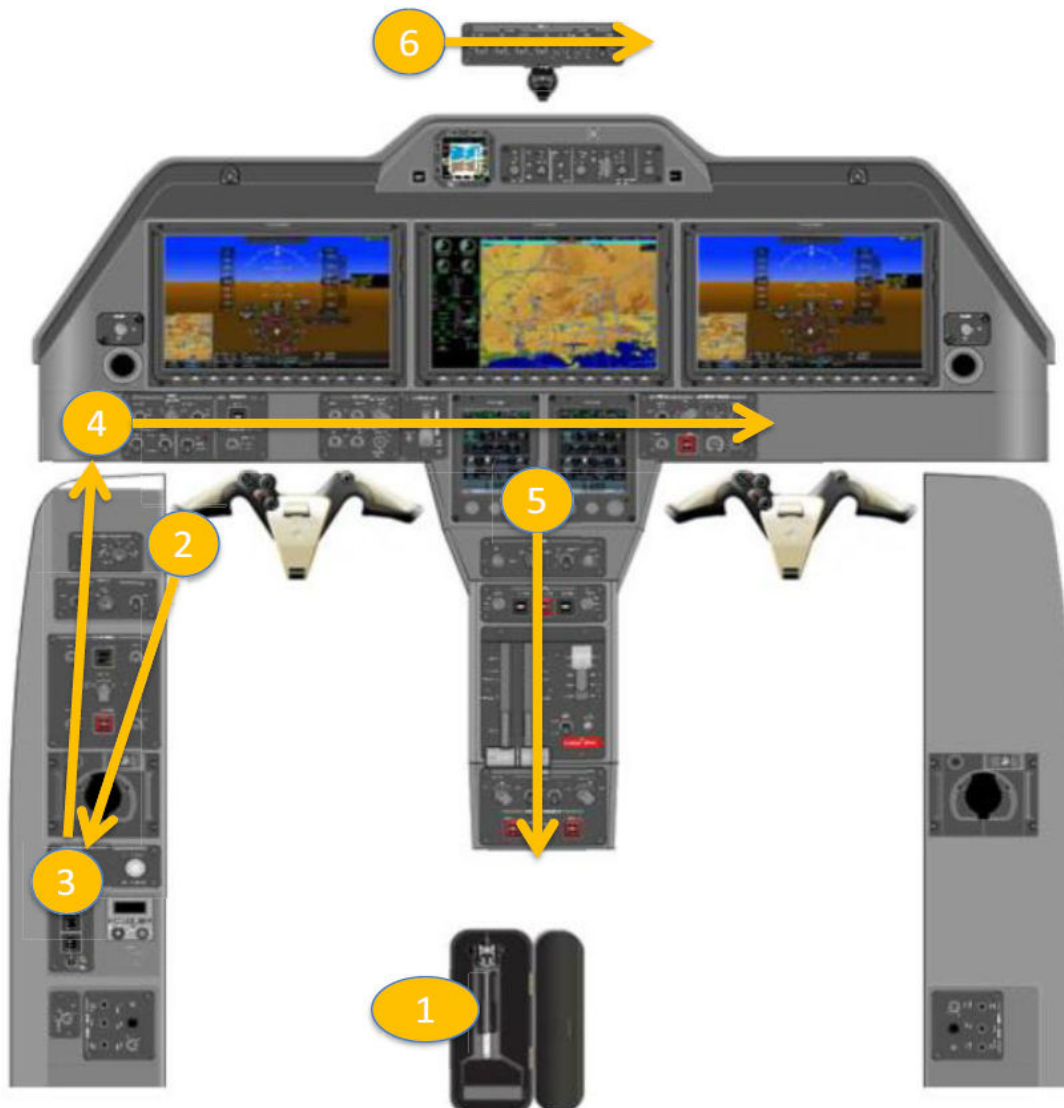
External power should be used if available and appropriate, enabling cockpit set-up to be completed prior to starting engines. For all revenue legs, if a GPU is available, cockpit set-up should be completed 30 minutes prior to scheduled departure time.

If no external power is available, caution should be exercised to avoid draining the batteries.

2.3.2.1 Cockpit Preparation Flow

The purpose of this flow is to ensure the aeroplane is correctly configured to safely apply electrical power.

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Expanded Cockpit Preparation Flow

1. Emergency Equipment

WATER BARRIER **CHECK**

SMOKE GOGGLES (X2) **CHECK**

Not damaged, strap serviceable, stowed correctly.

FLASHLIGHT (X2) **CHECK**

Check operation.

CRASH AXE **STOWED**

PBE **CHECK**

Correctly stowed, blue indicator in window.

FIRE EXTINGUISHER **STOWED**

CREW LIFE JACKETS (X2) **CHECK**

+EMER LIGHTS **CHECK**

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Confirm correct operation of the Emergency Lights including the exit signs with the batteries off and select back to OFF unless carrying out the Power-up checklists immediately.

2. **CIRCUIT BREAKERS CHECK**

Check all circuit breakers on LH, RH, and AFT panels in, except those circuit breakers that are tie wrapped for the purpose of complying with MEL maintenance procedures.

3. **IFE / REFRESHMENT CENTRE ON**

FLIGHT CONTROLS RESET NONE

OXY BOTTLE VALVE HANDLE RESTORE

SUPPLY CONTROL KNOB PAX AUTO

OXYGEN MASK REGULATOR CHECK 100%

BATT 1 and 2 SWITCHES OFF

ELEC EMER BUTTON PUSHED OUT

BUS TIE KNOB AUTO

GEN 1 and 2 SWITCHES AUTO

GPU BUTTON PUSHED OUT

BLEED 1 and 2 SWITCHES AUTO

XBLEED KNOB AUTO

TEST PANEL OFF

4. **DC PUMP 1 and 2 SWITCHES AUTO**

XFEED KNOB OFF

SOV 1 and 2 SWITCHES OPEN

ELT SWITCH ARM

PUSHER CUT OUT BUTTON PUSHED OUT

SIGNS / OUTLET OFF / ON

GUST LOCK PIN (STOW IN LH PILOT STOWAGE POCKET) STOW

WSHLD 1 and 2 SWITCHES OFF

ADS PROBES KNOB AUTO

ENG 1 and 2 SWITCHES OFF

WING STAB SWITCH OFF

PRESN MODE SWITCH AUTO

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ECS KNOB BOTH

DUMP BUTTON PUSHED OUT

AIR CONDITIONING PANEL AS REQUIRED

5. **PFD MODE Switches AUTO**

TAWS buttons Pushed Out

FLAP Lever Match Flap Position

If flap position is anything other than 0, match flap lever to observed flap position and ensure personnel remain clear of flaps when applying power.

SPEED BRAKE SWITCH CLOSE

PARKING BRAKE SET

START / STOP 1 and 2 STOP

IGNITION 1 and 2 AUTO

SHUTOFF 1 and 2 BUTTONS PUSHED OUT

BOTTLE SWITCH OFF

6. **Confirm all lights are selected off before Power-up Checklist.**

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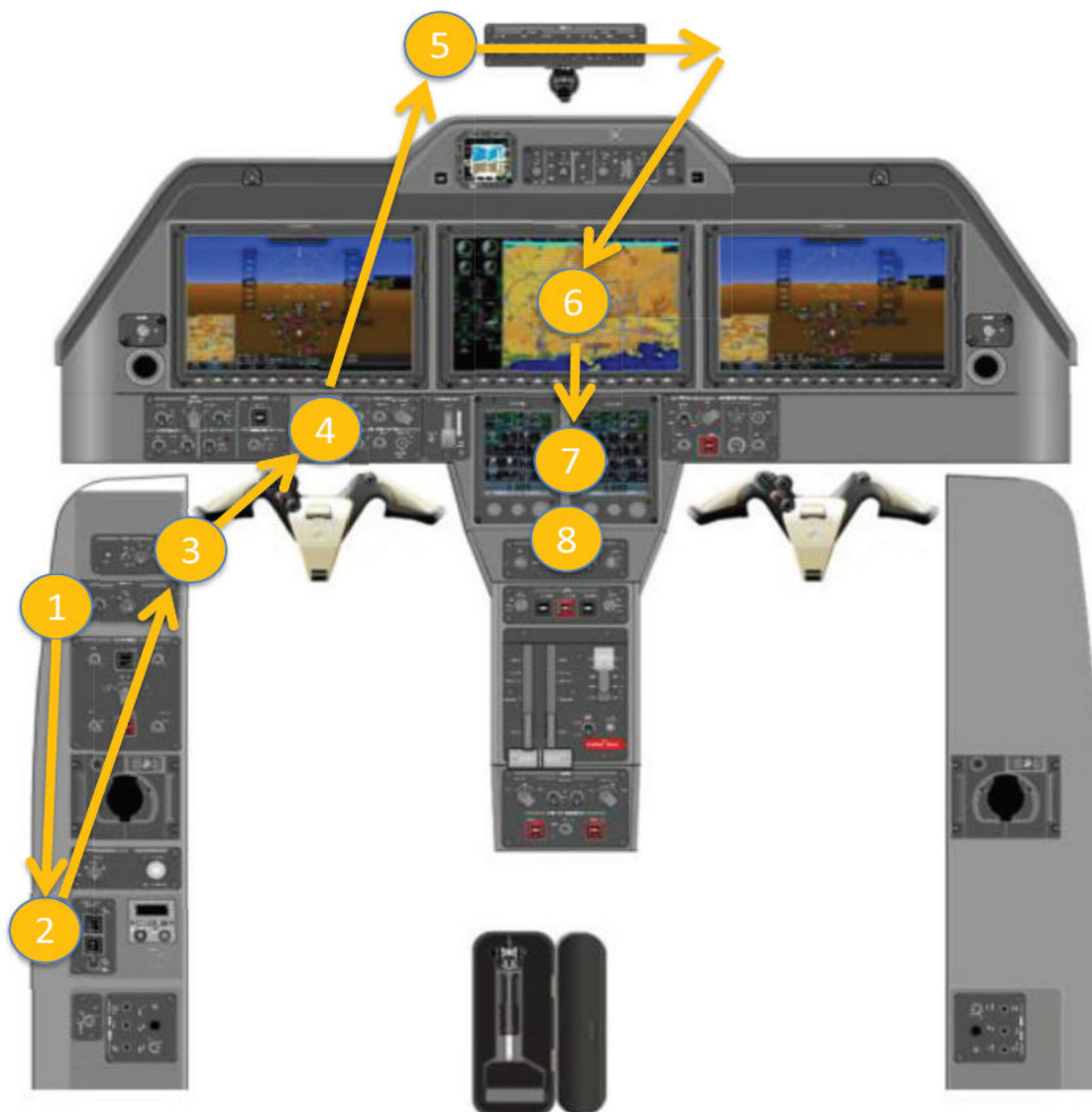
2.3.2.1.1 Power-Up Flow

The purpose of this flow is to prepare the cockpit for flight.

Trigger: The flow should be commenced once the cockpit preparation flow is complete.

If the aeroplane batteries are the power source, expedite the flow to prevent battery drain.

After initial battery power ON, do not touch any other switches or buttons while the aircraft is going through the initial power-up cycle (this is to avoid interrupting any internal Built-in Tests). You may acknowledge the splash page on the MFD but wait until all screens are showing normal flight indications before continuing with the Power-up Flow.



Expanded Power-up Flow

1. **BATT 1 and 2 SWITCHES** **ON**

Verify batteries voltage at or above 24 V. If battery voltage is below 19 V, it must be replaced. If battery voltage is below 24 V, report to the maintenance personnel or recharge the affected battery by using an external power supply (GPU).

Power up causes FD to become active, regardless of the last FD status selected.

Check for 'Aural Warning Test Ok' a few seconds after power-up.

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Automatically power-up tests: AP, YD, TAWS, Radar Altimeter.

Check for 'TAWS System Test Ok' after power-up.

NOTE

The aeroplane must not be moved during the first 90 seconds after powerup, while the IESI is undergoing alignment. Moving the aeroplane during this period can cause in-flight attitude indication errors that are not noticeable on ground.

GPU Button AS REQ

Confirm AVAIL light illuminated in GPU button.

When GPU is not available or is not necessary, maintain GPU button pushed out.

Confirm GPU voltage.

The minimum GPU voltage for batteries charging is 27 V. If the GPU voltage is less than 27V, it will still provide power to the aeroplane but the CAS message 'Batt Discharge' may be displayed because the batteries will not be charging.

If the GPU is outside the battery charging limits (27-29v), it is permissible to turn the batteries off until engine start and power the aeroplane from the GPU.

2. **# CVDR PANEL TESTED**

Set the toggle switch to TEST HOLD 5 SEC position and hold for 5 seconds. Verify no fail indication on CVDR panel annunciators.

O2 MASK FLOW and MIC TESTED

On AUDIO JACKS panel set MASK MIC switch to the ON position and speaker volume turned up.

With the mask stowed, keep the TEST / RESET button pressed to test the oxygen mask and activate the microphone. The flow indicator star momentarily illuminates and oxygen flow will be audible through cockpit speaker.

Release the button to terminate the test. Set MASK MIC switch to the OFF position. Deselect speaker on GTC and ensure mask regulator set to 100%.

Repeat for RHS mask.

O2 PRESS PSI

Check oxygen cylinder pressure on the MFD.

If in normal operating range (GREEN), the aeroplane has the minimum oxygen pressure for dispatch.

If in advisory operating range (WHITE), the minimum pressure for dispatch must be checked according to oxygen dispatch pressures table in POH.

If in cautionary operating range (YELLOW), no dispatch.

3. **+TEST PANEL ANTI-CLOCKWISE TESTED**

Set the TEST knob to the desired position and press start button.

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FIRE PROTECTION (FIRE) CHECK

Message 'FIRE, FIRE' sounds.

FIRE message shown on ITT dials.

ENG 1 and ENG 2 SHUT-OFF push buttons red and white lights illuminate. E1 and E2 FIRE CAS messages are displayed.

(SMK DET) CHECK

The CAS message 'BAG SMK' shall be displayed while the START button is pressed.

Message 'FIRE, FIRE' sounds.

Annunciator (ANN) CHECK

All push buttons illuminate and the parking brake light illuminate.

Check for aural warnings and CAS messages. Ensure all Push Buttons lights are extinguished on completion of test (parking brake light will remain on if parking brake is set ON).

The systems operate normally if not tested, but a CAS message may be displayed.

It is not possible to test the systems in flight.

4. SIGNS / OUTLET CHECKED and BELTS / ON

Cabin indications should be checked with switch in (# PED-) BELTS / OFF position before being set to BELTS / ON. This check should be carried out in conjunction with the lights check detailed below.

5. LIGHTS PANEL +TESTED AND A/R

For the first flight of the day, turn ALL lights on and carry out a visual check using the other crew member if they are carrying out the external check. On completion of the lights check, turn all OFF except NAV ON, EMER ARM, and COCKPIT lighting (if required).

6. +MFD DATABASES CHECKED

Ensure all databases are current. If any updates are required, see Phenom Fleet Handbook for updating procedures.

7. TRANSPONDER STBY

The Transponder defaults to the last code and AUTO on Power-up. Make sure 2000 and STBY is selected.

8. +TRIMS TESTED

Verify that yaw roll and pitch (NORM and BKP) trims are operating properly in both directions. Adjust yaw and roll trims to the neutral position.

Verify yoke pitch trim does not operate unless both halves operated, and ensure pressing TRIM DISC button stops pitch trim operation.

NOTE

If the two halves of the switch are operated in opposite directions, then the trim motor will still activate.

2.3.2.1.2 Power-Up Checklist

Power-Up Checklist

Trigger: The power-up checklist should be carried out prior to flight guidance programming.

COCKPIT PREPARATION FLOW..... COMPLETED

BATT 1 and 2 SWITCHES..... ON

GPU BUTTON..... AS REQUIRED

CVDR PANEL..... TESTED

O2 SYSTEM..... +TESTED /__PSI

+ TEST PANEL ANTI-CLOCKWISE..... TESTED

+TRIMS.....TESTED and CENTRED

NOTE

1. If external power is not available and it is necessary to reduce electrical loads on the batteries, the ELEC EMER button may be pushed in to reduce loads but still allow radio comms and all systems on the EMER BUS - FOL PHE505-001/13.
2. Do not move or tow the aeroplane while the AHRS is initialising. Allow 60 seconds for AHRS initialisation to avoid an ATT flag message appearing on the IESI as well as preventing heading drift to be displayed on the PFD prior to take-off.
3. Whenever possible, avoid powering up the aircraft near metallic buildings or objects that may cause magnetic interference. This will prevent magnetic heading drift and the ATT flag on the IESI. If this cannot be avoided, monitor the systems for correct operation during taxi.
4. If connected to external power and the output voltage is less than 27V, the batteries will not be charged and a BATT DISCHARGE CAS will be displayed. In this case, the batteries may be turned OFF (BATT 1 (2) OFF BUS CAS will appear) and the aeroplane can be powered by the GPU. Turn the batteries back on before Engine Start. Refer to FOL PHE505-001/13.

2.3.2.1.3 Flight Guidance Programming

If GPU is...	Then...
Available or battery voltage is sufficient.	Accomplish Flight Guidance programming.
Not available or insufficient battery voltage.	Delay until after 1st engine start.

NOTE

If the Flight Guidance Programming is completed prior to engine start, the flow above may be completed by a single crew member.

Phenom 300E - Any of the following items marked with § will be automatically accessed via the Initialisation process, and should be treated as a PF responsibility.

PF:**§ NETJETS DEFAULT CREW PROFILE ACTIVATE**

The NetJets Default Profile should load automatically (normally shown by both PFDs having split screens). If it does not load, select the 'NetJets Default Profile' on the GTC under the 'Crew Profile' button and 'Activate'.

§ FLIGHT PLAN INSERT

Insert the flight plan (consider saving the plan in the store if there is a possibility that the power may be removed from the aeroplane before start).

Missing or incorrect altitude constraints may be created for some procedures in the flight plan. Once the flight plan is inserted in the avionics, the pilot shall check the parameters loaded, including the altitude constraints, with the respective current chart procedure and in case of discrepancy, manually enter the correct value (ACFOG).

§ ENGINE DATA (OAT) SET

Set outside air temperature in the ENG SETTINGS page on GTC.

§ LANDING FIELD ELEVATION SET

Enter manually or use the FMS value.

WEATHER RADAR STANDBY

Select the radar to standby to initiate the warm up sequence. If carrying out a battery start, consider leaving the radar off until after first engine start to reduce load on the batteries.

§ WEIGHT PLANNING PROGRAM

Refer to [Mass and Balance](#) for further guidance.

AFCS CONTROL PANEL SET

Select PF using CPL button, then set desired FD modes.

TO / GA PRESS

Press to select take-off guidance on the PFD.

PFD SET

Ensure the PFD is set up for the departure. Consider setting the PM PFD to Green Needles to back up the FMS in case of failure.

PITCH TRIM SET FOR TAKE-OFF

Adjust pitch trim to take-off (green band) according to CG position.

ALTITUDE ALERT SET (WHEN CLEARANCE RECEIVED)

Leave AAD blank until departure clearance is received from ATC.

PORTABLE EFB PREPARED AND SECURE

Ensure both iPads have the flight loaded and Jeppesen charts available.

PM:

NAV 1 and 2 SET

COM 1 and 2 SET

ALTIMETERS SET x 3

§ TAKE-OFF AND RETURN SPEEDS SET and ON

Set and activate speed bugs for V_1 , V_R , V_2 and V_{FS} as per runway analysis, and set return speeds in case of an emergency return.

TRANSPONDER AND FLIGHT ID C / S and 2000 SET and STBY

Insert the Flight ID using the call sign shown on the OFP. Unless ATC clearance has already been received, insert European Unassigned IFR squawk 2000 and set mode to SBY. Once clearance and squawk are received, set code in the transponder and set to AUTO as aircraft taxis out.

BARO TRANSITION ALERT SET

Set the published Transition Altitude and ensure the Alert is Enabled.

TCAS § TESTED and A / R

Test the TCAS, (Aircraft Systems, System Tests on the GTC), and set either NORM or ABOVE depending on departure environment.

PFD SET

Ensure the PFD is set up for the departure. Consider setting the PM PFD to Green Needles to back up the FMS in case of failure.

2.3.2.1.3.1 Suggested Avionics Setup

The table below is a guide of how the screens should be set up for the various stages of flight. During start, taxi, and shutdown, the pages below should be displayed.

For other stages of flight the crew may display any screen or flight information as long as it is relevant to that particular stage.

During the flight, the PF needs to have control of the MFD to make sure all flight information is correctly displayed.

Any changes of display configuration on the MFD should be briefed between crew members.

Flight Phase	PFD 1		MFD		PFD 2	
Start	SID	Normal Flight Display	Elec	Status	Taxi / Parking chart	Normal Flight Display
Taxi Out	SID		Safe Taxi ²		See Note ¹	
Departure ³	Chart ³		Flight Plan Map		Chart ³	
Cruise	-		Flight Plan Map		-	
Descent	Arrival / Approach chart		Flight Plan Map		Arrival / Approach chart	
Taxi in	-		Safe Taxi		Taxi / Parking chart	
Shutdown	-		Status		-	

NOTE

The table above assumes that the PF is the LH pilot.

1. The Taxi / Parking chart shall be displayed on the PM side to aid monitoring of the taxi route. At a later stage in the taxi the SID chart should be displayed to aid with the Departure Review.
2. If safe taxi is not available, use any available taxi chart.
3. Display SID or other appropriate chart.

The map display on the MFD should be adjusted to show a suitable degree of detail for each departure, arrival and approach. If airspace adjacent to an airfield could have an impact on any prescribed manoeuvre then the map DETAIL scale should be adjusted to display the airspace restrictions on the MFD.

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2.3.3 Departure Briefing

Refer to OM-A, Operating Procedures.

Ensure the brief highlights and emphasises any hazards or threats to the departure. Consider briefing the technique to be used during low level offs, placing emphasis on preventing flap overspeed and avoidance of high rates of climb. Conventional navigation back-up shall be considered for each departure.

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2.4 Engine Start

2.4.1 Normal Start

2.4.1.1 Before Start

Both Pilots shall visually verify that no hazards exist that could affect the engine start. The safety area fore and aft of the engines must be clear, all doors and panels must be properly closed and locked where applicable.

Both flight crew members shall be on the flightdeck for the engine start sequence, the engine start procedure shall be carried out by a two-member crew.

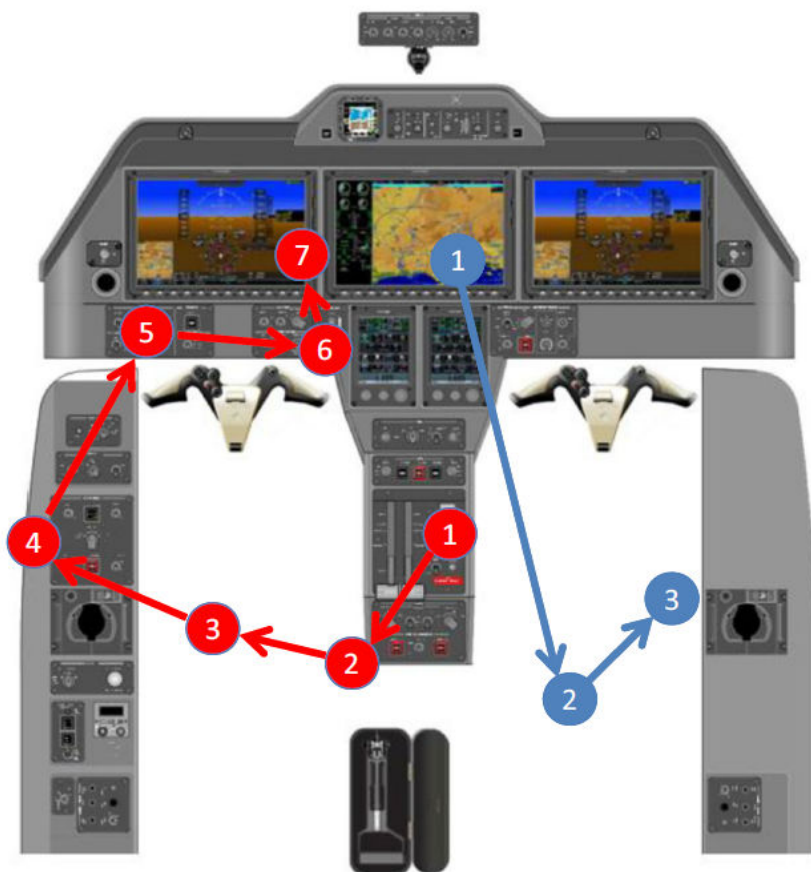
CAUTION

Items not addressed during pre-flight preparation (e.g. chocks, doors, parking brake, etc.) due to varying situations (e.g. towing, refuelling, etc.) must be accomplished prior to starting engines. If pushback is required, the procedures detailed in OM-A, Operating Procedures and OM-B, [Pushback Operations](#) shall be observed. Additionally the commander shall ensure that the ground personnel are fully aware how to reconnect the NLG Torque Link.

2.4.1.1.1 Before Start Flow

The purpose of this flow is to prepare for imminent engine start.

Trigger: The flow is actioned when both crew members are in the cockpit and start clearance has been received.



Expanded Before Start Flow**LHS:**

1. **PARKING BRAKE** **ON**
2. **START / STOP 1 or 2** **RUN**
Selects anti-collision beacon ON.
3. **SHOULDER HARNESSSES** **SECURE**
4. **MOBILE PHONES** **OFF**
5. **SIGNS / OUTLET** **BELTS / OFF**
..... **# ON / OFF**
6. **ADS PROBES** **AS REQUIRED**
If OAT 10°C or below and icing conditions exist, ADS PROBES should be set to ON before engine start.
7. **CAS** **A / R**

NOTE

Whenever a Checklist item requires a check of the CAS messages, the response shall be 'CLEAR' only if no CAS messages are displayed. Any visible CAS messages should be summarised verbally.

RHS:

1. **MFD STATUS AND ELEC PAGES (SPLIT SCREEN)** **SET**
 - Status on PM side - doors closed, no status messages.
 - Fuel - on board fuel matches required fuel for the intended flight.
 - Elec on PF side - display for monitoring engine start.
2. **SHOULDER HARNESS** **SECURE**
3. **MOBILE PHONES** **OFF**

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2.4.1.1.2 Before Start Checklist

Trigger: The before start checklist should be called for by the LHS once the flow is complete.

FLIGHT LOG / RELEASE.....COMPLETE
 PORTABLE EFB.....PREPARED and SECURE
 + PINS AND COVERS..... COUNTED and STOWED
 TORQUE LINK.....A / R
 SIGNS OUTLET SWITCH.....BELTS / OFF
ON / OFF
 FUEL..... __ LBS / BALANCED
 PARKING BRAKE.....ON
 ADS PROBES.....A / R
 CAS..... A / R

NOTE

Pins and covers need only to be checked for the first flight unless they have been replaced during the day before the current flight. The pitot covers must be removed and, along with the gear pins, shown to the other crew member prior to stowing in the nose with the gear pins.

LHS shall confirm the status of the Torque Link, either 'CONNECTED' or 'REMOVED FOR PUSHBACK'.

Starting the second engine causes the running engine to produce significant jet-blast as it achieves 72% N2. It is imperative that a good visual check of the area behind and on both sides of the aircraft is made before starting the second engine.

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2.4.1.2 Engine Start

LHS will start the engines.

The engine start sequence and parameters shall be silently monitored. The following standard callouts must be used:

SITUATION	LHS	RHS
When ready to start the right engine.	'STARTING 2'	
LHS checks left side and announces.	'LEFT CLEAR'	
RHS confirms right is clear.		'RIGHT CLEAR'
LHS starts engine 2 and guards ENGINE RUN switch. ^{1,2,3} Both pilots monitor start.	'STABLE'	Starts timing at FF indication for ITT rise within ten seconds.
When ready to start left engine.	'STARTING 1'	
LHS checks left side and announces.	'LEFT CLEAR'	
RHS confirms right is clear.		'RIGHT CLEAR'
LHS starts engine 1 and guards ENGINE RUN switch. ^{1,2} Both pilots monitor start.		Starts timing at FF indication for ITT rise within ten seconds.
After a successful start on both engines RHS check both engines are started and verbally state the respective N1's (no decimals or percentage).		'XX', 'XX'
	'START COMPLETE'	

¹ Starting engine, to be performed for each individual engine start.

LHS 1/2 ENG START / STOP Knob.....START / RUN

LHS 1/2 ENG START / STOP Knob.....GUARD

² During engine start:

ENGINE PARAMETERS..... MONITOR

³ Trigger: The stable call following the completion of Engine 1 start is the 'Trigger' for the PF to 'brief the glass' (if performed between starting engines).

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NOTE

1. For an external power assisted start, the No. 2 Engine should be started first and the external power source should be disconnected before the other engine is started. If, for any reason, the No.1 engine is started with external power connected, then the ground crew must be informed.
2. The FADEC will abort a ground start automatically for no light-up after 10 sec. of fuel flow initiation, a hot start or a hung start. However if a false start is recognised by either pilot, the start shall be aborted manually.
3. A tail wind above 10kt can lengthen the start-up and also a higher ITT temperature can be observed. In the event of an unsuccessful engine start a tailwind can also contribute to the accumulation of fuel in the combustion chamber which could lead to a hot start or tail pipe fire during a subsequent start. It is recommended, whenever possible, that the start-up be performed with headwind or cross wind conditions.

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2.4.1.3 After Start

2.4.1.3.1 After Start Flow

The purpose of this flow is to prepare the aeroplane for taxiing.

Trigger: The after start flow should be actioned after LHS calls 'START COMPLETE'.

After the Flight Guidance set-up is completed the PF shall 'brief the glass' to ensure the set-up is correct.

This can be done during the engine start in the two-minute wait between starting engines or after the engine start before the checklist is called for. An example of the brief would be:



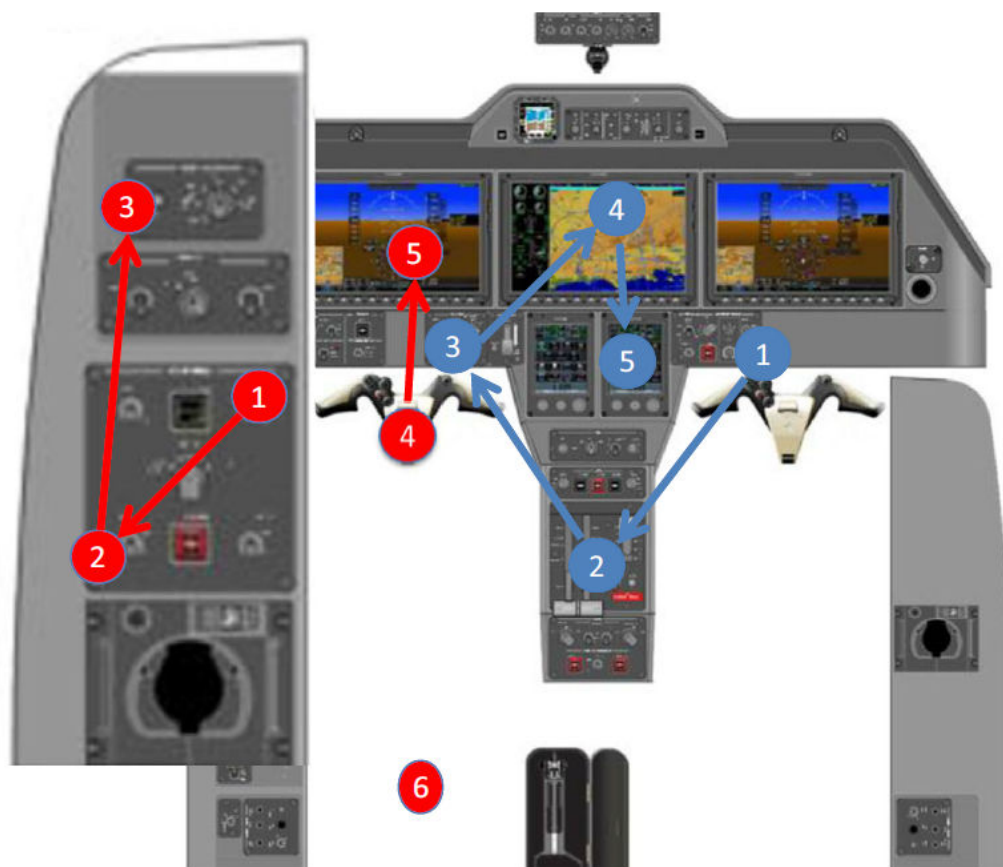
- Take-off Speeds ON.
- Target speed 150 kts set.
- FMS Pink, coupled right, TO Green ALTS white.
- COM 1 active.
- Initial Altitude 5000', Altitude 770' with QNH 1028 Set and verified.

NOTE

The verified part (QNH) could be either from ATIS, ATC, CONNEXT or other appropriate source.

- Compass HDG 186, HDG Bug 271.

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Expanded After Start Flow LHS:

1. **GPU BUTTON** **PUSH OUT**

Press GPU button, verify disconnected on ELEC synoptic, IN USE extinguished, AVAIL illuminated.

Signal Ground Crew to disconnect.

Verify GPU symbol removed from ELEC synoptic, AVAIL light extinguished and ground crew confirm disconnect.

2. **ELEC EMER** **TESTED**

Press ELEC EMER button and check:

DC BUS 1 disconnected from EMER BUS.

DC BUS 2 disconnected from CENTRAL BUS .

HOT BUS 2 connected to EMER BUS.

Allow time for voltage to stabilise.

BATT 1 and 2 minimum 23.5V.

Press again (out) and check:

DC BUS 1 connected to EMER BUS.

DC BUS 2 connected to CENTRAL BUS.

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HOT BUS 2 disconnected from EMER BUS.

3. TEST PANEL CLOCKWISE

Move yoke aft away from forward limit and hold in position.

STALL PROT **TEST**

Set the TEST knob to **STALL PROT** and press the start button.

Aural message 'STALL, STALL' sounds 4 times through cockpit speaker. On the 3rd callout, the stick pusher actuates.

Check for aural warnings and CAS messages.

The system operates normally if not tested, but a CAS message may be displayed.

It is not possible to test the systems in flight.

+ICE PROT **TEST**

Set the TEST knob to ICE PROT and press the start button. ICE CONDITION CAS message is displayed.

ICE DET FAIL CAS message may appear temporarily. If this message persists, consider the ice detector has failed.

NOTE

ICE CONDITION CAS may remain illuminated on completion of the test, but should clear after a short period of time.

4. **FLIGHT CONTROLS** **CHECKED**

Elevator: full up, full down.

Aileron: full left, neutral, full right and neutral.

Rudder: full left, neutral, full right and neutral.

(FFT) Both pilots should check visually for correct movement of ailerons and spoilers.

5. **CAS** **A / R**

6. **CABIN** **SECURE**

Ensure pax seated with belts on and cabin secure for taxi / departure.

RHS:

1. **AIR CONDITIONING** **A / R**

2. **FLAP** **SET for T / O**

3. **HEATING and ICE PROTECTION Panels** **+TEST**

- **ADS PROBES**..... **AUTO**

Return to AUTO position if ON used for engine start.

- Select WSHLD 1 and 2, ENG 1 and 2 and WING STAB ON.
- Check EICAS for the following:
 - WSHLD 1(2) HTR FAIL message not displayed;
 - A-I 1(2) message displayed after 10 seconds;
 - A-I WINGSTB ARM or A-I WINGSTB INHB message may be displayed.

All items should be selected OFF and all CAS messages must disappear to complete the TEST prior to setting up for ambient conditions below.

Select WSHLD 1 and 2, ENG 1 and 2 and WING STAB as required. If SAT is between 5° C and 10° C with visible moisture:

ENG 1 and 2 Switches.....ON

WINGSTAB Switch.....OFF

WSHLD 1 and 2 Switches..... OFF

If SAT is below 5°C with visible moisture:

WSHLD 1 and 2 Switches..... ON

ENG 1 and 2 Switches.....ON

WINGSTAB Switch.....ON

NOTE

A-I WINGSTB INHB CAS may be displayed if the test is done outside the icing envelope.

4. **Screens** **Set for Taxi**

Ensure ELEC EMER test is complete before changing MFD display.

5. **TRANSPONDER** **CODE SET and AUTO**

Check the ATC code and select to AUTO.

NOTE

1. After completion of STALL TEST RHS will deselect speaker on BOTH GTCs.
2. Select the Transponder to AUTO when ready to taxi or push-back.

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2.4.1.3.2 After Start Checklist**After Start Checklist**

Trigger: The after start checklist should be called for by the LHS once the flow is complete.

GPU (IF USED).....DISCONNECTED
 ELEC EMER..... TESTED
 STALL PROTECTION.....TESTED
 FLIGHT CONTROLS.....CHECKED
 ICE PROTECTION..... + TESTED and A / R
 FLAPS..... A / R
 PITCH TRIM....._
 ALTIMETERS..... QNH__ SET and BLUE
 FLIGHT GUIDANCE.....SET and CHECKED
 TRANSPONDER.....CODE SET and AUTO
 CAS..... A / R
 CABIN.....SECURE

2.4.2 Alternative Start Procedures

Reserved.

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2.5 Taxi

Before taxiing in or out of the parking area, crews shall identify any potential hazards on or adjacent to the planned taxi route, discussing mitigation as necessary. If at any stage a hazard is identified which encroaches within, or is likely to encroach within 3m (10ft) of any part of the aeroplane, the aeroplane shall be stopped and one or more of the following measures shall be taken:

- Request removal of the hazard;
- Request towing;
- Request ground personnel guidance;
- Use an alternative taxi route.

The Commander remains ultimately responsible for the safety of the aeroplane.

Taxi speed shall not exceed 30 knots (based on a Ground Speed readout) and shall be reduced as necessary for, passenger comfort, weather, surface conditions, or traffic, to ensure the aeroplane can be stopped safely.

2.5.1 Taxi Procedure

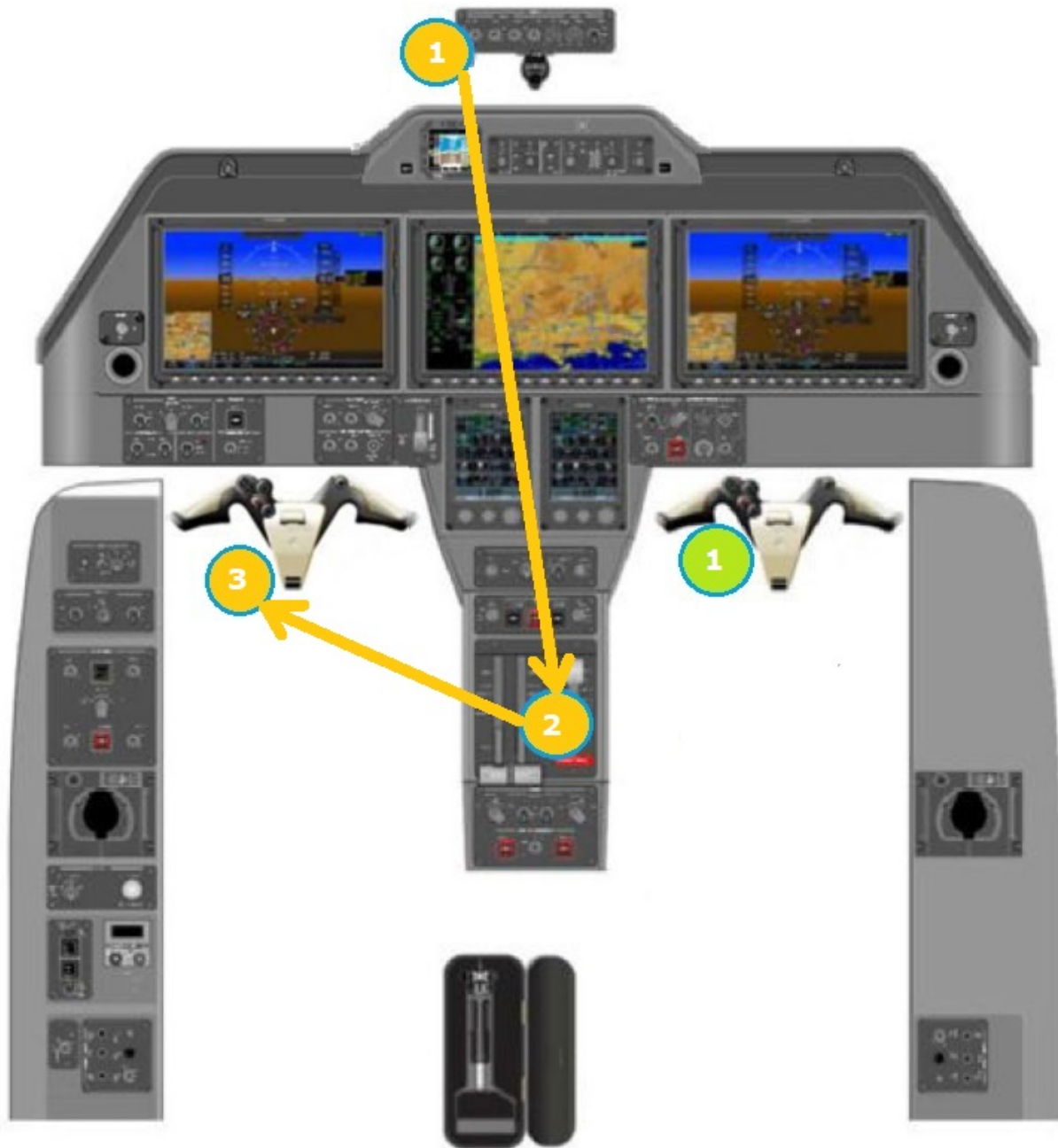
The following standard callouts and procedures must be used:

Situation	LHS	RHS
Cleared to taxi. PF: switches LDG / TAXI Light to TAXI		
Verify area on either side is clear	'LEFT CLEAR'	'RIGHT CLEAR'
	PF	PM
Parking Brake release by PF	'PARKING BRAKE OFF'	
Brake Check	'CHECK YOUR BRAKES'	'BRAKES CHECKED'

2.5.2 Taxi Flow

The purpose of this flow is to ensure the aeroplane is safe to taxi for departure.

Trigger: The flow is initiated when taxi clearance is received.



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Expanded Taxi Flow**PF:**

1. **LDG / TAXI LIGHT** **TAXI**
2. **PARKING BRAKE** **OFF**
3. **BRAKES** **CHECKED**

Brakes shall be checked by both pilots.

PM:

1. **BRAKES** **CHECKED**

Brakes shall be checked by both pilots.

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2.6 Line-Up

2.6.1 Line-Up Flow

The purpose of this flow is to ensure the aeroplane is correctly configured for take-off.

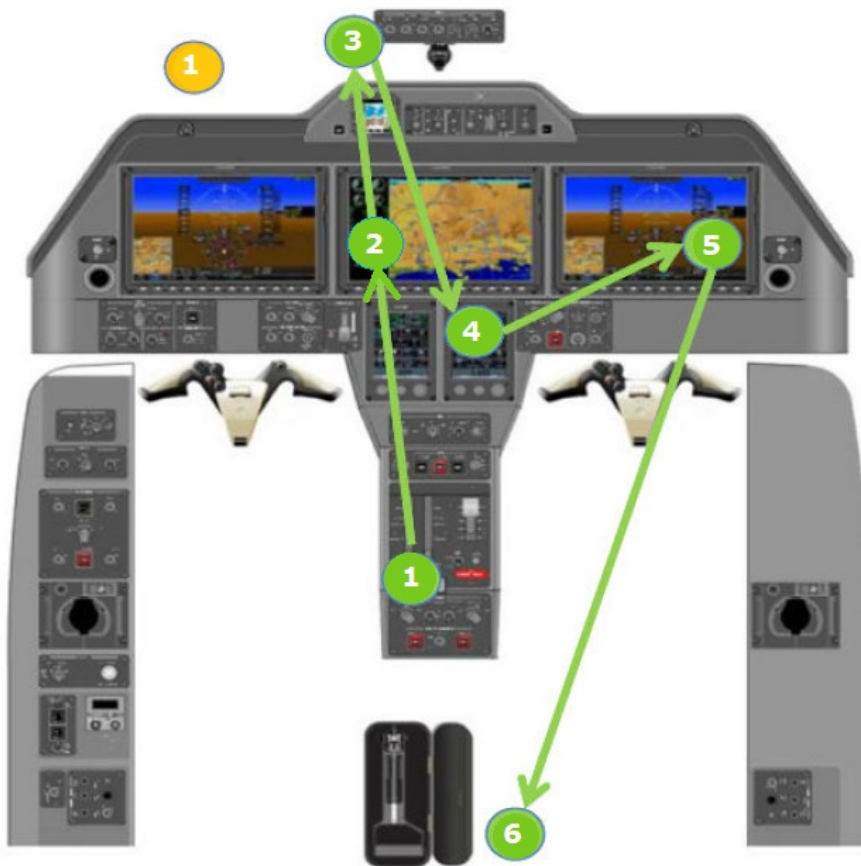
The PF will call for the Departure Review at a suitable point prior to the holding point.

Trigger: The trigger for the remaining flow items is receipt of a line-up clearance.

Both pilots shall confirm and verbally announce the runway and approach are clear before initiating line-up.

The Landing Lights will be selected on by the PM only after take-off clearance is received.

The PFDs / MFD should be set up appropriately for departure.



Expanded Line-up Flow**PF:**

1. **RUNWAY FOR DEPARTURE** **VERIFIED**

The correct runway shall be identified prior to crossing the holding point.

PM:

- DEPARTURE REVIEW** **COMPLETE**

The departure review shall be completed before takeoff to ensure both pilots share a clear understanding of the initial departure and aircraft setup.

The review should be initiated by the PF and shall be completed by the PM at a suitable point on the taxiway prior to the holding point—it must cover the initial heading or track, first turning point, initial assigned altitude or flight level, and any emergency or special procedures in accordance with OM-A. As part of the review, the PM must also confirm the active lateral Flight Director mode (HDG or FMS).

1. **TAKE-OFF CONFIGURATION** **OK**

Before entering the runway, press T / O CONFIG button on the central console and check the aural 'TAKE-OFF OK' sounds.

If any other announcement is heard, the aeroplane is out of the normal configuration and must be properly configured prior to take-off.

2. **ENGINE INDICATIONS** **CHECK**

Ensure matched indications within normal limits before entering runway.

3. **STROBE** **ON**

4. **RADAR** **A / R**

5. **CAS** **A / R**

6. **PASSENGERS** **ADVISE**

Advise pax of imminent departure, ensuring they are seated with belts on and tables stowed.

When take-off clearance is received, the PM will select the LDG / TAXI light to LDG.

NOTE

When lining up to the runway, avoid using differential thrust for ground manoeuvring. An N1 difference of more than 20% detected between the two engines may trigger the T / O Reserve (TO RSV) when the thrust levers reach TO / GA. If differential thrust is required for ground manoeuvring, allow time for N1 stabilisation before advancing thrust levers to TO / GA (ref: FOL PHE 505-010/11).

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2.6.2 Line-Up Checklist

Trigger: The Line-up checklist should be called for by the PF on completion of the flow.

DEPARTURE REVIEW..... COMPLETE
T / O CONFIG.....OK
ENGINE INDICATIONS..... CHECKED
STROBE..... ON
CAS..... A / R

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2.7 Take-Off

2.7.1 Normal Take-Off

The following standard callouts must be used:

SITUATION	PF	PM
Take-off clearance received and checklists complete. PF: Sets Thrust to the TO / GA detent. PM: Checks N_1 equal to N_1 target.	'THRUST SET'	'CHECKED'
Airspeeds Alive PM: Verifies all IAS indications are alive, crosschecks speed values.		
60 KIAS (see Note 1)	'CHECKED'	'60 KNOTS'
V_1		'V1'
V_R		'ROTATE'
Airborne PM: Verifies positive rate. PF: Verifies positive rate. PM: Moves LDG GEAR lever UP PM: Selects FLC, verifies FLC indication on PFD and ensures speed set at 150 KIAS. PF: Verifies FLC indication on PFD and speed 150 KIAS selected. PM: Verifies gear is indicating UP. PF: Verifies Gear is indicating UP.	'GEAR UP'	'POSITIVE RATE' 'FLC GREEN / PINK, SPEED 150' 'GEAR UP'
1000 ft AAL PF: Verifies IAS is within limits. Reduces thrust to CLB, or below, to adjust rate of climb or maintain airspeed as required. PM: Verifies IAS within limit, calls out present IAS and moves flap lever. PM: After selecting Flap Zero, selects and verifies YD ON, LDG / TAXI light to TAXI. PM: Verifies flap indications, calls new configuration. Checks thrust set to CLB or below. PF: Checks Flaps indicate zero.	'FLAP ZERO'	'1000 FEET' 'XXX' 'YD GREEN' 'FLAP ZERO'
Transition Altitude		

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SITUATION	PF	PM
PF: Sets Standard. PM: Sets Standard on own PFD and Stby Altimeter and crosschecks the altimeter indications.	'SET STANDARD'	'STANDARD SET AND BLUE'

NOTE

1. The '60' call is a combination of airspeed cross check and transition from low to high energy. At 60 KIAS all but the most urgent CAS are inhibited until 400 ft agl. The '60' call also act as an incapacity check in accordance with the standard 2 call rule and should be acknowledged by the PF. However, common sense should be applied before stopping the aircraft at high speed if the call was simply missed.
2. Minimum AP engage height 600' AGL (1000' AGL if OEI).
3. Set thrust levers to TO / GA for Take-off. If thrust levers are inadvertently set to MAX position, this will engage the TO RSV which will reduce the lifetime of engine components. In addition it may trigger ECS 1 (2) OFF, which may result in higher cabin altitude. Ref FOL PHE505-010/11.

2.7.1.1 Take-Off Technique

Rotate in a smooth and controlled manner to approx. 15 degrees nose-up initially.

After retraction of the landing gear, pitch up to a maximum of 20 degrees nose up to maintain airspeed between $V_2 + 10$ and 150 KIAS.

NOTE

Crew should limit the pitch angle to 20 degrees nose up for passenger comfort. In addition, the FD will not command higher than 20 degrees nose up, therefore a reduction in thrust may be necessary to control IAS during take-off.

Target speed at 1000 ft is 150 KIAS.

At 1000 ft AAL, and above $V_2 + 10$, select flaps 0.

NOTE

If a flap 2 take-off is performed, ensure that when flap 0 is requested the flap selector is moved via the flap 1 detent to Flap 0.

Below 1000 feet AAL movement of any switches or levers should be restricted to those that are necessary for safe flight (e.g. landing gear and AFCS selections).

The aeroplane performance will require proactive energy management to achieve a controlled rate of climb approaching level off.

2.7.1.2 Gear and Flaps Overspeed Protection

Prior to completion of the Flap Zero closed loop procedure, remain in FLC mode and do not select the speed target bug above 170 KIAS (160 KIAS for flap 2 take-off). VS mode should not be engaged until the Flap Zero closed loop procedure has been completed.

If a flap 2 departure is flown, crew should be aware that the flap retraction time will be significantly increased.

Refer to [Exceeding Speed Limitations](#) for actions in case of an inadvertent Gear or Flap overspeed.

2.7.2 Crosswind Take-Off

The maximum demonstrated crosswind for take-off is 25 kts. This is to be regarded as limiting.

The crosswind limit for STATIC Take-off is 18 kts.

Refer to OM-A, Operating Procedures for reduced crosswind limits when braking action is less than 'good'.

2.7.3 Take-Off on Wet Runway

No specific techniques apply to taking off on a wet runway.

2.7.4 Noise Abatement

The departure procedure as depicted in [Normal Take-off](#) and the profile 2-100 A Normal Take-off and Noise Abatement meets the requirements for a combined NADP 1/ NADP 2 in accordance with ICAO Doc.8168.

2.7.5 Departure with level-off at or below 3000 ft AAL

A standard noise abatement profile should be followed until reaching 1000' AAL. At 1000' AAL the following technique is recommended:

- Retract the flaps as normal;
- Reduce thrust to approx. 65% N1;
- Lower the nose to approx. 10 degrees nose-up.

NOTE

1. The power setting and pitch attitude above are approximate and will vary with differing conditions and aeroplane weights.
 - Do not make any AFCS vertical mode changes once 'ALTS' is green as this may prevent capture of the selected altitude.
2. Should there be a requirement to level off significantly below 3000' AAL, the thrust may need to be reduced before reaching 1000'AAL in order to control the aeroplane's energy. This early power reduction should be briefed in the departure briefing and in all cases the minimum height for thrust reduction is 400' AAL.
3. Similarly, should there be a requirement to level off significantly below 3000' AAL, crew may consider an earlier flap retraction altitude as long as all performance considerations are taken into account and that the flap retraction altitude is clearly briefed during the departure briefing. In all cases, the minimum height for flap retraction is 400' AAL.

2.7.6 Wake Turbulence Separation - Departure

Refer to OM-A.

Under current wake turbulence standards for departure, the Phenom 300 is classified as follows:

- ICAO - Medium (M);
- RECAT EU - Light (F).

Table 1: ICAO Wake Turbulence Time Based Separation on Departure

Leading aircraft	Departure Separation Time	
	Departing from same position, or from a parallel runway separated by less than 760 m (2500 ft).	Departing from an intermediate point on the same runway, or from an intermediate point of a parallel runway separated by less than 760 m (2500 ft).
A380-800	3 minutes	4 minutes
Heavy	2 minutes	3 minutes

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Table 2: RECAT-EU Wake Turbulence Time Based Separation (time in seconds) on Departure

Leading aircraft	Departure Separation Time(s)	Departure Separation Time(s)
	Departing from same position, or from a parallel runway separated by less than 760 m (2500 ft).	Departing from an intermediate point on the same runway, or from an intermediate point of a parallel runway separated by less than 760 m (2500 ft).
Super heavy	180	240
Upper heavy	140	200
Lower heavy	120	180
Upper medium	120	180
Lower medium	100	160
Light	80	140

The minima specified in both tables 1 and 2 above apply when the aeroplanes are using:

- The same runway;
- Parallel runways separated by less than 760 m (2500 ft);
- Crossing runways if the projected flight path of the second aeroplane will cross the projected flight path of the first aeroplane at the same altitude or less than 300 m (1000 ft) below;
- Parallel runways separated by 760 m (2500 ft) or more, if the projected flight path of the second aeroplane will cross the projected flight path of the first aeroplane at the same altitude or less than 300 m (1000 ft) below.

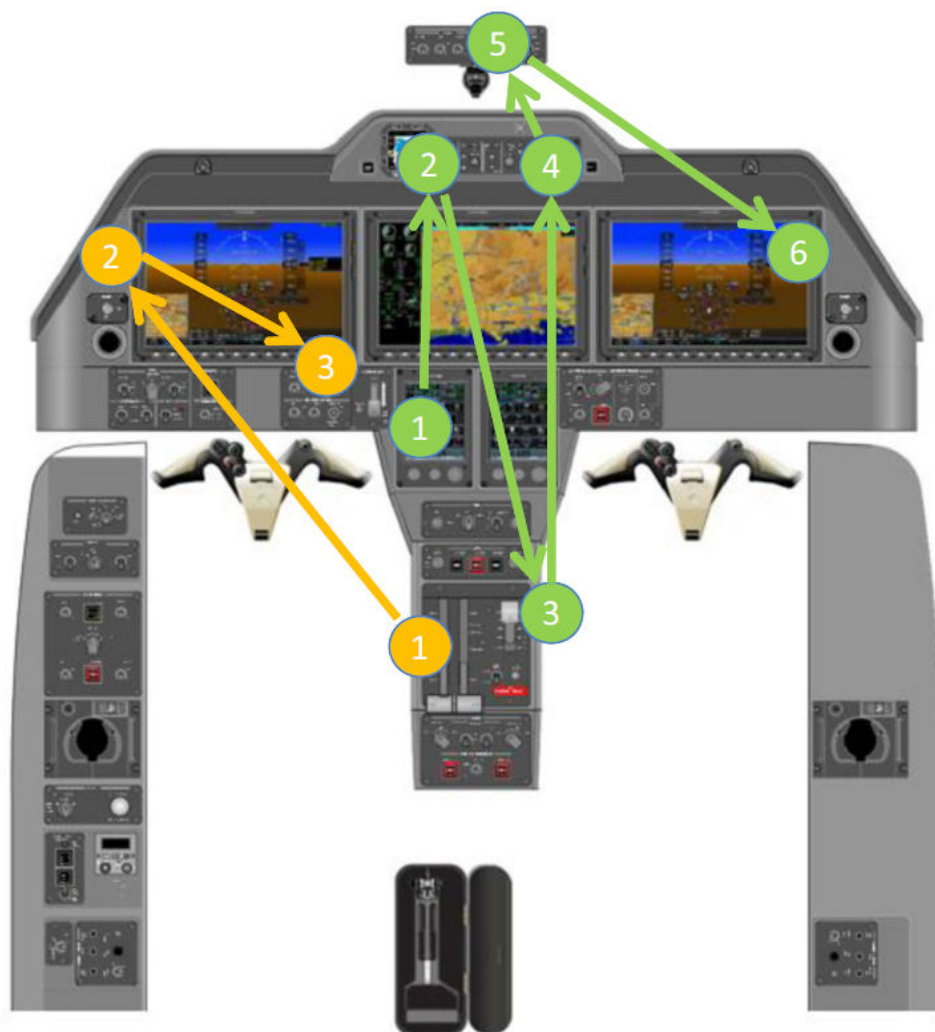
2.7.7 After Take-Off / Go-Around

2.7.7.1 After Take-Off / Go-Around Flow

The purpose of this flow is to configure the aeroplane for the climb, and it is embedded within Take-off Procedure.

After Take-off / Go-around Flow

Trigger: The After Take-off Flow is commenced when the PM calls 'positive rate' and the PF calls for 'Gear Up'.



Expanded After Take-off / Go-around Flow

PM:

1. **LANDING GEAR** **UP**
 PM commands landing gear UP after PF has requested.
 PM confirms landing gear indicates up and locked.
 While the gear is traveling up, PM selects FLC.
2. **FLC** **ENGAGE**
 Set / confirm speed at 150.
 If conducting a Go Around, PM selects / verifies SPEEDBRAKE switch to CLOSE and verifies indication on MFD.
3. **FLAP** **0**
 Speed shall be maintained below 170 KIAS (160 KIAS for flap 2 take-off) until Flaps are confirmed 0.
4. **YAW DAMPER** **ON**

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Press YD button on AFCS control unit or engage AP on PF command. AP / Yaw Damper should not be engaged below 600 ft AAL.

5. **LDG / TAXI LIGHT** **TAXI**

6. **CLIMB THRUST**

Checks that climb thrust is set to CLB or below.

7. **ALTIMETERS** **STD SET AND BLUE**

At transition altitude, or when cleared to a flight level, set own PFD and Stby Altimeter to standard pressure.

PM crosschecks altimeter settings and indications.

PF:

1. **THRUST LEVERS** **CON / CLB**

Thrust is controlled by PF to control rate of climb, with CON / CLB as a maximum setting.

2. **ALTIMETER** **STD SET AND BLUE**

3. **ICING CONDITIONS** **VERIFY**

2.7.7.2 After Take-Off / Go-Around Checklist

Trigger: The After Take-off Checklist should be called for by the PF upon reaching the first level-off altitude or at a suitable point during the initial departure.

LANDING GEAR.....UP

FLAP..... 0

YAW DAMPER.....GREEN

ALTIMETERS..... STD SET and BLUE

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2.8 Climb

The Embraer POH CLIMB data, that is used to generate our OFP, is calculated using 225 KIAS. Therefore, under normal conditions, it is recommended to fly 225 KIAS until transition to Mach number at nominally 29,600 ft, followed by M 0.60. Other climb speed profiles are permitted to comply with ATC and regulatory airspeed restrictions.

The normal Flight Director modes for climbing are FLC and VS. Basic modes should only be used if the aircraft is not following the correct flight path using either FLC or VS.

Maximum thrust setting during climb is CON / CLB.

These speeds may be varied to suit operational requirements as long as all the aircraft limits are observed.

2.8.1 Climb Flow

The purpose of the climb flow is to ensure the aeroplane is correctly configured for climb to higher levels.

Climb Flow

Trigger: The flow is initiated when passing FL 100 or Transition Altitude, whichever is higher.



Expanded Climb Flow**PM:**

1. **LDG / TAXI LIGHT** **OFF**
2. **PRESSURISATION** **CHECKED**
 Checks Cabin Altitude.
 Checks Cabin rate of change indication.
 Checks Differential Pressure reading displayed in green. No AMS CAS Messages are displayed.
3. **WINDSHIELD HEAT** **ON**
4. **COM 2** **121.5**
5. **RVSM CHECK** **PERFORM IF APPLICABLE**
 Check primary altimeters read within 200 ft of each other for RVSM compliance.
6. **CPDLC LOGIN** **IF REQUIRED**

PF:

1. **SIGNS / OUTLET** **BELTS / ON**
 **# ON / ON**

2.8.2 Climb Checklist

Trigger: The Climb Checklist should be called for by the PF on completion of the flow.

PRESSURISATION.....CHECKED

WINDSHIELD HEAT.....ON

SIGNS OUTLET SWITCH.....BELTS / ON

..... # ON / ON

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2.9 Cruise

Cruise speed may be Long Range Cruise, M 0.70 Cruise, M 0.74 Cruise or Maximum Speed Cruise. Performance data for these speeds is provided in the POH.

At the top of climb the Signs Outlet Switch may be selected to OFF / ON at the discretion of the Commander. However, it shall be recommended that passengers keep their seat belts fastened even if the FASTEN SEAT BELT sign has been switched off, in case unexpected turbulence (weather or wake) is encountered.

2.9.1 Cruise Flow

The purpose of the cruise flow is to ensure that the aeroplane is correctly configured for high level cruise and is also achieving the expected performance parameters.

The flow should be routinely repeated in the cruise as an airmanship consideration.

Cruise Flow



Expanded Cruise Flow**PF:**

1. **Thrust** **SET**

NOTE

Even though the CON / CLB setting is not a time limited rating, it produces more thrust at a higher ITT when compared to the MAX CRUISE setting. In order to decrease engine wear, a reduction to MAX CRUISE or below should be achieved within 5 minutes. For reference, see FOL PHE505-010/11.

PM:

1. PRIMARY ALTIMETERS..... CHECK

Check primary altimeters read within 200 ft of each other for RVSM compliance.

2. PRESSURISATION..... CHECK

Delta-P and Cabin Alt indications Green.

3. FUEL..... CHECK and NOTE

Enter the actual fuel quantity on the OFP and compare to the expected fuel quantity. Update the fuel quantity in the Weight and Fuel page on the GTC using the FOB SYNC, if necessary.

4. TCAS..... A/R

Select whichever mode offers the best SA assistance.

These routine system checks are to be carried out at regular intervals during the cruise, as defined in OM-A Chapter 8.

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2.10 Arrival Preparation and Briefing

Refer to OM-A, Operating Procedures.

Only those approaches that are available will be displayed when attempting to insert an approach into the flight plan. If certain GPS parameters (SBAS, RAIM, etc.) are not available, some published approach procedures for the desired airport may not be displayed. It is, therefore, advisable to programme the approach with adequate time to consider contingencies should this issue arise.

The flap configuration to be used for landing must be briefed during the arrival preparation.

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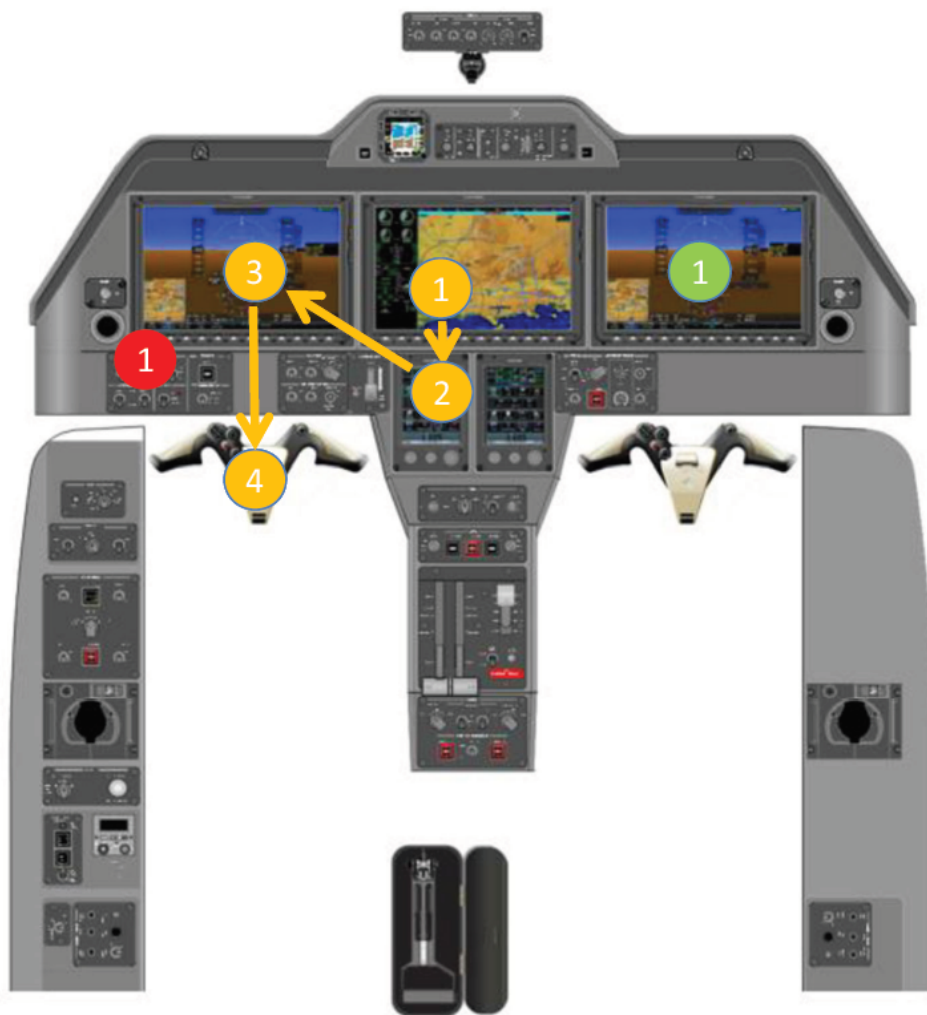
2.11 Descent

The Embraer POH Descent Data, that is used to generate our OFP, is calculated using M0.75 above 35,600 ft and 250 kts below until 1500 ft. These speeds may be varied to suit operational requirements as long as all aircraft limits are observed.

2.11.1 Descent Flow

The purpose of the Descent flow is to prepare the aeroplane for arrival.

Trigger: The flow should be carried out prior to leaving the en-route cruise altitude.



LHS:

1. **SIGNS OUTLET SWITCH** **BELTS / ON**
..... **# ON / ON**

PF (may be delegated):

1. **LFE** **SET**
Verify landing field elevation (LFE) is set in the Aircraft Systems page.
2. **LANDING SPEEDS** **SET and ON**
Set and activate speed bugs for V_{AP} , V_{REF} , V_{AC} , and V_{FS} as per runway analysis.

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3. **FLIGHT GUIDANCE / PFD** **SET A/R**

Flight Guidance includes the approach PROC being loaded, NAVAIDS set to correct frequency, BARO MIN being set and BARO TRANSITION ALERT set to destination Transition Alt / Level.

4. **ARRIVAL BRIEFING** **COMPLETE**

As per OM-A, Operating Procedures.

PM:

1. **PFD** **SET A/R**

2.11.2 Descent Checklist

Trigger: PF should call for the Descent Checklist on completion of the descent flow.

LFE.....SET
 FLIGHT GUIDANCE.....SET and CHECKED
 ARRIVAL BRIEFING..... COMPLETE
 SIGNS / OUTLET SWITCH..... BELTS / ON
 # ON / ON

2.11.3 Holding

Refer to Jeppesen Airway Manual for entry procedures and speeds. Normal Holding Speed is 170 KIAS.

In order to achieve maximum fuel economy, holding should be performed in accordance with the Holding Configuration Tables in the POH. Note the speeds shown are based on zero bank angle, a buffet margin of 1.3, and Ice Protection OFF. This configuration must be taken into account if using these lower speeds for fuel saving. A guide to the % of fuel increase for Anti-ice ON is shown at the bottom of the POH Holding tables.

The Final Reserve Fuel of 30 mins shown on the OFP is taken from these tables in the POH.

CAUTION

When holding in icing conditions the minimum airspeed should be 150 KIAS provided the WHSAIS is ON or 165 KIAS if it is inhibited or armed and consideration should be given to flying an even higher speed to minimise the ice build-up on unprotected surfaces - this is likely to increase the holding fuel usage to above the figures quoted in the POH. At 150 KIAS with WHSAIS ON the aircraft has the capability to fly a 40° AOB turn without stall warning activation.

During winter operations it may be prudent to consider additional fuel uplift to account for extra fuel burn in icing conditions.

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2.12 Approach General

2.12.1 Approach Category

The normal approach category for the EMB-505 is Category B; however, at higher weights and with WINGSTAB ON the aeroplane may enter Category C classification.

The Garmin Navigation database is programmed with BOTH Category B and C procedures.

2.12.2 Approach Manoeuvring Speeds

For manoeuvring prior to final approach, the following minimum airspeeds should be respected to provide adequate margin above the stall:

Flaps 0:	150 KIAS
Flaps 1:	140 KIAS
Flaps 2/3:	$V_{ref} + 10$ KIAS
Flaps Full:	$V_{ref} + 10$ KIAS

These speeds provide the following stall protection:

- Flaps 0 and 1: A coordinated turn using a maximum of 40° bank without the stall warning activation.
- Flaps 2/3 and Full: 1.3 x Stall Speed at 30° angle of bank.

2.12.3 Configurations Gates

The following table details the standard low drag, low noise arrival and approach configuration gates.

The approach configuration gates are the latest the airplane shall be configured to minimise the risk of an unstable approach. The crew may configure at any time earlier in order to best manage aeroplane energy. In any event, any more restrictive speed restriction shall be respected.

Latest Position	Configuration	Max IAS (kts)
LOC Intercept		210
GS / VP		180
6 NM to threshold	F1	160
5 NM to threshold	Gear Down, F2	160
4 NM to threshold	F3 / Full	150

At or approaching the configuration gates specified in the table above, the PM shall call deviations from the intended flight path to the attention of the PF in accordance with the callout table in OMA 8.3.19.3 'Monitoring and Crosschecking', and hereby applying the OMA 8.3.19.2 'By exception Rule Policy'.

NOTE

Call by Exception:

At each configuration gate, the PM shall make a call only if the aircraft is not in the required configuration or not within the applicable target speed.

If the aircraft is not in the correct configuration □ 'Check configuration.'

If the aircraft speed exceeds the target □ 'Check speed.'

These calls prompt the PF to verify and, if necessary, correct the aircraft configuration or speed.

2.12.4 Final Approach Target Speed and Gust Corrections

In order to provide an adequate margin to stall and to ensure landing performance, the following target speeds shall be used during the final approach.

TARGET SPEEDS

	Fully Configured – 500' AAL	500' AAL - Threshold
V_{target}	V_{ap} + GC	V_{ref} + GC

Gust Correction GC

SPEED BUG SETTING

VREF, VAc and VFS speed bugs shall be set in accordance with the expected landing weight.

VAP shall normally be set to VREF + 10 kt.

When wind or gust corrections are applied, the speed bug shall be set to the applicable target speed.

WIND AND GUST CORRECTION

Wind and Gust corrections to VAP shall be calculated as follows:

50% of the steady headwind component.

100% of the gust increment.

The maximum total correction shall not exceed VREF + 20kt.

If the calculated correction results in a VAP greater than VREF + 10 kt, PM airspeed deviation callouts shall be referenced to the revised VAP, the speed bug shall be sent to the revised VAP, and programmed speeds shall remain unchanged.

SPEED MANAGEMENT BELOW 500 FT AAL

VAP + GC shall be maintained until approaching 500 ft AAL.

At 500 ft AAL, or upon PF command, the PM shall set the speed bug to VREF + GC.

When the 'Continue' call is made at minimums, all subsequent airspeed deviation callouts shall be referenced to VREF + GC and speed shall be adjusted to ensure to runway threshold is crossed at VREF+ GC, with a maximum gust correction of 10 kt at 50 ft above TDZE.

GUST CORRECTION AND LANDING DISTANCE

When a target speed greater than VREF is required at the runway threshold, the landing distance required (LDR) shall be corrected accordingly and shall not exceed the landing distance available (LDA), in accordance with OM-A.

VISUAL APPROACHES

For a visual approach, the target speed of VREF + GC shall be set at 500 ft unless a different altitude is briefed by the PF.

2.12.5 Approach Automation Modes and Settings

APPROACH AUTOMATION & SETTINGS									
APP CLASSIFICATION	PRIOR FAF	NAV AID		LAT MODE (FMA)		VERT MODE (FMA)		HSI ANNUN	MINIMUMS
APP TYPE		LAT	VERT	PRIOR TO FAF	INSIDE FAF	PRIOR TO FAF	INSIDE FAF		
CAT 1		LOC	GLIDESLOPE	FMS or HDG	LOC	ALT	GS	LOC 1 (2)	DA
ILS									
CAT 1		RADAR-ATC	RADAR-ATC	HDG	HDG	ALT	VS		DA
PAR									
NPA		LOC	BARO MAN	FMS or HDG	LOC	ALT	VS	LOC 1 (2)	DA/MDA +50
LOC									
NPA		RADAR-ATC	RADAR-ATC	HDG	HDG	ALT	VS		DA/MDA +50
SRA									
NPA		FMS	BARO MAN	FMS	FMS	ALT	VS	FMS LNAV	DA/MDA +50
VOR (DME)									
NPA	GREEN NEEDLES	VOR ²	BARO MAN	VAPP	VAPP	ALT	VS	VOR 1 (2)	DA/MDA +50
VOR (DME)									
NPA		FMS	BARO MAN	FMS	FMS	ALT	VS	FMS LNAV	DA/MDA +50
NDB (DME)									
NPA	GREEN NEEDLES	NDB ²	BARO MAN	HDG	HDG	ALT	VS		DA/MDA +50
NDB (DME)									

APPROACH AUTOMATION & SETTINGS									
APP CLASSIFICATION	PRIOR FAF	NAV AID		LAT MODE (FMA)		VERT MODE (FMA)		HSI ANNUN	MINIMUMS
APP TYPE		LAT	VERT	PRIOR TO FAF	INSIDE FAF	PRIOR FAF	INSIDE FAF		
APV		FMS ³	FMS	FMS	FMS	VPATH or ALT	GP	FMS LPV	LPV DA
LPV									
APV		FMS ³	FMS Or BARO VNAV	FMS	FMS	ALT	GP	FMS L/VNAV	LNAV/VNAV DA
LNAV/VNAV									
NPV		FMS ³	VNAV Or BARO MAN	FMS	FMS	VPATH or ALT	GP /VS	FMS LNAV	LNAV DA +50'
LNAV (See Note 4)									

NOTE

1. If circling is to be flown following an ILS the automation shall be set up as for a LOC and the AAD set to circling altitude.
2. If an approach is not selectable in the FMS procedure list the PF shall have his navigation source selected to green needles. If an approach is selectable from the FMS procedure list it may be flown in overlay and the PF shall have FMS as his navigation source and shall have the relevant navigation aid displayed on a bearing pointer.
3. For all RNAV approaches, a suitable display shall be selected to show aircraft position relative to the lateral and vertical (if appropriate) paths.
4. For LNAV approaches the Garmin can show LNAV+V in the HSI. The +V designation adds vertical guidance for assistance in maintaining a constant vertical glidepath. The guidance is internally generated by the Garmin but can be coupled to if LNAV+V is annotated in the HSI prior to the FAF. If just LNAV is annotated in the HSI then VS will need to be used.

2.12.6 Approach Configuration Callouts

The following standard callouts must be used:

SITUATION	PF	PM
Passing 10,000 ft or FL100 in the descent PF: States current speed and if above 250 KIAS takes positive action to reduce below 250 KIAS.	'XXX'	'FL100' or '10,000 FEET'
Altimeter setting to QNH / QFE PF: Verifies conditions for setting altimeters to QNH / QFE are met. PM: Verifies conditions for setting altimeters to QNH / QFE and crosschecks the altimeter pressure setting.	'SET QNH / QFE'	'QNH / QFE XXXX SET AND BLUE'
Flaps configuration change PF: Verifies IAS within limit. PM: Verifies IAS within limit, calls out present IAS and moves flap lever. PM: Verifies flap indications and calls new configuration. PF: Verifies flaps position indication.	'FLAP [X]'	'XXX' 'FLAP [X]'
Lowering landing gear PF: Verifies IAS within limits. PM: Verifies IAS within limits, calls out present IAS and moves LDG GEAR lever DN. PM: Verifies gear is indicating down. PF: Verifies gear is indicating down.	'GEAR DOWN'	'XXX' 'GEAR DOWN'

2.12.6.1 Setting of Missed Approach Altitude

When in ALT mode prior to intercepting the final approach glidepath, PF or PM to set MAA. If not completed prior, PM shall set MAA at initiation of final descent.

2.12.6.2 Callouts at DA / MDA

Crews should set Barometric MINIMUMS for all instrument approaches. This will provide an automatic callout of 'MINIMUMS, MINIMUMS' at the selected Altitude. Should the callout not be made automatically, the PM shall call 'DECIDE' at DA / MDA.

On hearing the callout of 'MINIMUMS' / 'DECIDE', PF shall state whether to 'CONTINUE' or 'GO-AROUND, FLAP 1 or 2'.

2.12.7 Wake Turbulence Separation - Final Approach

Under current wake turbulence standards for departure, the Phenom 300 is classified as follows:

- ICAO - Medium (M);
- RECAT EU - Light (F).

Table 1: ICAO Distance-based Wake Turbulence Separation on Final Approach

Leading Aeroplane	Wake turbulence Separation Minima Distance (NM)
Super	7
Heavy	5
Medium	3
Light	3

Table 2: RECAT-EU Distance-based Wake Turbulence Separation on Final Approach

Leading Aeroplane	Wake Turbulence Separation Minima Distance (nm)
Super heavy	8
Upper heavy	7
Lower heavy	6
Upper medium	5
Lower medium	4
Light	3

The minima specified in both tables above shall be applied when:

- An aeroplane is operating directly behind another aeroplane at the same altitude or less than 1000 ft below; or
- An aeroplane is crossing behind another aeroplane at the same altitude or less than 1000 ft below; or

- c. Both aeroplanes are using the same runway or parallel runways separated by less than 760 m (2500 ft).

Time-based separation (TBS) for final approach:

The TBS minimums provide consistent time separation which is equivalent to the existing distance-based separation minimums in a light headwind - in stronger winds this will result in a smaller physical distance between aircraft.

Accurate speed control is essential for TBS to work, therefore crew should ensure strict adherence to ATC speed instructions and advise ATC immediately in the event they are not able to comply with a given speed restriction.

The TBS minimums for the UK are shown in the table below. Irrespective of headwind conditions, an absolute minimum separation of 3 NM remains.

Leading Aircraft	Wake Turbulence Separation Time(s)
Super heavy A380	158
Upper heavy >136000kg	158
Upper medium 104000-136000kg	135
Lower medium 40000-104000kg	113
Small 17000-40000kg	90
Light 17000kg or less	# ¹

¹ The '#' signifies that separation for wake turbulence reasons alone is not necessary.

2.12.8 Flaps and Gear Extension

In order to prevent flap and gear overspeeds, a 10 kt buffer should be applied to the respective limits when making gear and flap selections, while respecting minimum manoeuvring speeds.

2.12.9 Navigation Aids

NAVAIDs used for departure, arrival, and approach will normally be identified by the correct station ID being displayed in green on the PFD or NAV 1/2 selection on the GTC. Checking this annotation is sufficient to confirm the navigation aid. However, if required, the aural Morse code ident may still be used.

NAVAIDS required for the descent and expected approach should ideally be set, briefed and crosschecked before the TOD.

When cleared to establish on an instrument approach, the PM shall check that the appropriate NAVAIDS are tuned and identified, calling any discrepancies by exception.

2.12.10 Warning Flags

A flashing yellow mode annunciation indicates loss of sensor (ADC) or navigation data (VOR, LOC, GPS, VNV) required to compute FD commands. When such a loss occurs, the system automatically begins to roll the wings level (enters ROL Mode) and / or maintains the pitch angle (enters Pitch Mode), depending on the affected axis.

The flight director is automatically disabled if the attitude information required to compute the selected flight director modes becomes invalid or unavailable.

Should a sensor or input fail, a miscompare alert may also be displayed on the PFDs - all indications should be included within any subsequent fault analysis.

2.12.11 Approach Plate

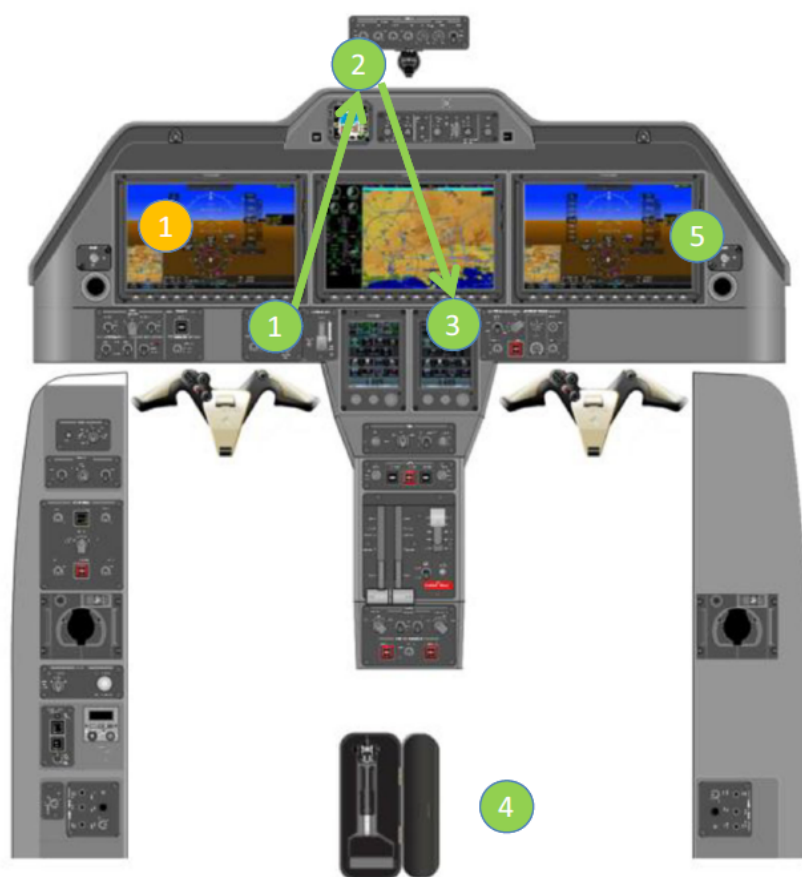
When flying a published instrument approach, the approach plate shall be displayed on the iPad EFB or PFD, as appropriate. When using the iPad EFB, it shall be used by the PM only and secured appropriately using the kneeboard.

The iPad is the primary source for the Jeppesen Charts. However, the charts may be displayed on the Garmin screens, as long as the Jeppesen database has been confirmed as in date.

2.12.12 Approach Flow

The purpose of the Approach Flow is to prepare the aeroplane and its occupants for the approach.

Trigger: This flow should be carried out when passing FL100 or when cleared to an altitude, whichever occurs first during the descent.



Expanded Approach Flow**PF:**

1. **Altimeters** **QNH_SET & BLUE**

Both pilots set QNH when cleared to an altitude, cross-checked by the PM. The PM also sets the standby altimeter QNH.

2. **SIGNS / OUTLET** **BELTS / OFF**
..... **# ON / OFF**

PM:

1. **WSHLD 1 and 2** **A/R**

Unless required windshield should be selected OFF to avoid visual distortion on final approach.

2. **LDG / TAXI Light** **TAXI**

3. **NAVAIDS** **SET UP and CHECK**

The NAVAIDS should auto tune for the approach but must be checked for correct indent code and green on the PFD with PM calling any deviations from what is expected by exception.

4. **CABIN** **SECURE**

Ensure pax seated with belts on and cabin secure for landing.

5. **ALTIMETERS** **SET and X-CHECKED**

Both pilots set QNH when cleared to an altitude, cross-checked by the PM. The PM also sets the standby altimeter QNH.

2.12.13 Approach Checklist

Trigger: The Approach Checklist should be called for by the PF when the Approach Flow has been completed.

WINDSHIELD HEAT..... A/R

CABIN..... SECURE

ALTIMETERS..... QNH___ SET and BLUE

SIGNS OUTLET SWITCH..... BELTS / OFF

..... # ON / OFF

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2.12.14 Stabilised Approach Concept

Refer to OM-A, [Operating Procedures](#).

For all straight-in IFR approaches the aeroplane shall be stabilised for landing, with the exception of speed, at the latest when passing 1000 ft above the landing runway threshold elevation. Speed stabilisation shall be achieved no later than 500 ft above the landing runway threshold elevation.

For VFR, visual approach, and circling approaches, the aeroplane shall be stabilised when passing 500 ft above the landing runway threshold elevation. In the case of circling approaches, the aeroplane shall be stabilised in the circling configuration no later than passing 1000 ft above the landing runway elevation.

Compliance with configuration gates stipulated in the approach profiles will aid in the correct configuration of the aeroplane to comply with the stabilised approach criteria.

Stabilised Approach Criteria Not Achieved

A go-around shall be initiated if:

- the stabilised approach criteria are not met at the required altitude / height;
- it is apparent that the stabilised approach criteria cannot safely be achieved by the required altitude / height;
- the approach becomes destabilised after the required altitude / height.

Stable Approach Call Outs

If the stable approach criteria are not met at the required altitude / height, the PM shall call 'UNSTABLE-GO-AROUND'.

Stabilisation Gates		
Approach Type	Height above Landing RWY THR (ft)	Stable criteria
IFR Straight-in-Approach	1000	Stable except Speed
	500	Fully stable
VFR, Visual	500	Fully stable ¹
Circle to land	1000	Circling configuration
	500	Fully stable ¹

¹ Where visual manoeuvring is required below 500 ft AAL, wings-level by 300 ft above landing runway threshold.

2.13 IFR Approach

2.13.1 Precision Approach – PA

2.13.1.1 ILS - to CAT I minimum (3D)

Refer to Flight Profile Diagram 2-100 B, [CAT I Approach Profile](#).

The following standard callouts must be used:

SITUATION	PF	PM
Cleared Approach PF: Selects APR mode. PM: Verifies selected modes.	'LOC WHITE, GS WHITE'	
LOC Capture PM: Verifies active mode.	'LOC GREEN'	
Glideslope capture PM: Verifies active mode and sets / verifies missed approach altitude. PF: Verifies correct missed approach altitude set.	'GS GREEN' 'CHECKED'	'[XXXX] FT SET'
Glideslope check PM: At any appropriate place on the approach until the OM or 4 NM. PF: Reads passing altitude from altimeter. PM: Checks against the approach plate whether established on the correct glidepath and calls out any deviation.	'[XXXX] FT'	'GLIDESLOPE CHECK'
Stable Approach Check If STABLE criteria are NOT met at the required altitude / height.	'GO-AROUND, FLAP 1 (or 2)'	'UNSTABLE, GO-AROUND'
100 ft above DA PF: Starts scanning outside for visual references if appropriate.		'ONE HUNDRED ABOVE'
DA PF: Makes decision to continue or go-around.	'CONTINUE' or 'GO-AROUND FLAP 1 (or 2)'	'MINIMUMS, MINIMUMS'

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2.13.1.2 RNP - to LPV minimum (3D)

Refer to the [Approach Automation Modes and Settings Table](#) and APV Approach Flight Profile diagram.

SITUATION	PF	PM
Cleared Approach PF: Selects APPR mode. PM: Verifies selected modes.	'FMS PINK, GPWHITE'	
PM confirms HSI Annunciation 'FMS LPV' is displayed prior to FAP, announcing to PF any variation from expected annunciation.		
Glidepath Intercept PM: Verifies active mode and sets / verifies missed approach altitude. PF: Verifies correct missed approach altitude set.	'GP PINK' 'CHECKED'	'[XXXX] FT SET'
Stable Approach Point If STABLE criteria are NOT met at the required altitude / height.	'GO-AROUND, FLAP 1 (or 2)'	'UNSTABLE, GO-AROUND'
100 ft above DA PF: Starts scanning outside for visual references if appropriate.		'ONE HUNDRED ABOVE'
DA PF: Makes decision to continue or goaround.	'CONTINUE' or 'GO-AROUND FLAP 1 (or 2)'	'MINIMUMS, MINIMUMS'

This approach is defined by LPV minimums.

Prior to the FAP:

- **FMS LPV** must be annunciated on the HSI.
- Ensure automatic sequencing of waypoints has not been suspended (no **SUSP** annunciation on the HSI).

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If SBAS becomes unavailable prior to one minute to the FAF on an LPV approach, LPV will be shown in amber, but the VDI will continue to be shown. At one minute to the FAF, the system will downgrade to either:

1. LNAV / VNAV service level (Baro VNAV) (L / VNAV shown in magenta / # amber), and an APR DWNGRADE system message will be generated (VDI will be flagged NO GP until the message is acknowledged);

or

2. LNAV service level (LNAV shown in magenta / # amber), the VDI will be flagged NO GP, and an APR DWNGRADE message will be generated.

If SBAS becomes unavailable after one minute to the FAF the system will downgrade to LNAV service level when past the FAF (LNAV shown in magenta), with no downgrade message generated.

CAUTION

Crew must acknowledge the message in the GTC to allow the system to automatically downgrade to the next available approach.

If **'APR DWNGRADE'** occurs before FAF, crews should either carry out a missed approach or revert to a non SBAS approach using either LNAV / VNAV or LNAV minima as appropriate.

If **'APR DWNGRADE'** occurs after FAF, a missed approach is to be carried out unless the crew have the required visual references to continue the approach or, if the aeroplane is still above 1000 ft, the approach can be continued to the LNAV minima if the downgrade option was briefed and it is supported by the system or available.

If GPS integrity is lost before required visual references are acquired, the approach shall be discontinued.

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2.13.1.3 PAR (3D)

For detailed information on PAR approaches, refer to OM-C.

NTA pilots shall use the minima published on the Jeppesen approach chart or 200' and 550 m RVR / Visibility, whichever is higher.

The flight profile for the PAR approach shall be identical to that used for the CDFA approach with the exception that HDG is to be substituted for FMS. The different callouts are detailed in the table below.

A database-stored approach may be inserted if the missed approach procedure is identical to the PAR / SRA Missed Approach Procedure.

SITUATION	PF	PM
Radar vectors for the approach. PF: Selects HDG. PM: Verifies selected mode.	'HDG GREEN'	
ATC: 'APPROACHING DESCENT POINT' PF: Sets AAD to 300' above current altitude or MAA if higher. PF: Selects VS Mode. PM: Verifies VS Green.	'VS GREEN'	
ATC: 'BEGIN DESCENT NOW' PF: Selects an appropriate ROD. PM: Verifies VS setting and selects MAA once below MAA. PF: Verifies MAA.	'CHECKED'	'[XXXX]FT SET'
Stable Approach Check If STABLE criteria are NOT met at the required altitude / height. PF: When executing a Go-around.	'GO-AROUND, FLAP 1 (or 2)'	'UNSTABLE, GO-AROUND'
100 ft above DA PF: Starts scanning outside for visual references if appropriate.		'ONE HUNDRED ABOVE'
DA PF: Makes decision to continue or go-around.	'CONTINUE' or 'GO-AROUND FLAP 1 (or 2)'	'MINIMUMS, MINIMUMS'

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NOTE

The MAA shall only be set when the MAA is above the current altitude to avoid unwanted capturing of the MAA.

2.13.2 Approach with Vertical Guidance - APV

2.13.2.1 RNP - to LNAV/VNAV minimum (3D)

Refer to the [Approach Automation Modes and Settings Table](#) and APV Approach Flight Profile diagram.

SITUATION	PF	PM
Cleared Approach PF: Selects APPR mode. PM: Verifies selected modes.	'FMS PINK, GP WHITE'	
PM confirms HSI Annunciation 'FMS L/VNAV' is displayed prior to FAP, announcing to PF any variation from expected annunciation.		
Glidepath Intercept PM: Verifies active mode and sets / verifies missed approach altitude. PF: Verifies correct missed approach altitude set.	'GP PINK' 'CHECKED'	'[XXXX] FT SET'
Stable Approach Check If STABLE criteria are NOT met at the required altitude / height.	'GO-AROUND, FLAP 1 (or 2)'	'UNSTABLE, GO-AROUND'
100 ft above DA PF: Starts scanning outside for visual references if appropriate.		'ONE HUNDRED ABOVE'
DA PF: Makes decision to continue or goaround.	'CONTINUE' or 'GO-AROUND FLAP 1 (or 2)'	'MINIMUMS, MINIMUMS'

This approach is defined by LNAV / VNAV minimums.

Prior to the FAF:

- 'FMS L/VNAV' must be annunciated on the HSI.

- Ensure automatic sequencing of waypoints has not been suspended (no '**SUSP**' annunciation on the HSI).

The vertical profile in the FMS shall be checked against the approach plate to ensure the profile and altitude constraints are accurate. It is prohibited to insert or change any lateral or vertical element of a waypoint that forms part of the approach procedure in the FMS database.

If executing an LNAV / VNAV approach and SBAS is unavailable, Baro VNAV is used for vertical guidance. This situation will be recognised by the appearance of the Vertical Deviation Indicator (VDI):

- SBAS provides a magenta Diamond.
- Baro VNAV provides a magenta pentagon.

If a BARO VNAV approach is to be flown, when QNH is set for the approach the difference in altimeter readings must not exceed 100 feet. Due consideration should be given to the published minimum temperature for the approach.

(Phenom 300 only) If the SBAS becomes unavailable prior to one minute to the FAF L / VNAV will be shown in amber, the system will switch to LNAV / VNAV (Baro VNAV) (L / VNAV shown in magenta) and an APR ADVISORY message will be generated. 'NO GP' will be seen until the System message has been acknowledged.

(Phenom 300 only) If SBAS becomes unavailable after one minute to the FAF the system will downgrade to LNAV service level with no downgrade message generated. If SBAS becomes unavailable and there is no LNAV minima the approach will abort when past the FAF.

If '**APR ADVISORY**' occurs before FAF, crew may continue the approach using the BARO VNAV indications because the minima will be the same or revert to a Non-precision Approach using LNAV minima - refer to the [NPA Callouts Table](#).

If '**APR ADVISORY**' occurs after FAF, a missed approach is to be carried out unless the crew have the required visual references to continue the approach or, if the aeroplane is still above 1000 ft, the approach can be continued to the LNAV minima if the downgrade option was briefed and it is supported by the system or available.

If GPS integrity is lost before required visual references are acquired, the approach should be discontinued.

2.13.2.2 RNP AR (3D)

2.13.3 Non-Precision Approach - NPA

Following the decision to continue on an NPA and the FD commands are no longer being followed the FD should be switched off.

2.13.3.1 RNP - to LNAV minimum (2D)

Refer to the [Approach Automation Modes and Settings](#) Approach Automation Modes and Settings Table and NPA - CDFA Approach Flight Profile diagram.

SITUATION	PF	PM
Cleared Approach PF: If LNAV+V selects APR mode. or PF: If LNAV confirms. PM: Verifies selected modes.	'FMS PINK, GP WHITE' 'FMS PINK'	
PM: confirms HSI Annunciation 'FMS LNAV+V or LNAV' is displayed prior to FAP, announcing to PF any variation from expected annunciation.		
On the vertical path PM: Calls out distance versus Altitude at the specified point. When within 50 ft or, otherwise PF: Responds to HIGH or LOW. PM: Advises next gate altitude.	'CORRECTING'	'[X] DME at [XXXX] FEET' 'ON PROFILE' '[XXX] FEET HIGH or LOW' 'NEXT [X] NM, [XXXX] FEET'
Stable Approach Check If STABLE criteria are NOT met at the required altitude / height.	'GO-AROUND, FLAP 1 (or 2)'	'UNSTABLE, GO-AROUND'
100 ft above DA PF: Starts scanning outside for visual references if appropriate.		'ONE HUNDRED ABOVE'
DA		'MINIMUMS, MINIMUMS'

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SITUATION	PF	PM
PF: Makes decision to continue or go-around.	'CONTINUE' or 'GO-AROUND FLAP 1 (or 2)'	

NOTE

1. If a vertical discrepancy is observed during an LNAV+V approach the vertical mode must be changed to VS to make any required corrections and remain in VS for the rest of the approach.

This approach is defined by LNAV minimums.

Prior to the FAF, 'LNAV+V' must be displayed in the HSI Annunciation if VNAV is to be used beyond the FAF.

If VNAV is to be used on final approach, the vertical profile in the FMS shall be checked against the approach plate to ensure the profile and altitude constraints are accurate. It is prohibited to insert or change any lateral or vertical element of a waypoint that forms part of the approach procedure in the FMS database.

When QNH is set for the approach, the difference in altimeter readings must not exceed 100 feet.

If the programmed approach is 'LNAV' only, 'LNAV' must be displayed in the HSI annunciation and VS shall be used to follow the published descent profile.

If GPS integrity is lost before required visual references are acquired, the approach should be discontinued.

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2.13.3.2 FMS Overlay (2D)

Refer to NPA - CDFA Approach Flight Profile Diagram 2-100 C.

This approach is flown using FMS and VS modes, with the FD coupled to FMS. However, the PRIMARY AID (VOR / NDB) shall be displayed at all times, either through the PM PFD display or by having the relevant RMI needle displayed.

CDFA Non-precision Approach Callouts

The following standard callouts must be used:

SITUATION	PF	PM
Cleared Approach PF: Selects FMS mode. PM: Verifies FMS mode.	'FMS PINK'	
Approaching FAF PF: Select VS 0.3 NM before FAF. PM: Verifies active mode and sets MAA when the aeroplane is lower than the MAA. PF: Verifies MAA.	'VS GREEN' 'CHECKED'	'[XXXX] FT SET'
On the vertical path PM: Calls out distance vs. altitude at the specified point. When within 50 ft or, otherwise PF: Responds to HIGH or LOW with PM: Advises next gate.	'CORRECTING'	'[X] DME at [XXXX] FEET' 'ON PROFILE' '[XXX] FEET HIGH or LOW' 'NEXT [X] NM, [XXXX] FEET'
Stable Approach Check If STABLE criteria are NOT met at the required altitude / height.	'GO-AROUND, FLAP 1 (or 2)'	'UNSTABLE, GO-AROUND'
100 ft above DA PF: Starts scanning outside for visual references if appropriate.		'ONE HUNDRED ABOVE'
DA PF: Makes decision to continue or go-around.	'CONTINUE' or 'GO-AROUND FLAP 1 (or 2)'	'MINIMUMS, MINIMUMS'

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2.13.3.3 Non-FMS Overlay (2D)

- Localiser based non-precision approaches shall always be flown as a Non-FMS Overlay approach with the flight director coupled to the Localiser.
- VOR approaches may be flown using Non-FMS Overlay with the flight director coupled to the VOR using APR (VAPP annunciation) mode.
- NDB approaches may be flown as a Non-FMS Overlay using HDG mode.
- SRA approaches shall be flown in HDG mode.

Non-FMS Overlay approaches should be flown using the same procedures, profiles and callouts as used for the FMS Overlay approaches above, except that the lateral mode callouts shall be as follows:

SITUATION	PF	PM
Cleared Approach PF: Arms appropriate mode; for example: PM: Verifies selected mode.	'LOC WHITE' or 'VAPP WHITE'	
Intercepting final approach PM: Verifies active mode.	'LOC GREEN' or 'VAPP GREEN'	

2.13.3.3.1 Surveillance Radar Approach (SRA)

A Surveillance Radar Approach is conducted in the same manner as a PAR approach but no Glidepath control is available. Instead, the controller gives azimuth corrections and target altitudes vs. range (as with a CDFA). The PF responds by increasing or decreasing the rate of descent to match the target altitudes.

For the SRA approach 50' shall be added to the approach minima given by ATC or on the Jeppesen plate and the approach shall be flown as a CDFA using HDG and VS modes.

2.13.3.4 IGS (2D)

Refer to OM-A, Operating Procedures.

The IGS is a system that uses ILS components. However, the IGS procedure is not an ILS procedure. The IGS procedure may be flown as an ILS procedure, but the published altitudes at the published DME / LOC steps shall strictly be observed and complied with.

2.13.3.5 Non-CDFA

Reserved.

2.13.4 Circling Approach

Refer to [Circling Approach Flight Profile](#) Diagram 2-101 A.

A circling approach, if authorised by ATC, may follow any authorised instrument approach.

If circling follows an approach with vertical guidance, the approach shall be flown using VS mode to ensure automatic altitude capture at the MDA / circling altitude.

If conducting a Noise Abatement Circling Pattern, the procedures in the OM-B should be followed. Similarly, if conducting a visual pattern following prescribed tracks, the published speed and configuration requirements shall be observed.

However, if weather conditions require descent to the circling MDA, the aeroplane must be fully configured for landing by the time it reaches the MDA, and maximum use of automation should be made to ensure optimum situational awareness.

NOTE

1. Max. Cat B circling speed is 135 KIAS. At maximum landing weight the manoeuvring speed will be:

- Flaps 1 - 140 KIAS
- Flaps 2/3 - $V_{ref} + 10$ KIAS
- Flaps Full - $V_{ref} + 10$ KIAS

See [Approach Manoeuvring Speeds](#) for additional details.

Therefore it may be necessary or desirable to select flaps 2 prior to the circling in order to be able to comply with the Cat B maximum circling speed and give a larger margin above the stall.

If weather and obstacle clearance permit, use of Cat C circling minima will permit the circling to be flown at a maximum speed of 180 KIAS.

2. If the Jeppesen plate does not define a break-off point, PF should brief where it is intended to break-off into the visual manoeuvring part of the approach. Whether briefed in this way or defined on the approach plate, the break-off point is to be regarded as the Decision Point in accordance with the procedure below.
3. Setting of the BARO MIN to the Circling minima permits the AAD to be set accurately to the same figure if necessary. BARO MIN should be set to the circling minima and the AAD set to the desired circling altitude. If necessary to set both to the same figure, the automatic 'MINIMUMS' call is to be treated as a decision point. If circling above the BARO MIN setting, during the final descent to land the automatic 'MINIMUMS' call will be triggered and should be treated in the normal way. The decision to 'CONTINUE' at this point is valid provided stable criteria are achieved for the visual landing.

The following standard callouts must be used:

SITUATION	PF	PM
Cleared Approach PF: Selects appropriate mode.	'LOC / VAPP WHITE'	
Intercepting final approach	'LOC / VAPP GREEN'	
Approaching FAF PM: Sets AAD to Circling Altitude. PF: Checks AAD Setting. PF: Selects VS 0.3 NM before FAF. PM: Verifies selected mode.	'CHECKED' 'VS GREEN'	'[XXX] FT SET'
On the vertical path (NPA) PM: Calls out distance vs altitude at the specified points. When within 50 ft or otherwise PF: Responds to HIGH or LOW with PM: Advises next gate.	'CORRECTING'	'[X] DME at [XXXX] FEET' 'ON PROFILE' '[XXX] FEET HIGH or LOW' 'NEXT [X] NM, [XXXX] FEET'
On the vertical path (Precision Approach) PM: At any appropriate place on the approach until the OM or 4 NM. PF: Reads passing altitude from altimeter. PM: Checks against the approach plate whether established on the correct glidepath and calls out any deviation.	'[XXXX] FEET'	'GLIDESLOPE CHECK'
100 ft Above Circling Altitude PF: Starts scanning outside for visual references.		'ONE HUNDRED ABOVE'
Reaching Circling Altitude PF and PM: Monitor the level off and speed. PM: Sets missed approach altitude. PF: Verifies correct missed approach altitude set.	'ALT GREEN' 'CHECKED'	'[XXXX] FEET SET'
At Circling Minima or Break-off Point, whichever is reached first		'MINIMUMS, MINIMUMS'

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SITUATION	PF	PM
PF: Decides to continue approach or go-around. If decision to CONTINUE PF: Selects HDG mode. PM: Confirms AFCS selection and starts clock as required.	'CONTINUE' or 'GO-AROUND FLAP 1 (or 2)' 'HDG GREEN'	
Stable Approach Check If STABLE criteria are NOT met at the required altitude / height.	'GO-AROUND, FLAP 1 (or 2)'	'UNSTABLE, GO-AROUND'

NOTE

- The above table assumes A/P ON throughout approach.
- The aeroplane shall be stabilised in the appropriate circling configuration not later than passing 1000 ft above the runway elevation.

Noise Abatement Circling

If the cloud ceiling is >1500 ft and the visibility is >5KM then provided the procedure in use permits, a noise abatement circling procedure should be used. This will follow the same ground tracks and the procedure as outlined above, however it shall be flown at 1500 ft AAL or normal circling altitude whichever is higher.

Consideration should be given as to where to begin the descent from 1500 ft AAL so as to avoid high rates of descent during the turn to final. If it is considered that a noise abatement circling is not compatible with a given procedure, then the normal circling procedure should be flown.

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2.14 VFR Approach

Refer to [Visual Approach Flight Profile](#) Diagram 2-100 D.

A Visual Approach may be loaded from the PROC page (if available) to assist with visual manoeuvring to the runway.

The following standard callouts must be used:

SITUATION	PF	PM
Stable Approach Check If STABLE criteria are NOT met at the required altitude / height.	'GO-AROUND, FLAP 1 (or 2)'	'UNSTABLE, GO-AROUND'

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SITUATION	PF	PM
If missed approach procedure is to be flown in HDG or VOR		
PM: Confirms TOGA button pressed and PF CDI selected to appropriate source, selects desired lateral mode.		'HDG / VOR GREEN'
PF: Verifies stated mode green.		
1000 ft AAL		'1000 FEET'
PF: Verifies IAS is within limits. Reduces thrust to CLB, or below, to adjust rate of climb or maintain airspeed as required.	'FLAP ZERO'	
PM: Verifies IAS within limit, calls out present IAS and moves flap lever.		'XXX'
PM: After selecting Flap Zero, selects and verifies YD ON, LDG / TAXI light to TAXI.		'YD GREEN'
PM: Verifies flap indications, calls new configuration. Checks thrust set to CLB or below.		'FLAP ZERO'
PF: Checks Flaps indicate zero.		

2.15.1.1 Missed Approach - After Take-Off Checklist

The after take-off checklist should be called for by the PF when established at the Missed Approach altitude - this will be the altitude as published on the Jeppesen plate, an ATC prescribed altitude or circuit pattern altitude following a visual approach.

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2.15.2 Balked Landing

A balked landing is defined as a landing maneuver that is discontinued at any point below the Obstacle Clearance Altitude / (DA/MDA). The normal Missed Approach callouts and actions, as per 2.15.1, shall apply.

When the aircraft is in a low-energy regime, the Low Energy Balked Landing procedures detailed below shall be applied.

2.15.2.1 Low Energy Balked Landing

A Low Energy Balked Landing is a low energy go-around initiated from a very low height above the runway or, potentially, even after touchdown has occurred.

The procedure to be followed is:

- PF: Simultaneously pushes the yoke-mounted AP trim disconnect button and TOGA button, apply TOGA thrust, maintaining aircraft attitude (runway contact may be made);
- Allow engines to 'spool up' and aircraft to accelerate to V_{ref} – maintain aircraft configuration (Take-off Config warning may sound);
- At V_{ref} rotate or initiate a climb while accelerating to V_{ac} prior to selecting GA flap, then continue climb at V_{ac} / V_{fs} (depending on terrain);
- With a positive climb rate and airspeed at or above V_{ac} , continue with the actions for a missed approach / go around.

CAUTION

Once wheel braking has commenced following touchdown, no attempt shall be made to initiate a Low Energy Balked Landing; the landing shall be continued.

2.16 Landing

2.16.1 Normal Landing

2.16.1.1 Before Landing Flow

The purpose of this flow is to ensure the aeroplane is correctly configured for landing.

Trigger: The flow commences when the first configuration change is initiated by PF.



Expanded Before Landing Flow

LHS:

1. **FUEL XFEED** **OFF**

PM:

1. **LDG GEAR** **DN**
2. **FLAP** **LANDING FLAP**

When landing clearance received:

3. **LDG / TAXI** **LDG**

PF:

1. **AP / YD** **DISCONNECT**

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At or above AP minimum disengage height, disconnect using AP disconnect button on yoke. PM verifies AP and YD disengaged.

NOTE

The latest disengagement height for the AP/YD is 195 ft AEO or 220 ft OEI.

2.16.1.2 Before Landing Checklist

Trigger: The Before Landing Checklist should be called for once the flow is complete. The checklist should be completed prior to the STABLE point.

LANDING GEAR..... DOWN

FLAPS..... A/R

CAS..... A/R

2.16.1.3 Landing Procedure

SITUATION	PF	PM
When cleared to land PM: Switches LDG / TAXI Light to LDG.		
Before Touchdown PM: Verifies YD is OFF.		
Following Touchdown PM: Verifies GND Spoiler deployment.		'60 KNOTS'

NOTE

1. Minimum AP disengage height 195' AGL (220' AGL if OEI).
2. For optimum braking efficiency, smoothly apply constant brake pressure after touchdown of the nosewheel. DO NOT pump brakes.
3. Any non-essential actions required during rollout shall only be performed below 60 KIAS.

2.16.2 Crosswind Landing

The crosswind limit is 28 kts, to be considered as limiting.

For crosswind landings, the 'de-crab' technique shall be accomplished according to the AFM.

Refer to OM-A, Operating Procedures for reduced crosswind limits when the braking action is less than good.

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2.16.3 Reduced Flap Landing

Flap positions 3 and Full are both considered normal landing flap positions.

The preferred landing flap for the Phenom is flap 3 due to lower drag characteristics, lower noise signature and reduced fuel consumption.

Flap Full is normally used to assist speed control for approaches steeper than 3° and to allow higher throttle settings when the anti-icing systems are ON to ensure good bleed air flow.

When deciding which landing configuration is to be used, crew shall take into consideration the benefits and limitations of either flap setting and brief accordingly.

FCM must ensure that the correct speeds are flown for whichever configuration the aircraft is in (e.g. Flap 3, STEEP, or in icing conditions).

2.16.4 Landing on Wet / Contaminated Runway

The recommended procedure to be followed for landing on wet runway is:

- Anticipate approach procedures and speeds.
- Conduct a positive landing to ensure initial wheel spin-up and initiate firm ground contact on touchdown.
- Lower nose wheel immediately to decrease lift and increase main gear loading.
- Apply brakes using moderate to firm pressure, allowing anti-skid to function.
- If no retardation evident, hydroplaning could be a cause - do not apply Emergency / Parking Brake as it removes anti-skid protection. Maintain runway centreline and continue braking until aeroplane decelerates.

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2.17 After Landing

2.17.1 After Landing Flow

The purpose of this flow is to configure the aeroplane for taxiing to the parking position.

Trigger: The After landing Flow should be commenced after vacating the landing runway. No unnecessary movement of switches or levers shall take place whilst on the runway.



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Expanded After Landing Flow**PM:**1. **FLAP** **0**2. **ICE PROTECTION** **A/R**

If WSHLD Heat and / or ENG / WINGSTAB were required ON for the approach, select OFF unless required by ambient conditions to taxi in.

3. **LDG / TAXI** **TAXI****STROBE** **OFF**

Lights configuration changed after runway vacated.

2.17.2 After Landing Checklist

Trigger: The After Landing Checklist should be called for by the PF at an appropriate place during taxi in.

STROBE..... OFF

FLAPS.....0

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2.18 Shutdown

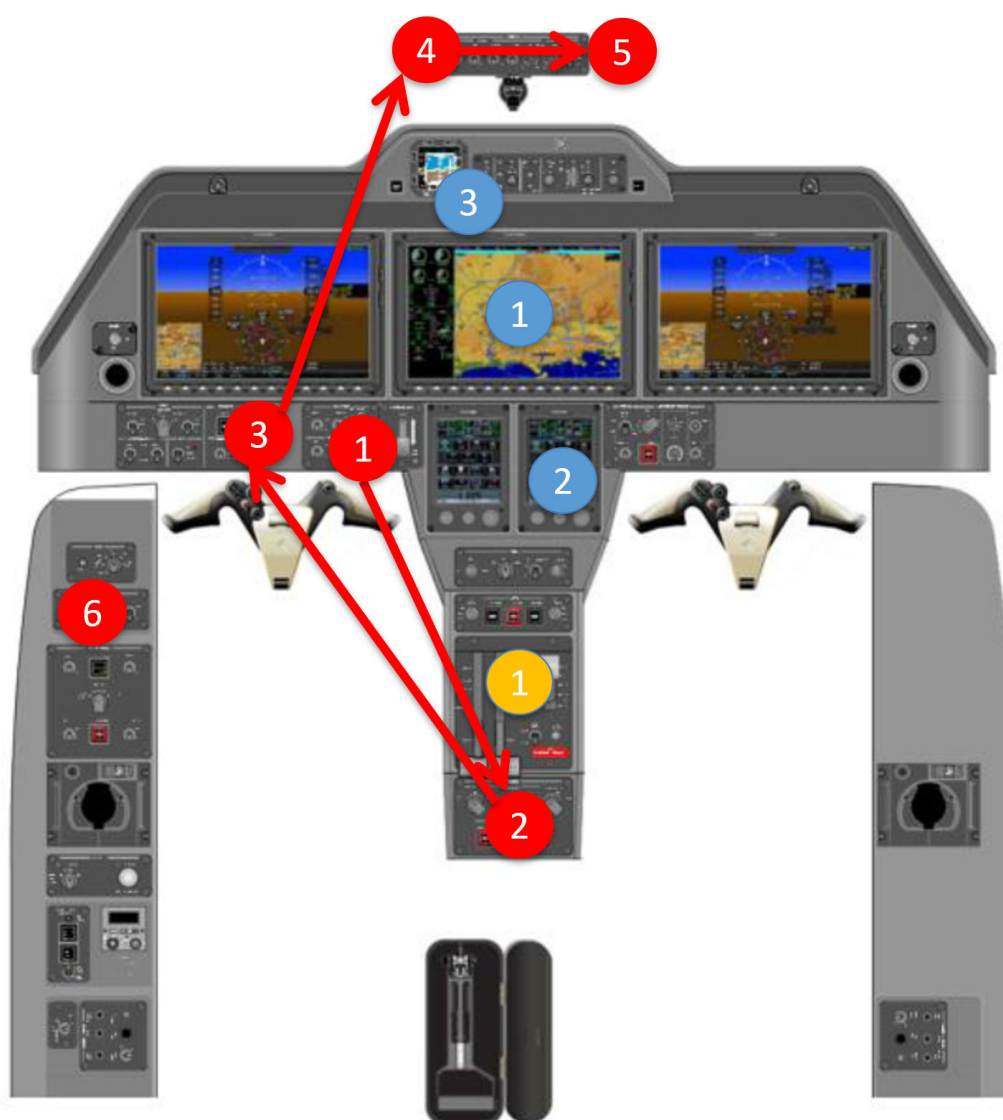
2.18.1 Before Shutdown

It is recommended to keep the engines running at idle for 2 minutes to permit engine thermal stabilisation prior to engine shutdown. Time of operation at or near idle, such as taxiing, is included in this 2 minutes period.

2.18.2 Shutdown Flow

The purpose of the shutdown flow is to prepare the aeroplane for safe disembarkation.

Trigger: The Shutdown Flow is commenced when at final parking position and the Parking Brake is applied.



Expanded Shutdown Flow**PF:**

1. **THRUST LEVERS** **IDLE**

PARKING BRAKE **SET**

Remains ON until engine shutdown complete and suitable chocks in place.

RHS:

1. **STATUS PAGE** **DISPLAY ON MFD**

Before moving the START / STOP knobs to STOP, STATUS page should be displayed on the MFD and monitored for any maintenance messages. Engine instruments must be monitored to verify spool down (fuel flow goes to 0 and ITT reduces).

Hydraulic pressure should be monitored to ensure the accumulator pressure is normal. Observe rate of hydraulic pressure decay, and note pressure when rate increases.

2. **TRANSPONDER** **2000 SET AND STBY**

3. **IESI** **SET STD**

Set the Integrated Electronic Standby Instrument (IESI) barometric pressure to STD, verify that both PFDs' pressure displayed are amber to indicate a mismatch.

LHS:

1. **HEATING and ICE PROTECTION** **OFF**

WSHLD 1 and 2 OFF, ADS Probes Auto, ENG 1 and 2 OFF and WINGSTAB OFF.

2. **START / STOP KNOBS** **STOP**

3. **SIGNS OUTLET SWITCH** **OFF / ON**

4. **EXTERIOR LIGHTS** **OFF**

5. **EMER LIGHTS** **OFF**

6. **BATT 1 and 2 SWITCHES** **AS REQUIRED**

Batteries may remain ON to provide lighting until passengers have deplaned or as needed. However, only use the batteries for the minimum time and do not allow batteries to drop below 24V.

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2.18.3 Shutdown Checklist

Trigger: The Shutdown Checklist should be called for by the LHS pilot when the flow is complete.

PARKING BRAKE.....ON
HEATING and ICE PROTECTION..... OFF
SIGNS OUTLET..... OFF / ON
EXTERIOR LIGHTS..... OFF
EMER LIGHTS..... OFF
IESI..... STD

2.18.4 Releasing the Brakes

The Emergency / Parking brake may remain ON while passengers disembark, but the crew shall be certain that the brake is selected OFF before leaving the aeroplane once it is properly chocked.

Parking Brake status placard shall be placed on the left-hand side window.

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2.19 Post-Flight

Please refer to [2.2.4.4 Post Flight / Turnaround Exterior Inspection](#).

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2.20 Turnaround

Please refer to [2.2.4.4 Post Flight / Turnaround Exterior Inspection](#).

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2.21 Securing Aeroplane

2.21.1 Aeroplane Condition

On completion of duty, or overnight, the aeroplane shall be left with the Parking Brake OFF, Gust Lock Pin IN, all electrical switches and lights turned OFF, the Oxygen bottle valve handle pulled to cut-out, the emergency exit door pin inserted and the Tow / Do Not Tow placard displayed in the windshield.

The commander shall ensure the flight deck is neat and tidy, all appropriate post flight paperwork is complete and that numbers have been sent to the office.

The galley and coffee machine should be clean and tidy, including the inside of the coffee machine door.

The toilet shall be checked for cleanliness and odour - if necessary, servicing shall be performed.

If the aeroplane is likely to be subjected to freezing temperatures for a prolonged period, the coffee machine is to be purged of water or removed from the aeroplane to the FBO. Any stock likely to be affected by cold temperatures should be removed to a suitable storage location and consider draining the toilet to avoid freezing.

All necessary cleaning will be done and the main cabin door shall be locked, unless company-approved arrangements have been made with an FBO to leave the aeroplane open for later servicing.

ADS and pitot covers shall be fitted. The NLG Torque Link shall be disconnected.

If winds of 25 kts or stronger are forecast, see OM-A, Operating Procedures for additional chocking and securing guidance.

2.21.2 Expanded Securing Flow

PARKING BRAKE **OFF**

Confirm that adequately sized chocks are in place before releasing the brake.

OXYGEN BOTTLE VALVE HANDLE **CUT-OUT**

GUST LOCK PIN **INSTALL**

LIGHTS **OFF**

Check that external, cockpit and cabin lights are all off.

BATT 1 and 2 SWITCHES **OFF**

EMERGENCY DOOR PIN **INSTALL**

TORQUE LINK **DISCONNECTED**

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2.21.3 Securing Checklist

PARKING BRAKE.....OFF

OXYGEN BOTTLE VALVE HANDLE..... CUT-OUT

GUST LOCK PIN..... INSTALLED

LIGHTS..... OFF

BATT 1 and 2 SWITCHES..... OFF

EMERGENCY DOOR PIN..... INSTALLED

TORQUE LINK..... DISCONNECTED

This checklist should be performed prior to closing the aeroplane.

2.21.4 Battery Removal**NOTE**

The batteries do not need to be removed if a GPU will be available for start. The -18°C limitation is if the intention is to carry out a start using the batteries only, following their exposure to low temperatures.

When ambient temperature is expected to drop below -18°C, both main batteries shall be removed unless hangarage can be arranged. Removal of the main batteries is a maintenance action, therefore when such conditions are forecast, crew shall inform maintenance in advance in order to make the necessary coordination.

Exact limitations and additional guidelines are in the POH and FOLs.

2.21.5 Garmin and Navigation Database Update

See Phenom Fleet Handbook.

2.21.6 Oil Servicing

When Oil Servicing is required, it shall always be carried out between 10 and 20min after shutdown.

Any assessment of the oil level made after 20 mins could result in an inaccurate reading due to possible oil migration into the base of the engine.

This migration is completely normal but it could show a lower than true reading on the sight glass.

Only sufficient oil shall be added to achieve the desired level - if this only requires part of the can, then the remaining oil shall be disposed of appropriately and must NOT be added to the engine as this will result in overflow during the next flight.

The amount of oil added shall always be annotated in the AFL.

Engine Oil servicing shall be carried out in accordance with Section 5 of the POH 'Servicing and maintenance', block 5-25.

2.22 Cold Weather Operations

Icing conditions may exist whenever the Static Air Temperature (SAT) on the ground or for take-off or the Total Air Temperature (TAT) in flight is 10°C or below and visible moisture in any form is present.

Icing conditions may also exist when the SAT on the ground and for take-off is 10°C or below when operating on ramps, taxiways or runways where surface snow, ice, standing water or slush may be ingested by the engines, or freeze on engines, nacelles or engine sensor probes.

Additional Reading Material

- NetJets Europe Winter Operations and Awareness Guide (NTA-TS-9006-99- DOC)
- OM-A, De-Icing and Anti-Icing on the Ground
- POH, 2-15
- AFM, 3-04
- Phenom Fleet Handbook

2.22.1 Aeroplane Preparation

2.22.1.1 Pre-Flight

Make sure you know about the airfield procedures for winter operations and anti-icing before you proceed to the aeroplane. You need to check where the de-icing is normally carried out, whether the engines will be running or if it is done pre-start, how much notice the ground staff will require especially if the airfield is busy. Large airports will normally have winter operations well rehearsed and procedures will be established and widely publicised, however smaller airports may not be so efficient and it will be up to the crew to have everything planned and ready for the slot time.

Remember to refill all of the liquid stock and have a toilet service if the aeroplane was drained of fluids the previous night. Simple things such as finding out where the stock was placed can delay crews and pre-planning for normally simple tasks will be required. If the aeroplane was left in very cold conditions ensure that if the batteries were removed they are replaced as soon as possible. Request a GPU as soon as possible because even in temperatures above -10°C the batteries can be very sluggish especially when trying to power the start.

The minimum temperature for starting is -40°C Oil temperature and -37°C fuel temperature, in the very unlikely event that operations are being carried out in these extremes, following engine start, it is permissible to operate the engine up to 70% N2 in order to warm the oil to the minimum temperature for normal operation (above 10° C).

Refer to POH, Section II - Procedures and Techniques, Cold Weather Operation.

Refer to AFM, Section III - Normal Procedures, Operation in Icing Conditions.

2.22.1.2 Contamination Check

The pilot in command has the final responsibility for ensuring that the aeroplane is clear of ice, frost or snow. The crew is to ensure the aeroplane is clear by carrying out a visual inspection of wings and other critical surfaces prior to take-off. See OM-A.

Whenever possible, following de-icing visually check the wing, control surfaces, engines and fuselage prior to taxiing. Although not always possible, a physical (hands-on) inspection will ensure that there is no ice accumulation. Do not touch the surfaces with bare hands, as the skin may stick to a freezing surface.

During the pre-flight walk-around, ensure that the pitot tubes, pressurisation static ports, all inlets, outlets and vents are clear of ice and unobstructed.

Landing gear and associated doors shall be unobstructed and clear of ice, frost and snow. Accumulations such as blown snow may be removed mechanically or by a heat blower. However, where deposits have bonded to surfaces, they can be removed by the application of hot air or by spraying with hot de-icing fluids.

If the aeroplane has become cold soaked as a result of flight at very cold temperatures, fuel might be at a subfreezing temperature. This can cause ice accumulation if the aeroplane is subjected to high humidity, fog, drizzle or rain even when the outside air temperature is substantially above freezing.

On completion of a pre-flight walk-around, if ice, snow or frost is discovered, a de-icing procedure will be required. Unheated / heated water or Type I, II, III or IV de-icing fluid can be used.

The approved fluids are:

- Type I, that agree with SAE AMS 1424 specifications;
- Type II, III and IV, which agree with SAE AMS 1428 specifications.

Refer to OM-A, De-Icing and Anti-Icing on the Ground.

Refer to POH, Section II - Procedures and Techniques, Cold Weather Operation.

2.22.2 De-icing and Anti-icing

De-icing Before Engine Start

PRE-FLIGHT EXTERIOR INSPECTION.....COMPLETE

Carry out a normal pre-flight inspection before fluid application.

If de-icing is carried out before starting engines, ensure that the covers are all removed and all doors / panels are closed prior to fluid application.

If de-icing carried out before start, a thorough surface check must be made by the crew after de-icing to ensure that the aeroplane is free from snow and ice and is protected with the fluid if anti-icing fluid was applied. Be aware that as you open the door fluid may drip down and that the surface around the aeroplane will also be extremely slippery.

Engine Starting

Do not start the engine until all ice deposits have been removed from the intakes, if there is any doubt about the condition of the engine, try and check to make sure that the blades are free to turn and that the engine is not seized. If this check is to be carried out by the crew, ensure that correct and safe ground equipment is used and that crew are not carrying out intricate and dangerous balancing acts on the wing.

CAUTION

If no GPU available, do not allow battery voltage to drop below 24V.

Prior to starting the engines if ground icing conditions exist, select ADS Probes ON.

Start engines and accomplish the After Start flows.

During operation in cold conditions the ECS will use high flow mode for cabin conditioning (8°C and below). This can cause high cockpit and cabin noise due to the increased air flow. To reduce the noise, the ECS may be switched to ECS 2, advisory CAS will be displayed, and then reset to BOTH before take-off. FOL 011/13

Consider Flap setting for taxi - dependent on contamination on taxiways, leave flaps at 0 until line-up.

Complete the POST DE-ICING checklist.

De-Icing After Engine Start

Reposition if required

PARKING BRAKE.....ON

Thrust levers..... IDLE

De-Icing Configuration:

BLEED 1 and 2..... OFF

Set the bleed switches to the OFF position to close the PRSOV and keep the fumes out of the passenger cabin and cockpit.

AIR CONDITIONING MODE..... OFF

Set the air conditioning mode switch to the OFF position to turn off the VCS.

ECS.....OFF VENT

Set the ECS knob to the OFF VENT position to turn off all ACS lines and opens the RAV to emergency ventilation.

PITCH TRIM..... FULL NOSE UP

FLAPS.....0

YOKE.....FULL AFT AND HOLD

RUDDER AND AILERONS..... NEUTRAL

DE / ANTI-ICING..... ACCOMPLISH

Post De-icing Checklist

Note de-icing details in AFL.

ADS PROBES..... AUTO

PITCH TRIM..... SET

Wait at least one minute after fluid application or after second engine start:

BLEED 1 and 2.....AUTO

Wait at least 3 minutes after fluid application or second engine start:

ECS.....BOTH

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AIR CONDITIONING MODE..... AUTO

AFTER START FLOWS.....COMPLETE

Ensure no restrictions in flight controls or trims.

Ice Protection Check

With the engines running, check the ice protection systems as follows:

WSHLD 1 and 2 SWITCHES.....ON

The CAS messages WSHLD 1 (2) HTR FAIL must not be displayed.

WSHLD 1 and 2 SWITCHES.....OFF

ENG 1 and 2 SWITCHES..... ON

The CAS messages A-I E1 (2) ON must be displayed (after 10 seconds).

ENG 1 and 2 SWITCHES..... OFF

The CAS messages must disappear.

BLEED 1 and 2 SWITCHES.....AUTO

WINGSTAB SWITCH..... ON

The CAS message A-I WINGSTB ARM or A-I WINGSTB INHB may be displayed.

WINGSTAB SWITCH..... OFF

The CAS message A-I WINGSTB ARM or A-I WINGSTB INHB must disappear.

If ice detector is available:

TEST PANEL KNOB.....ICE PROT

START TEST BUTTON.....PRESS

The CAS messages ICE CONDITION must be displayed.

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After completing a successful test:

If SAT is between 5° C and 10° C:

ENG 1 and 2 SWITCHES..... ON

WINGSTAB SWITCH..... OFF

WSHLD 1 and 2 SWITCHES..... OFF

If SAT is below 5°C:

WSHLD 1 and 2 SWITCHES..... ON

ENG 1 and 2 SWITCHES..... ON

WINGSTAB SWITCH..... ON

Application Guidelines - If the de / anti-icing provider does not have own procedures, or the personnel are unfamiliar with the type of aeroplane, brief them on the following:

1. Do not spray fluid directly into ram air inlets, engine inlets or exhausts, engine pylons, and ADS probes.
2. Do not spray fluid directly onto windows or windshields. If necessary, spray fluid above them and allow the fluid to flow down, or use a windshield- approved cloth to wipe the surface.
3. Do not spray fluid onto wheel and brake assemblies.
4. Follow the spray pattern shown below.

2.22.2.1 Expanded De-Icing Procedures**De-icing Before Engine Start**

PRE-FLIGHT EXTERIOR INSPECTION..... COMPLETE

Carry out a normal pre-flight inspection before fluid application.

If de-icing is carried out before starting engines, ensure that the covers are all removed and all doors / panels are closed prior to fluid application.

If de-icing carried out before start, a thorough surface check must be made by the crew after de-icing to ensure that the aeroplane is free from snow and ice and is protected with the fluid if anti-icing fluid was applied. Be aware that as you open the door fluid may drip down and that the surface around the aeroplane will also be extremely slippery.

Engine Starting

Do not start the engine until all ice deposits have been removed from the intakes, if there is any doubt about the condition of the engine, try and check to make sure that the blades are free to turn and that the engine is not seized. If this check is to be carried out by the crew, ensure that correct and safe ground equipment is used and that crew are not carrying out intricate and dangerous balancing acts on the wing.

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CAUTION

If no GPU available, do not allow battery voltage to drop below 24V.

Prior to starting the engines if ground icing conditions exist, select ADS Probes ON.

Start engines and accomplish the After Start flows.

During operation in cold conditions the ECS will use high flow mode for cabin conditioning (8°C and below). This can cause high cockpit and cabin noise due to the increased air flow. To reduce the noise, the ECS may be switched to ECS 2, advisory CAS will be displayed, and then reset to BOTH before take-off. FOL 011/13

Consider Flap setting for taxi - dependent on contamination on taxiways, leave flaps at 0 until line-up.

Complete the POST DE-ICING checklist.

De-Icing After Engine Start

Reposition if required

PARKING BRAKE.....ON

Thrust levers..... IDLE

De-Icing Configuration:

BLEED 1 and 2..... OFF

Set the bleed switches to the OFF position to close the PRSOV and keep the fumes out of the passenger cabin and cockpit.

AIR CONDITIONING MODE..... OFF

Set the air conditioning mode switch to the OFF position to turn off the VCS.

ECS.....OFF VENT

Set the ECS knob to the OFF VENT position to turn off all ACS lines and opens the RAV to emergency ventilation.

PITCH TRIM..... FULL NOSE UP

FLAPS.....0

YOKE.....FULL AFT AND HOLD

RUDDER AND AILERONS..... NEUTRAL

DE / ANTI-ICING..... ACCOMPLISH

Post De-icing Checklist

Note de-icing details in AFL.

ADS PROBES..... AUTO

PITCH TRIM..... SET

Wait at least one minute after fluid application or after second engine start:

BLEED 1 and 2.....AUTO

Wait at least 3 minutes after fluid application or second engine start:

ECS.....BOTH

AIR CONDITIONING MODE.....AUTO

AFTER START FLOWS.....COMPLETE

Ensure no restrictions in flight controls or trims.

Ice Protection Check

With the engines running, check the ice protection systems as follows:

WSHLD 1 and 2 SWITCHES.....ON

The CAS messages WSHLD 1 (2) HTR FAIL must not be displayed.

WSHLD 1 and 2 SWITCHES.....OFF

ENG 1 and 2 SWITCHES.....ON

The CAS messages A-I E1 (2) ON must be displayed (after 10 seconds).

ENG 1 and 2 SWITCHES.....OFF

The CAS messages must disappear.

BLEED 1 and 2 SWITCHES.....AUTO

WINGSTAB SWITCH.....ON

The CAS message A-I WINGSTB ARM or A-I WINGSTB INHB may be displayed.

WINGSTAB SWITCH.....OFF

The CAS message A-I WINGSTB ARM or A-I WINGSTB INHB must disappear.

If ice detector is available:

TEST PANEL KNOB.....ICE PROT

START TEST BUTTON.....PRESS

The CAS messages ICE CONDITION must be displayed.

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After completing a successful test:

If SAT is between 5° C and 10° C:

ENG 1 and 2 SWITCHES..... ON

WINGSTAB SWITCH..... OFF

WSHLD 1 and 2 SWITCHES..... OFF

If SAT is below 5°C:

WSHLD 1 and 2 SWITCHES..... ON

ENG 1 and 2 SWITCHES..... ON

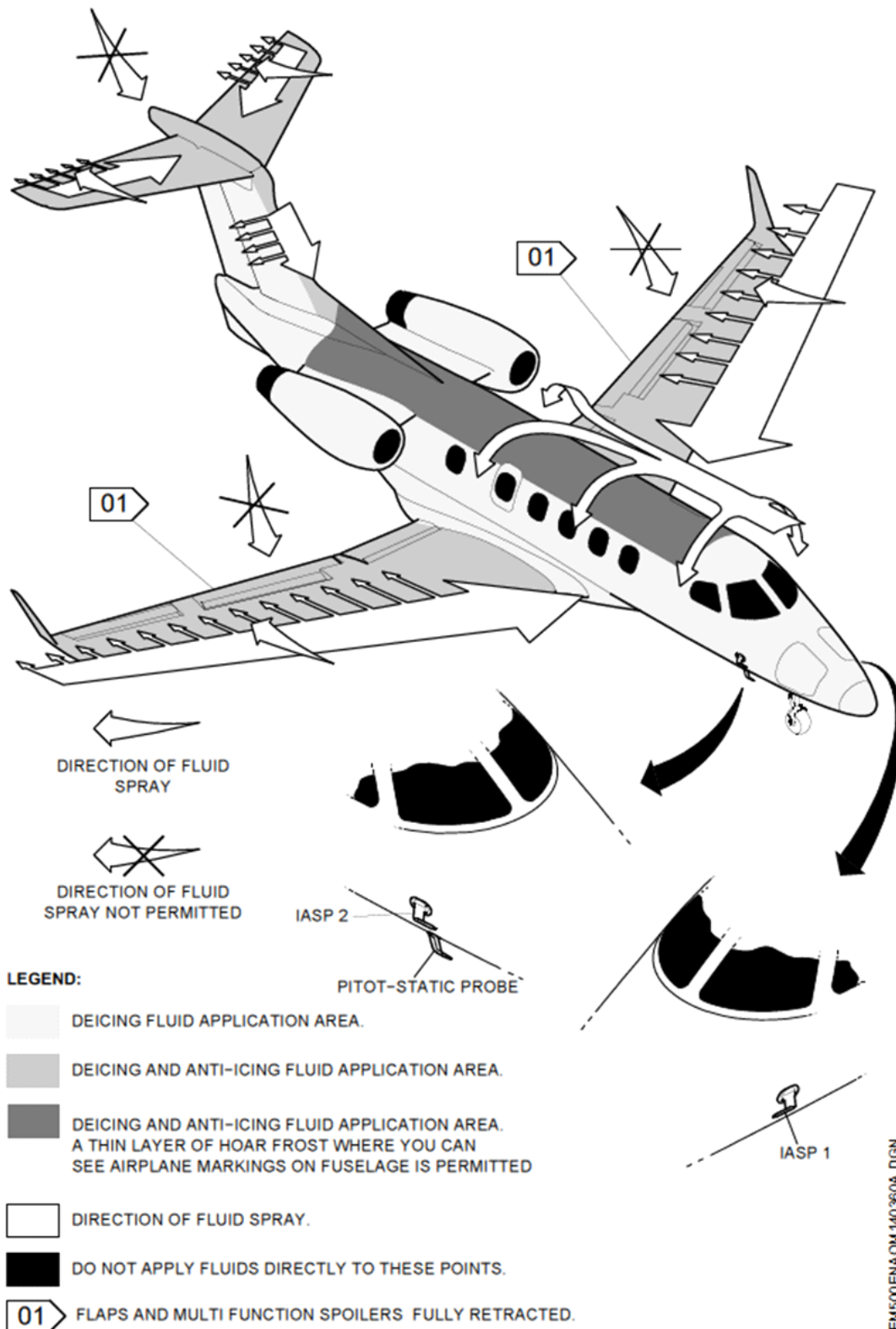
WINGSTAB SWITCH..... ON

Application Guidelines - If the de / anti-icing provider does not have own procedures, or the personnel are unfamiliar with the type of aeroplane, brief them on the following:

1. Do not spray fluid directly into ram air inlets, engine inlets or exhausts, engine pylons, and ADS probes.
2. Do not spray fluid directly onto windows or windshields. If necessary, spray fluid above them and allow the fluid to flow down, or use a windshield- approved cloth to wipe the surface.
3. Do not spray fluid onto wheel and brake assemblies.
4. Follow the spray pattern shown below.

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De / Anti-icing Spray Pattern



FLUID APPLICATION STRATEGY

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2.22.3 Operations on Contaminated Surfaces

2.22.3.1 Taxi

Before Taxi, carry out the Normal Checklist in Icing Conditions and make sure the ice protection systems are working.

Test the anti-ice systems according to the checklist and be aware that the WINGSTAB INHB CAS may be displayed depending on the SAT.

Consider Flap setting for taxi, if taxiways are contaminated, set flap 0.

Use minimum thrust for taxiing and try to avoid blowing snow and slush over other aeroplanes or ground personnel and equipment.

Maintain a low ground speed (<10 kts) when taxiing on snow or ice covered taxiways. This will also reduce the amount of slush thrown up onto the wheel and brake assembly. Remember that anti-skid protection is not available below 10 kts - this is to allow a locked wheel for a better turning radius, but in winter conditions this should be avoided where possible and a larger turning circle planned for.

Use firm braking where possible to warm up the brakes and help dry out any moisture in the disk stack.

Maintain a larger than normal gap behind other aeroplanes to avoid snow and slush blown up by jet wash.

If taxiing through slush or standing water, the flaps should remain retracted to avoid contamination and slush being thrown up by the main wheels onto the flap under surface. If you do this, make sure they are not forgotten when the pre-take-off checklist is completed, although it should be caught by the take-off Configuration test button.

Taxiways may not be cleared to the same standard as runways, some taxiways may be closed.

Be aware that taxiway markings may be obscured and braking action may be considerably worse than the stated braking action for the runway in use.

Under no circumstances will operations take place on runways where the braking action has been reported as POOR.

2.22.3.2 Pre-Take-Off

Hold Over Times - For Type 1 fluid, de-icing Embraer recommends using the Composite Hold Over Times (Table 1-c in OM-A, Chapter 8 Appendix 1) which are more conservative with regards to holdover times compared to table 1-b for normal aluminium surfaces. The reason for this is that the tail is made mostly from composite material and therefore it is considered safer to use the more restrictive composite table.

Check OM-A for x-wind limits with reduced braking action.

Ice Accumulation.....CHECK

A pre-take-off contamination check should be performed prior to take-off and within the holdover time. Aerodynamic surfaces must be confirmed free of all forms of frost, ice, snow and slush prior to entering the take-off runway or initiating take-off. This check is particularly important when the published holdover times are about to run out. When contamination is in evidence, the de-icing / anti-icing operation must be repeated. Visually inspect wing surfaces / leading edge and engine by looking through an appropriate window.

Ice Protection Check.....COMPLETE

Ice Protection Systems.....AS REQUIRED

Set the ice protection systems according to the operation in icing conditions procedures contained in the AFM.

Line-Up

FLIGHT CONTROLS.....CHECK

Before line-up the Flight Controls should be checked for full and free movement. If the flaps were left at 0 for taxi these should be set for take-off and a final check of all the surfaces should be made to ensure that the aeroplane does not have any accumulation of snow build up after the taxi.

2.22.3.3 Take-Off

Prior to take-off, the commander shall assess whether the applied holdover time is still appropriate.

OM-A states that de-icing and anti-icing must be repeated if the HOT has expired. However, if any doubt exists about the ice-free state of the aeroplane even within HOT Limitations, the aeroplane must return for a complete two-step procedure.

Do not apply static take-off techniques if the runway is slippery as the aeroplane may begin to slide as power is applied.

Apply light forward pressure on the control column to increase nosewheel steering effectiveness.

2.22.4 Climb

Operation in severe icing may induce ice build-up on the fan or spinner blades and this may result in fan vibration.

Do not reset the Ice Speed until you are certain all ice has been removed.

2.22.5 Descent

Whilst descending try to anticipate entry into cloud, a cold soaked aeroplane descending into wintry clouds will ice up quicker than perhaps normally expected.

2.22.6 Approach

Be aware that flying with the anti-ice systems on automatically increases the minimum power setting and the aeroplane may be reluctant to slow down for glideslope intercept and final approach.

If WingStab is expected to be ON below 1500 ft crew should use the higher Ice Speeds for the approach. Only if the aeroplane is above 1500 AAL and certain to be clear of ice should the SWPS Ice Speed be reset.

Caution should be made when selecting flap close to the limiting speeds and consideration may be given to selecting gear first to assist with speed control.

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2.22.7 Landing

Conduct a positive landing to ensure initial wheel spin-up and initiate firm ground contact upon touchdown.

Be aware of the factors that affect aquaplaning including high speed, standing water and poor runway surface.

Apply moderate to firm braking pressure and allow the anti-skid to assist with aeroplane stopping.

Slow down before exiting the runway and be wary of tight turns off the runway onto a possible slippery taxiway.

If landing on contaminated runway, flap should be left at landing setting until reaching final parking position where a visual inspection should be performed before raising the flaps.

2.22.8 Taxi In

Taxi slowly and take care, the taxiway may not be as well cleared or prepared as the runway - it could be more slippery with snow banks closer to the edge of the taxiway.

2.22.9 Parking and Post-Flight

Try to have chocks in position as soon as possible to allow brake release.

Carry out an external inspection carefully checking for any flap or gear damage from slush and ice.

If the aeroplane has to stay outside overnight, consider applying anti-icing fluid after landing if ground icing conditions are anticipated. If applied to a clean surface, this fluid layer prevents frozen contaminants from firmly adhering to the surfaces, and facilitates contamination removal during the prior-to-flight de-icing spray.

As guidance, if the temperature is forecast to be -10° C or below and you have to leave the aeroplane out for the night, make sure you take out all the items that may freeze (water bottles, soda cans, coffee and water containers, wine and champagne bottles).

The lavatory should be emptied of fluid and not refilled until prior to departure the next day.

2.23 Supplementary Normal Procedures

2.23.1 Pushback and Towing

Towing Operations

Refer to Section 5 of the POH 'Servicing and Maintenance' block 5-10.

There will be occasions when the aeroplane may not be taxied directly from its parking position, requiring a tow or pushback. In this case, the crew will ensure adequate marshalling personnel are available to move the aeroplane safely.

It is not required to insert the gear pins for towing or pushback for the purpose of positioning the aeroplane prior to start or after landing.

The use of gear pins for towing for maintenance purposes are the subject of separate maintenance procedures. If the aeroplane has come straight from maintenance, ensure that all 3 gear pins have been removed because their position on the landing gear can make them very difficult to see.

The battery No.1 switch shall be in 'ON' position and the Com No.1 shall be tuned to a frequency monitored by the tug driver (if available) or if required by ATC to the appropriate ATC frequency (which may also be monitored by the tug driver). If passengers are on board, then they must be briefed to be seated and fasten their seat belts prior to the towing operations commencing and stay seated until towing is completed.

Pushback Operations

An engine may be started before pushback as long as local procedures allow. Both FCM shall monitor the pushback until parking brake is applied. Second engine start should be conducted when pushback is complete and torque link has been verified as connected.

2.23.2 QFE Operations

Refer to OM-A, Operating Procedures.

See Garmin Service Alert 1538 - No QFE Operations on the G3000 until further notice. (This will be after a future software update).

2.23.3 Low Visibility Operations

Refer to OM-A, Operating Procedures for general procedures, limitations and task allocation during low visibility operations.

It is permissible for the PF to depart in FMS mode when conducting a LVTO.

2.23.4 Take-Off with Alternative Bleed Configuration

Reserved.

2.23.5 High Elevation Airport

Reserved.

2.23.6 Metric Altimetry

Refer to OM-A, Operating Procedures and OM-C.

The Phenom has the facility to select metric altimetry units. To enable the Meters display, select the 'PFD Settings' soft-key, 'Other PFD Settings' and then 'Altitude Units'. Select 'Meters' to add a meters window on the altitude tape.

2.23.7 Steep Approach

Refer to AFM Supplements.

The PIC or SIC can be Pilot Flying (PF) for landing providing they have completed the steep approach training and satisfy any airport or company specific qualification requirements.

Definition

A Steep Approach is one where the angle of approach to touchdown is 4.5° or greater.

Limitations

The steep approach must not be initiated with one engine inoperative or in emergency / abnormal situations.

The Steep Approach must not be flown with any failure leading to the loss of longitudinal or lateral primary flight controls.

Steep Approaches may be performed with an approach glidepath angle up to 5.5°.

Steep Approaches onto contaminated runways are prohibited. Flaps, Speedbrakes and Anti-Skid must be operative.

Configuration

The steep approach landing must be performed with flap surfaces at FLAP FULL position and Speed Brake switch set to the OPEN position.

Airspeeds

The approach must be flown at the Landing Reference Speed specific to Steep Approach (VREF STEEP).

Wind Components

Max Tailwind Component: 5 kts

Maximum Crosswind Component: 28 kts

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2.23.7.1 Steep Approach Callouts

The following standard callouts must be used:

SITUATION	PF	PM
Approaching Glideslope 1 NM before Glideslope (FAP) PF: Verifies IAS within limit and requests Flap FULL. PM: Verifies IAS within limit, calls out present IAS and moves flap lever. Verifies flap indications and calls new configuration. Sets missed approach altitude.	'FLAP FULL'	'XXX' 'FLAP FULL'
GS capture PF: Verifies glideslope capture. PM: Verifies active mode and selects speedbrakes switch to open. Verifies speedbrakes position on MFD and STEEP annunciation of PFDs. Sets / verifies missed approach altitude. PF: Verifies missed approach altitude.	'GS GREEN' 'CHECKED'	'STEEP GREEN' '[XXXX] FT SET'

NOTE

The remainder of the callouts are the same as a precision approach.

During the approach the PM must monitor that the speedbrakes do not retract due to the position of the thrust levers. In the event of inadvertent speedbrake retraction, the speedbrake switch must be moved to CLOSED and immediately to OPEN again.

In the event of a go-around, the PM closes the Speedbrakes as part of the go-around Flow; if this is not done, the Speedbrakes should retract automatically, but an advisory CAS message SPDBRK SW DISAG will illuminate. Setting the Speedbrakes switch to CLOSE will clear this CAS message.

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2.23.8 Non-ETOPS 180 Operations

Reserved.

2.23.9 High Latitude and Polar Operations

Refer to AFM Section II-Limitations, Navigation and Communication Equipment.

2.23.10 Hot Weather Operations

Refer to AFM Limitations and Route Performance Manual.

2.23.11 Operations in Volcanic Ash

Refer to POH, Section II 'Volcanic ash operation' and OM-A, Operating Procedures.

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2.24 Normal Checklist**2.24.1 Checklist Source**

Phase of Flight	Checklist Source
Cockpit / Cabin Safety and External Inspection	Phenom 300 POH, Section II Procedures and Techniques
Cockpit Preparation	NTA Normal Procedures Checklist
Power-up	
Before Start	
After Start	
Line-up	
After Take-off / Go-around	
Climb	
Cruise	
Descent	
Approach	
Before Landing	
After Landing	
Shutdown	
Securing	

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2.24.2 Normal Procedures Checklists

White Checklist - P300S

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White Checklist

POWER UP	
COCKPIT PREPARATION FLOW	COMPLETED
BATT 1 & 2 SWITCHES.....	ON
GPU BUTTON	AS REQUIRED
CVDR PANEL	TESTED
O2 SYSTEM	+TESTED / _PSI
+ TEST PANEL ANTI-CLOCKWISE	TESTED
+ TRIMS	TESTED & CENTRED

BEFORE START	
FLIGHT LOG / RELEASE	COMPLETE
PORTABLE EFB.....	PREPARED AND SECURE
+ PINS & COVERS	COUNTED & STOWED
TORQUE LINK	A/R
SIGNS OUTLET SWITCH.....	BELTS / OFF
FUEL	_LBS / BALANCED
PARKING BRAKE	ON
ADS PROBES	A/R
CAS	A/R

AFTER START	
GPU (IF USED)	DISCONNECTED
ELEC EMER	TESTED
STALL PROTECTION	TESTED
FLIGHT CONTROLS	CHECKED
ICE PROTECTION	+ TESTED & A/R
FLAPS	A/R
PITCH TRIM	_
ALTIMETERS	QNH _ SET & BLUE
FLIGHT GUIDANCE	SET & CHECKED
TRANSPONDER	CODE SET & AUTO
CAS	A/R
CABIN	SECURE

LINE UP	
DEPARTURE REVIEW	COMPLETE
TAKE-OFF CONFIG	OK
ENGINE INDICATIONS	CHECKED
STROBE	ON
CAS	A/R

AFTER TAKE-OFF / GO-AROUND	
LANDING GEAR	UP
FLAPS	0
YAW DAMPER	GREEN
ALTIMETERS	STD SET & BLUE

CLIMB	
PRESSURISATION	CHECKED
WINDSHIELD HEAT	ON
SIGNS OUTLET SWITCH	BELTS / ON

DESCENT	
LFE	SET
FLIGHT GUIDANCE	SET & CHECKED
ARRIVAL BRIEFING	COMPLETE
SIGNS OUTLET SWITCH	BELTS / ON

APPROACH	
WINDSHIELD HEAT	A/R
CABIN	SECURE
SIGNS OUTLET SWITCH	BELTS / OFF
ALTIMETERS	QNH _ SET & BLUE

BEFORE LANDING	
LANDING GEAR	DOWN
FLAPS	A/R
CAS	A/R

AFTER LANDING	
STROBE	OFF
FLAPS	0

SHUTDOWN	
PARKING BRAKE	ON
HEATING & ICE PROTECTION	OFF
SIGNS OUTLET SWITCH	OFF / ON
EXTERIOR LIGHTS	OFF
EMER LIGHTS	OFF
IESI	STD

SECURING	
PARKING BRAKE	OFF
OXYGEN BOTTLE VALVE HANDLE	CUTOUT
GUST LOCK PIN	INSTALLED
LIGHTS	OFF
BATT 1 & 2 SWITCHES	OFF
EMERGENCY DOOR PIN	INSTALLED
TORQUE LINK	DISCONNECTED

+: TEST FIRST FLIGHT OF THE DAY ONLY

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White Checklist

APPROACH SPEEDS AND LANDING DISTANCE REQUIRED (LDR)

MASS (LBS)	FLAP 3 LANDING		UNFACTORED LDR (Feet)		FLAP FULL LANDING		UNFACTORED LDR (Feet)		V _{FS}	STEEP LANDING	
	V _{REF}	V _{AC}	DRY	WET	V _{REF}	V _{AC}	DRY	WET		V _{REF}	V _{AC}
12400	101 (108)	104 (108)	2232	2952	100 (106)	100 (106)	2178	2849	113 (126)	112 (115)	112 (115)
12800	102 (110)	105 (110)	2279	3036	101 (108)	101 (108)	2222	2926	115 (128)	113 (116)	113 (116)
13200	104 (111)	107 (111)	2326	3123	103 (109)	103 (109)	2269	3009	117 (130)	114 (117)	114 (117)
13600	105 (113)	109 (113)	2374	3212	104 (111)	104 (111)	2313	3090	119 (132)	116 (118)	116 (118)
14000	107 (115)	110 (115)	2423	3304	106 (112)	106 (112)	2358	3172	121 (134)	117 (120)	117 (120)
14400	108 (116)	111 (116)	2469	3391	107 (114)	107 (114)	2403	3255	122 (136)	117 (121)	117 (121)
14800	110 (118)	113 (118)	2515	3481	109 (115)	109 (115)	2446	3336	124 (138)	118 (122)	118 (122)
15200	111 (119)	114 (119)	2562	3570	110 (117)	110 (117)	2492	3421	126 (140)	119 (123)	119 (123)
15600	113 (121)	116 (121)	2609	3657	111 (118)	111 (118)	2535	3498	127 (142)	120 (125)	120 (125)
16000	114 (122)	117 (122)	2657	3746	113 (119)	113 (119)	2577	3574	129 (143)	121 (127)	121 (127)
16400	115 (124)	119 (124)	2701	3830	114 (121)	114 (121)	2622	3655	130 (145)	122 (128)	122 (128)
16800	117 (125)	120 (125)	2746	3915	115 (122)	115 (122)	2663	3731	132 (147)	123 (129)	123 (129)
17200	118 (127)	121 (127)	2791	4002	117 (124)	117 (124)	2705	3808	133 (149)	124 (130)	124 (130)
17600	119 (128)	122 (128)	2838	4091	118 (125)	118 (125)	2751	3893	135 (150)	N/A	N/A
18000	121 (130)	124 (130)	2883	4180	119 (126)	119 (127)	2794	3974	136 (152)	N/A	N/A
18387	122 (131)	125 (131)	2967	4262	121 (128)	121 (128)	2839	4055	138 (154)	N/A	N/A

1. Darker shaded area exceeds **MAX LANDING MASS - 17042 Lbs.**
2. Values in brackets are **ENG** and **WINGSTAB ANTI ICE ON** figures
3. Landing Distance Required (LDR) figures are from 1000ft Altitude

RECAT - EU APPROACH AND DEPARTURE SEPARATION

DEPARTURE SEPARATION TIME (Secs)		Following Aircraft	APPROACH SEPARATION DIST (NMs)		Following Aircraft
		Light			Light
Leading Aircraft	Super Heavy	180	Leading Aircraft	Super Heavy	8
	Upper Heavy	140		Upper Heavy	7
	Lower Heavy	120		Lower Heavy	6
	Upper Medium	120		Upper Medium	5
	Lower Medium	100		Lower Medium	4
	Light	80		Light	3

TAKE-OFF PITCH TRIM SETTINGS

%MAC	FLAP 1	%MAC	FLAP 2
19	7.7	19	6.8
24	8.9	24	8.2
27	9.6	27	9.0
30	10.3	30	9.9
33	11.0	33	10.7
36	11.7	36	11.5
39	12.4	39	12.4
42	12.4	42	12.4

RECAT - EU WAKE TURBULENCE GROUPS

Category	Max Take off Mass (Tons)	Wing Span	Current ICAO Categories
Super Heavy (A)	Greater than 100T	Greater than 72 m	Super (J) A380
Upper Heavy (B)		60 - 72 m	Heavy (H) ≥136000lbs
Lower Heavy (C)		Greater than 52 m	
Upper Medium (D)	15 - 100T	Greater than 32 m	Medium (M) >7000lbs <136000
Lower Medium (E)	15 - 100T	Up to 32 m	
Light (F)	Up to 15T		Light (L) ≤7000lbs

MINIMUM MANOEUVRING SPEEDS

GEAR/FLAP SETTING	AIRSPEED
UP/ZERO	150
UP/1	140
* UP/2	V _{REF} + 10 KIAS
* DN/3	V _{REF} + 10 KIAS
* DN/FULL	V _{REF} + 10 KIAS

* MAX 30° AOB

White Checklist - P300E

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White Checklist

POWER UP	
COCKPIT PREPARATION FLOW	COMPLETED
BATT 1 & 2 SWITCHES.....	ON
GPU BUTTON	AS REQUIRED
CVDR PANEL.....	TESTED
O2 SYSTEM.....	+TESTED / _PSI
+ TEST PANEL ANTI-CLOCKWISE.....	TESTED
+ TRIMS	TESTED & CENTRED

BEFORE START	
FLIGHT LOG / RELEASE	COMPLETE
PORTABLE EFB.....	PREPARED AND SECURE
+ PINS & COVERS.....	COUNTED & STOWED
TORQUE LINK	A/R
SIGNS OUTLET SWITCH.....	ON / OFF
FUEL	_LBS / BALANCED
PARKING BRAKE	ON
ADS PROBES	A/R
CAS.....	A/R

AFTER START	
GPU (IF USED)	DISCONNECTED
ELEC EMER.....	TESTED
STALL PROTECTION	TESTED
FLIGHT CONTROLS	CHECKED
ICE PROTECTION	+ TESTED & A/R
FLAPS	A/R
PITCH TRIM	_
ALTIMETERS	QNH SET & BLUE
FLIGHT GUIDANCE	SET & CHECKED
TRANSPONDER	CODE SET & AUTO
CAS	A/R
CABIN	SECURE

LINE UP	
DEPARTURE REVIEW.....	COMPLETE
TAKE-OFF CONFIG	OK
ENGINE INDICATIONS	CHECKED
STROBE.....	ON
CAS.....	A/R

AFTER TAKE-OFF / GO-AROUND	
LANDING GEAR	UP
FLAPS.....	0
YAW DAMPER.....	GREEN
ALTIMETERS	STD SET & BLUE

CLIMB	
PRESSURISATION.....	CHECKED
WINDSHIELD HEAT	ON
SIGNS OUTLET SWITCH	ON / ON

DESCENT	
LFE.....	SET
FLIGHT GUIDANCE	SET & CHECKED
ARRIVAL BRIEFING	COMPLETE
SIGNS OUTLET SWITCH	ON / ON

APPROACH	
WINDSHIELD HEAT	A/R
CABIN	SECURE
SIGNS OUTLET SWITCH	ON / OFF
ALTIMETERS	QNH SET & BLUE

BEFORE LANDING	
LANDING GEAR	DOWN
FLAPS.....	A/R
CAS.....	A/R

AFTER LANDING	
STROBE.....	OFF
FLAPS.....	0

SHUTDOWN	
PARKING BRAKE	ON
HEATING & ICE PROTECTION	OFF
SIGNS OUTLET SWITCH	OFF / ON
EXTERIOR LIGHTS	OFF
EMER LIGHTS	OFF
IESI	STD

SECURING	
PARKING BRAKE	OFF
OXYGEN BOTTLE VALVE HANDLE.....	CUTOUT
GUST LOCK PIN.....	INSTALLED
LIGHTS	OFF
BATT 1 & 2 SWITCHES	OFF
EMERGENCY DOOR PIN.....	INSTALLED
TORQUE LINK	DISCONNECTED

+: TEST FIRST FLIGHT OF THE DAY ONLY

NETJETS

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White Checklist

APPROACH SPEEDS AND LANDING DISTANCE REQUIRED (LDR)

MASS (LBS)	FLAP 3 LANDING		UNFACTORED LDR (Feet)		FLAP FULL LANDING		UNFACTORED LDR (Feet)		V _{FS}	STEEP LANDING	
	V _{REF}	V _{AC}	DRY	WET	V _{REF}	V _{AC}	DRY	WET		V _{REF}	V _{AC}
12400	101 (108)	104 (108)	2216	2903	100 (106)	100 (106)	2116	2726	127 (129)	112 (115)	112 (115)
12800	102 (110)	105 (110)	2262	2987	101 (108)	101 (108)	2159	2800	126 (129)	113 (116)	113 (116)
13200	104 (111)	107 (111)	2310	3073	103 (109)	103 (109)	2205	2881	125 (131)	114 (117)	114 (117)
13600	105 (113)	109 (113)	2358	3162	104 (111)	104 (111)	2248	2958	123 (133)	116 (118)	116 (118)
14000	107 (115)	110 (115)	2407	3253	106 (112)	106 (112)	2292	3037	122 (135)	117 (119)	117 (120)
14400	108 (116)	111 (116)	2453	3339	107 (114)	107 (114)	2337	3117	123 (136)	117 (120)	117 (121)
14800	110 (118)	113 (118)	2500	3428	109 (115)	109 (115)	2378	3194	124 (138)	118 (121)	118 (122)
15200	111 (119)	114 (119)	2546	3517	110 (117)	110 (117)	2424	3277	126 (140)	119 (123)	119 (123)
15600	113 (121)	116 (121)	2594	3604	111 (118)	111 (118)	2466	3351	127 (142)	120 (124)	120 (124)
16000	114 (122)	117 (122)	2641	3692	113 (119)	113 (119)	2507	3423	129 (144)	121 (126)	121 (126)
16400	115 (124)	119 (124)	2686	3776	114 (121)	114 (121)	2551	3501	131 (145)	122 (127)	122 (127)
16800	117 (125)	120 (125)	2731	3861	115 (122)	115 (122)	2591	3575	132 (147)	123 (128)	123 (128)
17200	118 (127)	121 (127)	2776	3947	117 (124)	117 (124)	2632	3650	134 (149)	124 (130)	124 (130)
17600	119 (128)	122 (128)	2823	4036	118 (125)	118 (125)	2735	3841	135 (150)	125 (131)	125 (131)
18000	121 (130)	124 (130)	2868	4123	119 (126)	119 (126)	2779	3920	137 (152)	N/A	N/A
18400	122 (131)	125 (131)	2914	4208	121 (128)	121 (128)	2825	4005	138 (154)	N/A	N/A

1. Darker shaded area exceeds **MAX LANDING MASS - 17272 Lbs.**
2. Values in brackets are **ENG** and **WINGSTAB ANTI ICE ON** figures
3. Landing Distance Required (LDR) figures are from 1000ft Altitude

RECAT - EU APPROACH AND DEPARTURE SEPARATION

DEPARTURE SEPARATION TIME (Secs)		Following Aircraft	APPROACH SEPARATION DIST (NMs)		Following Aircraft
		Light			Light
Leading Aircraft	Super Heavy	180	Leading Aircraft	Super Heavy	8
	Upper Heavy	140		Upper Heavy	7
	Lower Heavy	120		Lower Heavy	6
	Upper Medium	120		Upper Medium	5
	Lower Medium	100		Lower Medium	4
	Light	80		Light	3

TAKE-OFF PITCH TRIM SETTINGS

%MAC	FLAP 1	%MAC	FLAP 2
19	7.7	19	6.8
24	8.9	24	8.2
27	9.6	27	9.0
30	10.3	30	9.9
33	11.0	33	10.7
36	11.7	36	11.5
39	12.4	39	12.4
42	12.4	42	12.4

RECAT - EU WAKE TURBULENCE GROUPS

Category	Max Take off Mass (Tons)	Wing Span	Current ICAO Categories
Super Heavy (A)	Greater than 100T	Greater than 72 m	Super (J) A380
Upper Heavy (B)		60 - 72 m	Heavy (H) ≥136000lbs
Lower Heavy (C)		Greater than 52 m	
Upper Medium (D)	15 - 100T	Greater than 32 m	Medium (M) >7000lbs <136000
Lower Medium (E)	15 - 100T	Up to 32 m	Light (L) ≤7000lbs
Light (F)	Up to 15T		

MINIMUM MANOEUVRING SPEEDS

GEAR/FLAP SETTING	AIRSPEED
UP/ZERO	150
UP/1	140
* UP/2	V _{REF} + 10 KIAS
* DN/3	V _{REF} + 10 KIAS
* DN/FULL	V _{REF} + 10 KIAS

* MAX 30° AOB

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Rev.12
29/Jun/26
Page 2 of 2

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Blue Checklist

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Blue Checklist

DE-ICING BEFORE ENGINE START

PRE-FLIGHT EXTERIOR INSPECTION.....**COMPLETE**ENSURE ALL COVERS ARE REMOVED AND ALL DOORS /
PANELS ARE CLOSED PRIOR TO FLUID APPLICATION.

DE-ICING CONFIGURATION

BATT 1&2 **ON**
 BLEED 1&2 **OFF**
 AIR CONDITIONING **OFF**
 ECS **OFF / VENT**
 PITCH TRIM..... **FULL NOSE UP**
 FLAPS..... **0**
 YOKE..... **FULL AFT AND HOLD**
 RUDDER AND AILERONS..... **NEUTRAL**
 BATT 1&2 **A/R**
 DE / ANTI-ICING..... **ACCOMPLISH**

APPLICATION GUIDELINES

SEE REVERSE OF CHECKLIST AND OM-B FOR FLUID
APPLICATION GUIDELINES AND SPRAY PATTERN
DIAGRAM.MOST DE-ICING OPERATORS WILL BE VERY FAMILIAR
WITH NORMAL PROCEDURES BUT CHECK IF ANY
DOUBT EXISTS.

POST DE-ICING & ENG START

CARRY OUT A NORMAL START BUT DELAY THE AFTER-
START CHECKLIST UNTIL ITEMS BELOW COMPLETE.ADS PROBES..... **AUTO**
PITCH TRIM..... **SET****AT LEAST 1 MIN AFTER FLUID APPLICATION.**BLEED 1 & 2..... **AUTO****AT LEAST 3 MINS AFTER FLUID APPLICATION.**ECS **BOTH**
AIR CONDITIONING MODE **AUTO**
AFTER START FLOWS **COMPLETE**

PERFORM THE AFTER-START CHECKLIST

CONSIDER DELAYING FLAP SELECTION UNTIL THE
HOLDING POINT, DEPENDING ON TAXIWAY CONDITION.

DE-ICING AFTER ENGINE START

CARRY OUT A NORMAL START BUT DELAY THE AFTER-
START CHECKLIST UNTIL POST DE-ICING ITEMS ARE
COMPLETEADS PROBES **AUTO**
FLAPS **0**

REPOSITION AS REQUIRED

BEFORE DE-ICING:

PARKING BRAKE..... **ON**
ENGINES **IDLE**CONTINUE WITH THE DE-ICING CONFIGURATION
BELOW.

DE-ICING CONFIGURATION

BLEED 1&2 **OFF**
 AIR CONDITIONING **OFF**
 ECS **OFF / VENT**
 PITCH TRIM **FULL NOSE UP**
 FLAPS **0**
 YOKE **FULL AFT AND HOLD**
 RUDDER AND AILERONS **NEUTRAL**
 DE / ANTI-ICING..... **ACCOMPLISH**

APPLICATION GUIDELINES

SEE REVERSE OF CHECKLIST AND OM-B FOR FLUID
APPLICATION GUIDELINES AND SPRAY PATTERN
DIAGRAM.MOST DE-ICING OPERATORS WILL BE VERY FAMILIAR
WITH NORMAL PROCEDURES BUT CHECK IF ANY
DOUBT EXISTS.

POST-DE-ICING

PITCH TRIM **SET****AT LEAST 1 MIN AFTER FLUID APPLICATION.**BLEED 1 & 2..... **AUTO****AT LEAST 3 MINS AFTER FLUID APPLICATION.**ECS **BOTH**
AIR CONDITIONING MODE **AUTO**
AFTER START FLOWS..... **COMPLETE**

PERFORM THE AFTER-START CHECKLIST

CONSIDER DELAYING FLAP SELECTION UNTIL THE
HOLDING POINT, DEPENDING ON TAXIWAY CONDITION.

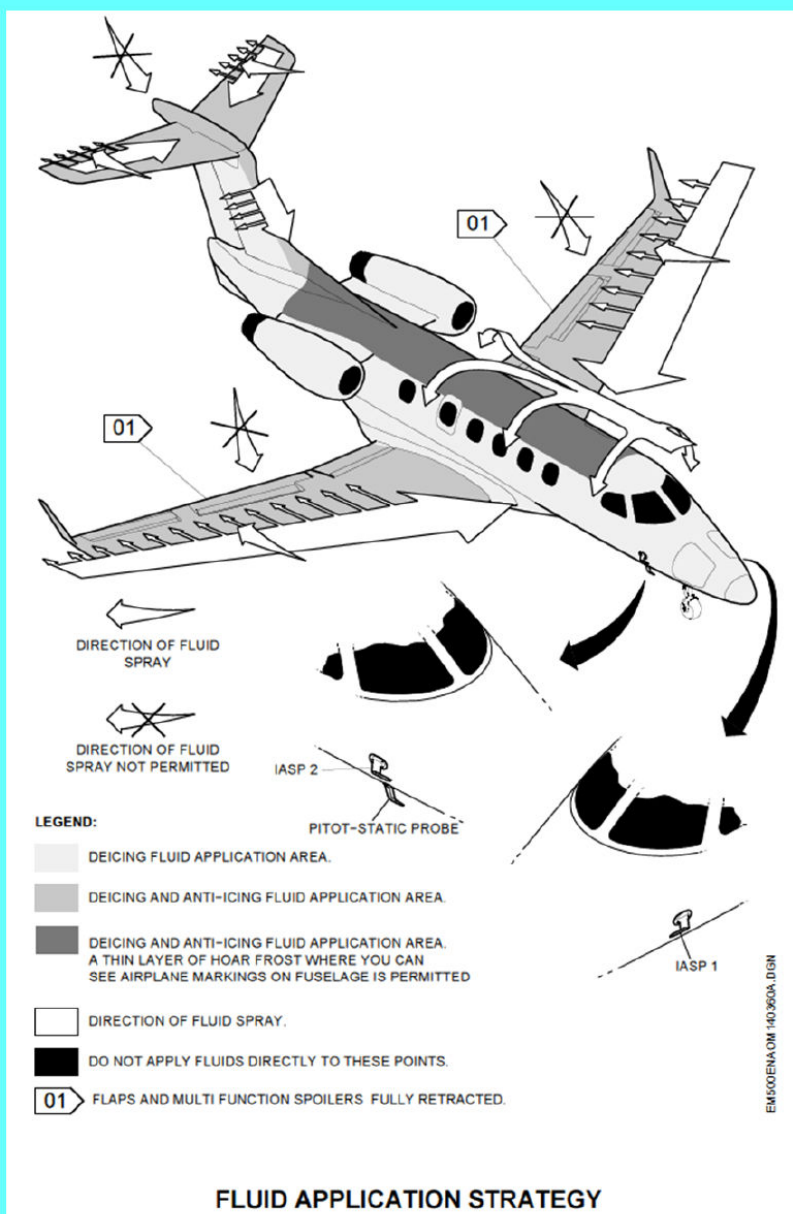
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Blue Checklist

Application Guidelines - If the de/anti-icing provider does not have own procedures, or the personnel are unfamiliar with the type of aeroplane, brief them on the following:

1. Do not spray fluid directly into ram air inlets, engine inlets or exhausts, engine pylons, and ADS probes.
2. Do not spray fluid directly onto windows or windshields. If necessary, spray fluid above them and allow the fluid to flow down, or use a windshield-approved cloth to wipe the surface.
3. Do not spray fluid onto wheel and brake assemblies.
4. Follow the spray pattern shown below.



2.24.3 Checklist Task Allocation

CHECKLIST	CALLED FOR BY	PERFORMED BY	TYPE
CABIN SAFETY and EXTERNAL INSPECTION	PF / PM		F
COCKPIT PREPARATION	PF		F + R/D
POWER-UP	PF		F + R/D
Blue Checklist	PF / PM		R/D
BEFORE START	LHS	RHS	F + C/R
AFTER START	LHS	RHS	F + C/R
TAXI	PF	PM	F
LINE UP			F + R/D
AFTER TAKE-OFF / GO-AROUND			F + R/D
CLIMB			F + R/D
DESCENT			F + R/D
APPROACH			F + R/D
BEFORE LANDING			F + R/D
AFTER LANDING			F + R/D
SHUTDOWN	LHS	RHS	F + C/R
SECURING	PM		F + R/D

R/D = Read and Do (Check)
C/R = Challenge and Response
F = Flow

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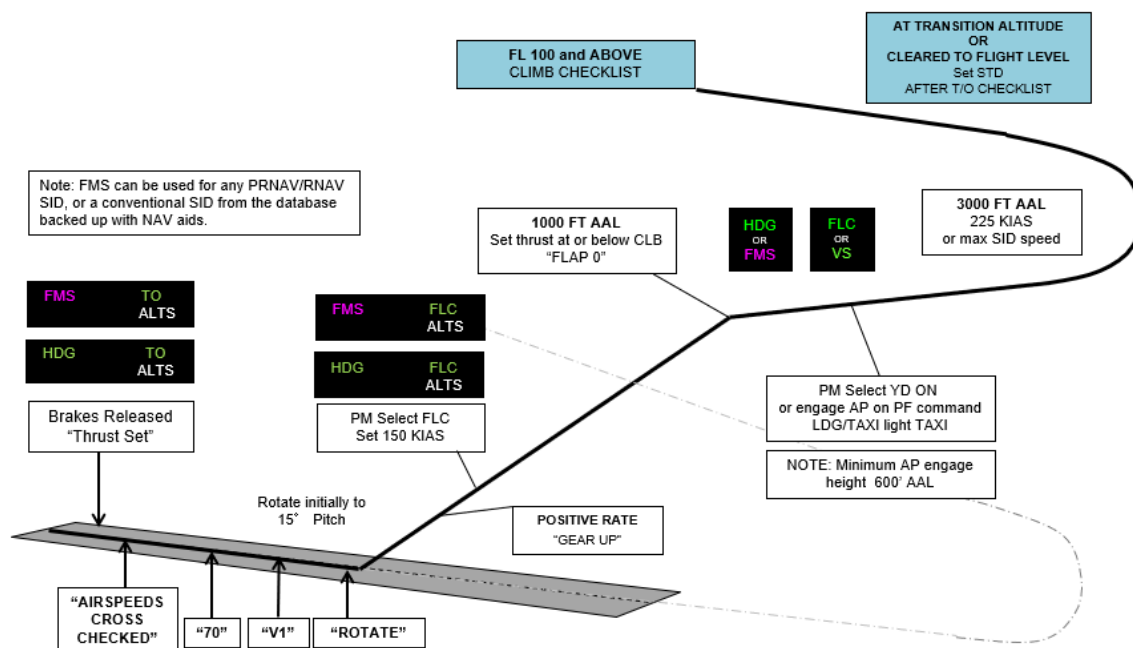
2.25 Expanded Flows and Checklists

The Expanded Checklists are incorporated in the respective flow sections in [Chapter 2](#).

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2.26 Flight Profiles

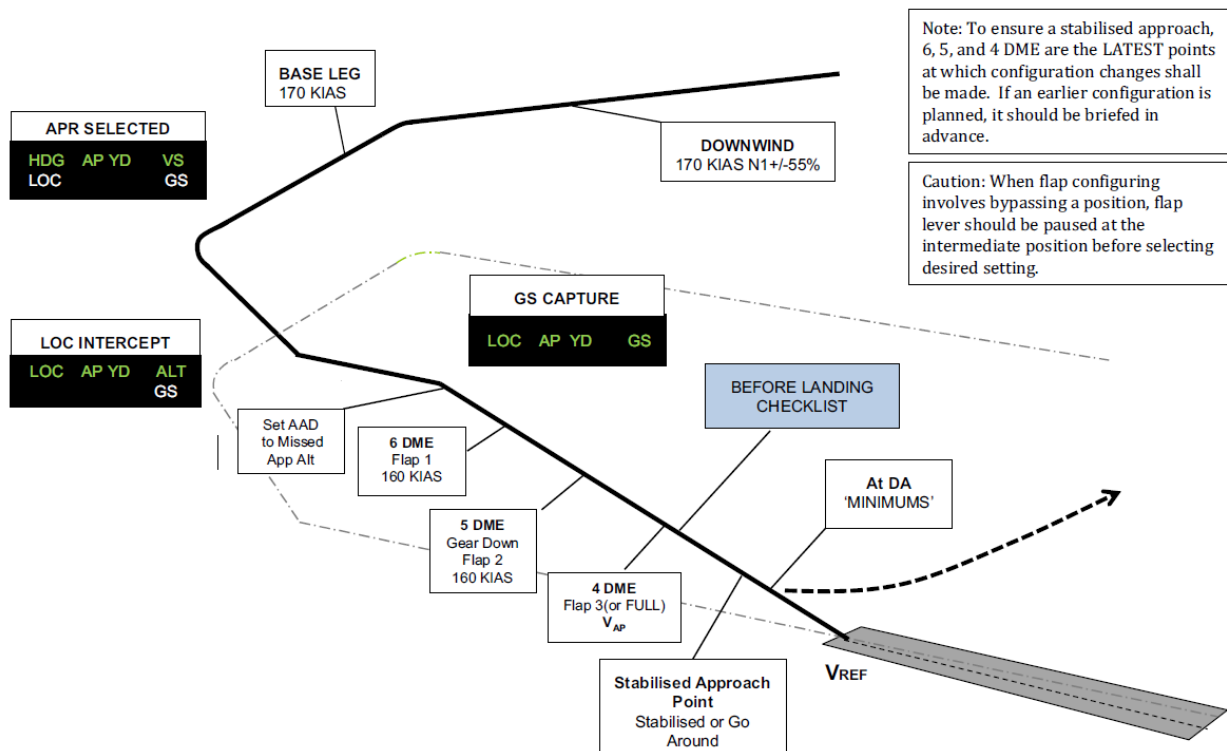
2.26.1 Normal Take-Off



2.26.2 Alternative Take-Off

Reserved.

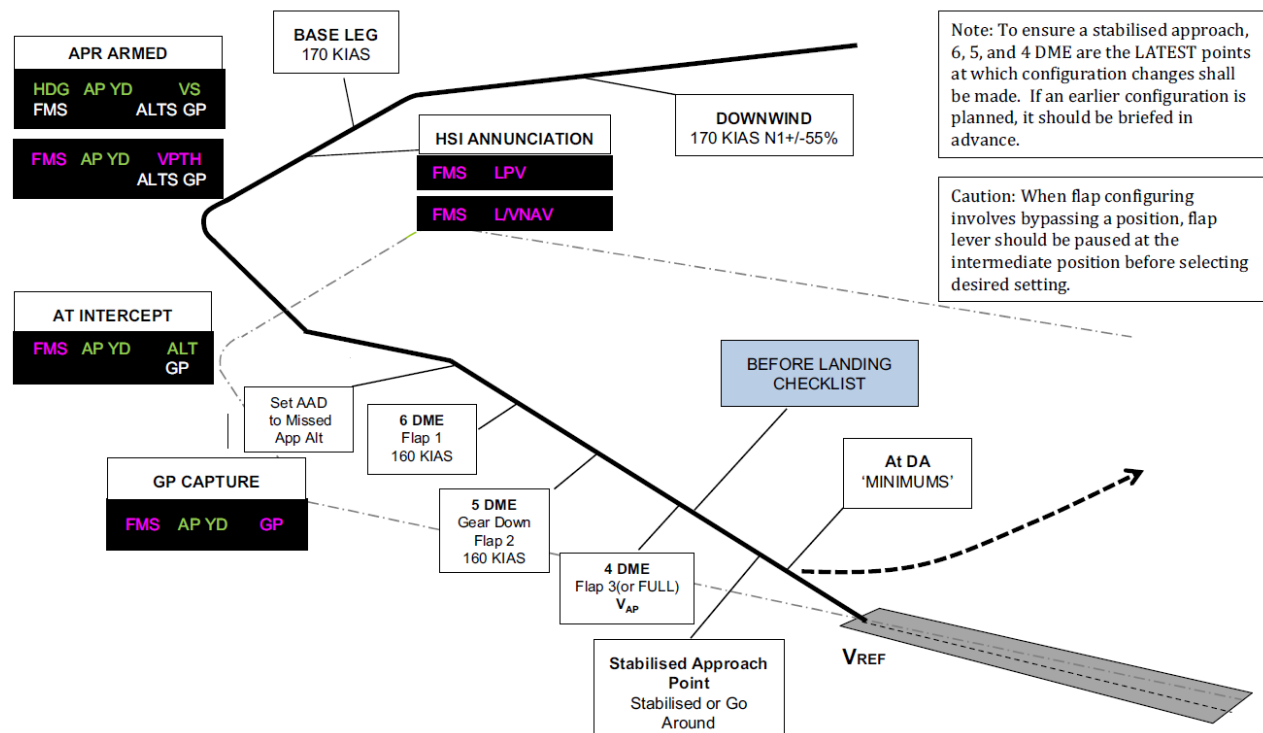
2.26.3 ILS – to CAT I minimum



NOTE

Setting of Missed Approach Altitude. When in ALT mode prior to intercepting the final approach glidepath, PF or PM to set MAA, if not completed prior, PM shall set MAA at initiation of final descent.

2.26.4 RNP – to LPV minimum



NOTE

Setting of Missed Approach Altitude. When in ALT mode prior to intercepting the final approach glidepath, PF or PM to set MAA, if not completed prior, PM shall set MAA at initiation of final descent.

2.26.5 PAR

Reserved.

2.26.6 RNP – to LNAV/VNAV minimum

Refer to [RNP – to LPV minimum](#).

2.26.7 RNP AR

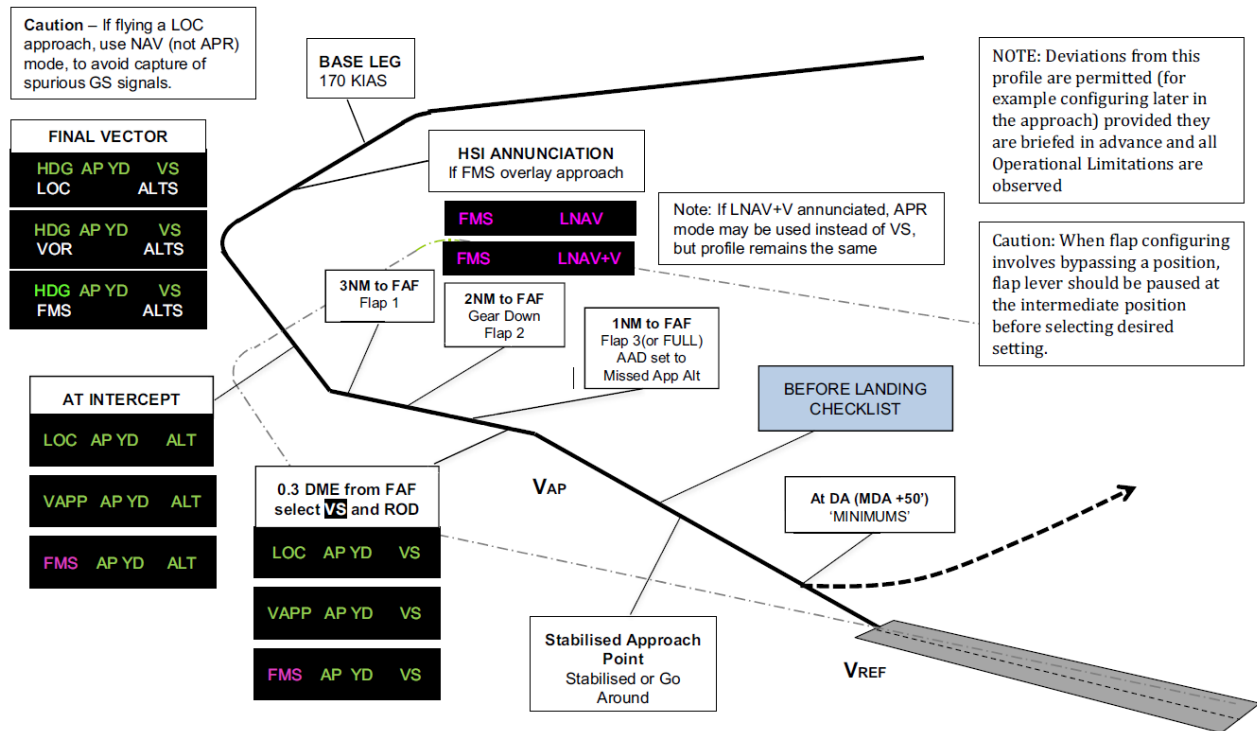
Reserved.

2.26.8 RNP – to LNAV minimum

Refer to [FMS Overlay](#).

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2.26.9 FMS Overlay



2.26.10 Non-FMS Overlay

Refer to [FMS Overlay](#).

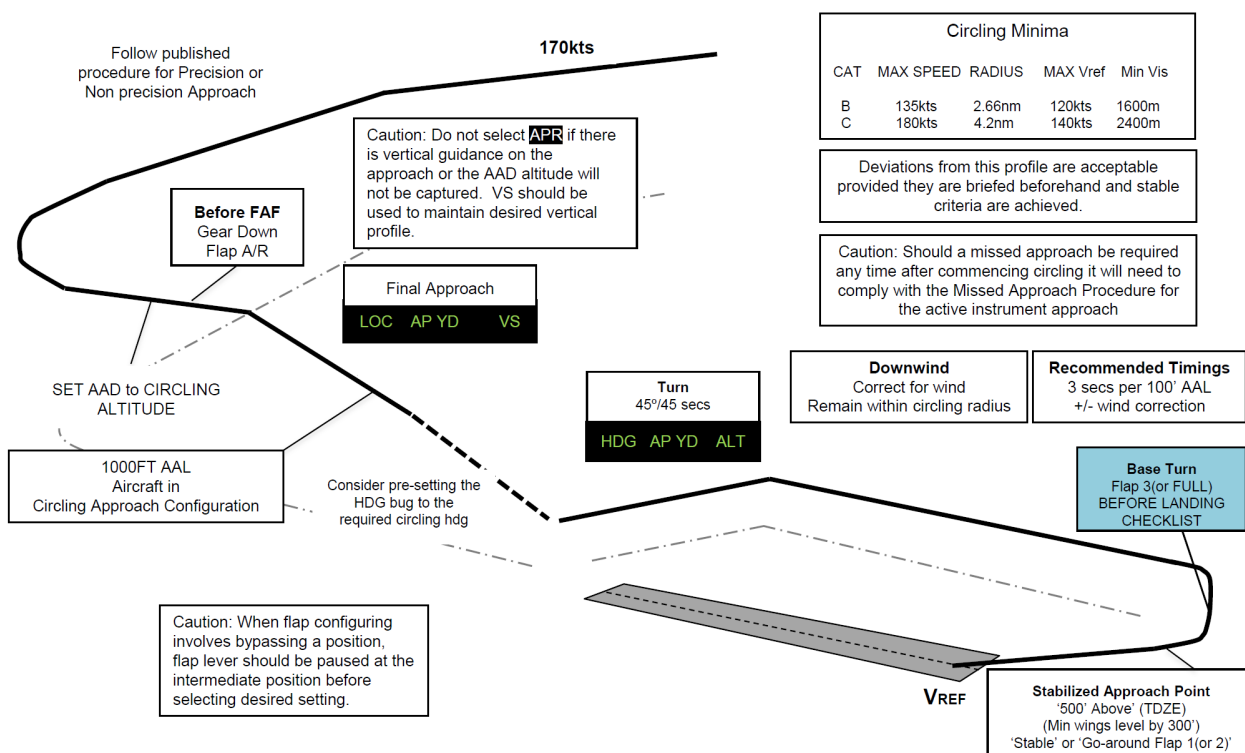
2.26.11 IGS

Reserved.

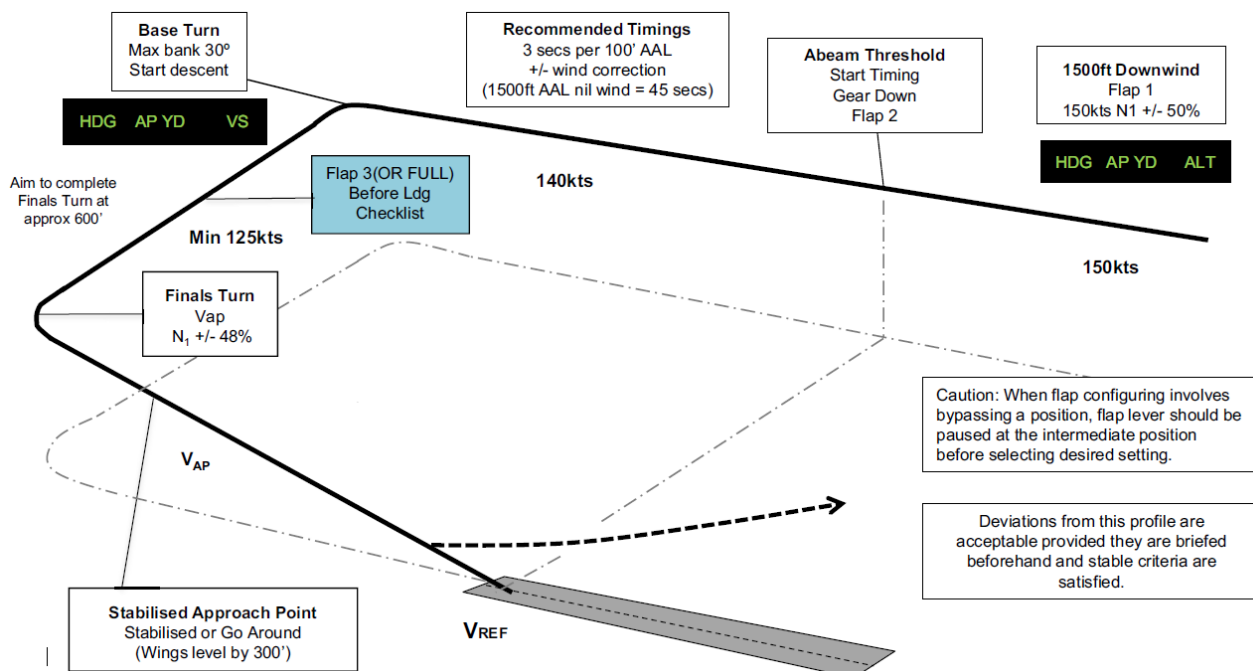
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2.26.12 Circling Approach

CIRCLING APPROACH

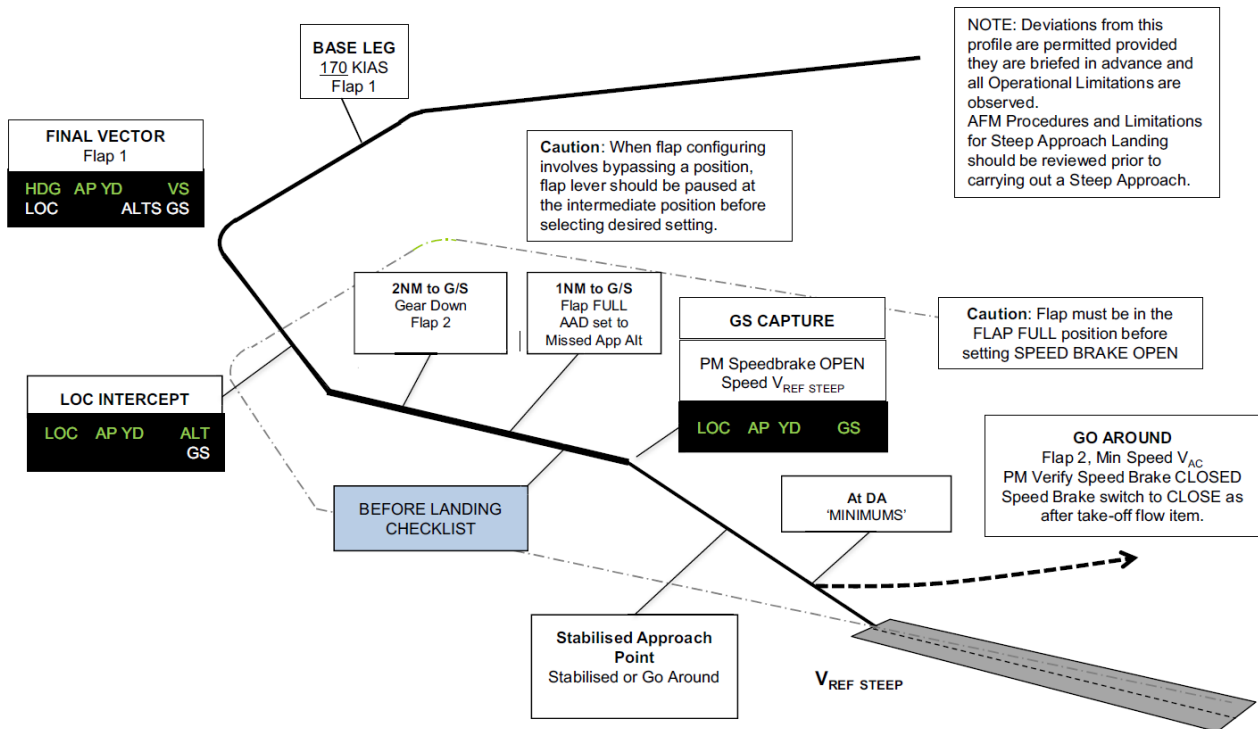


2.26.13 Visual Approach



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2.26.14 Steep Approach

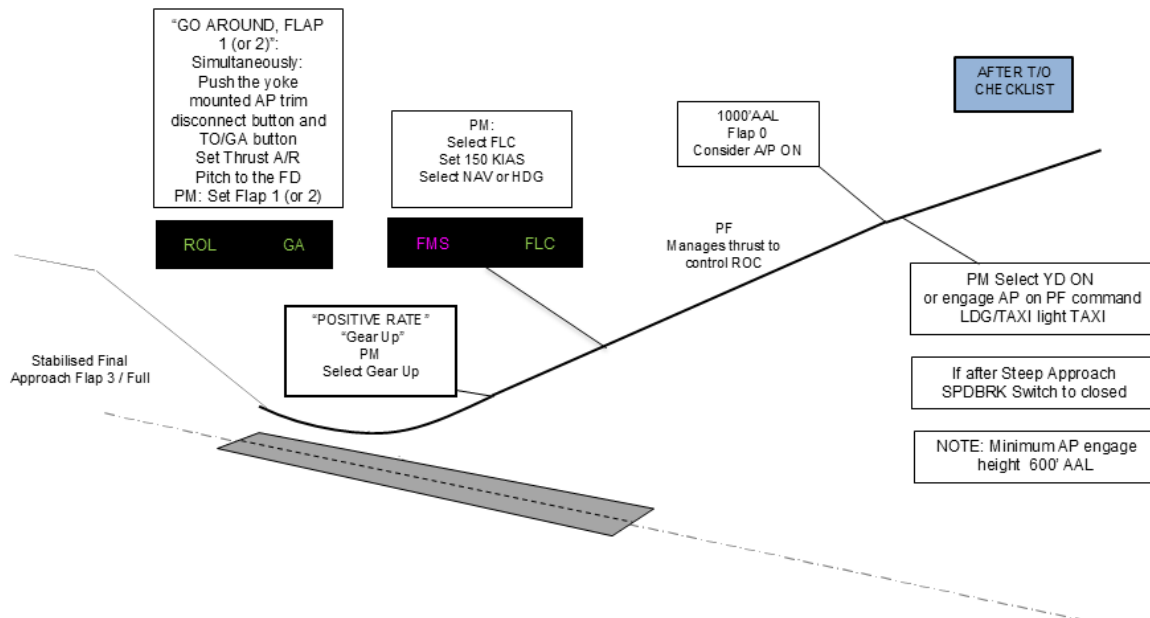


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2.26.15 All Engine Missed Approach

NETJETS

Phenom 300



NOTE

FMS mode should only be engaged if Missed Approach Procedure is correctly programmed into the FMS.

NOTE

TOGA Power should be applied for any go around requiring a climb. If the go around procedure has a low MAA, thrust may be reduced as soon as the aircraft is above 400 ft AAL, and established in a climb with the LDG selected up.

CAUTION

When configuring from flap 3 to flap 1, the selector lever must be paused in the flap 2 detent before continuing to 1.

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Chapter 3 - Abnormal and Emergency Procedures

3.1 Abnormal / Emergency Procedures Management

General

Refer to the AFM, QRH, and POH for the Abnormal and Emergency procedures.

3.1.1 Communication

Refer to OM-A, [Operating Procedures](#).

Time-critical: The nearest suitable airfield should be identified and its suitability determined. This needs to include Weather and NOTAM information. The decision should be based on the situation and the crew RSIC and DORDAR process. The company should only be contacted when time permits. The passengers' preference should not be considered.

Non-time-critical: The Company HOTLINE should be contacted and their view and guidance considered in the decision making. The Hotline should be called because of the Conference Call Facility and the fact that it is always manned. This input may be based upon passenger convenience (recovery option available) or the best maintenance facility. The office may be able to assist in the trouble shooting and in the overall handling of the situation. The passengers should be advised of the situation and course of action.

Additional Considerations: If time allows use the aircraft SMS / Email to explain any CAS messages in case voice communications are difficult or as an advanced notification before calling the HOTLINE.

QRH actions completed? Where are you heading to?

Make sure you try and add Lis-Phenom to any other office addresses so crew can get help from the duty pilot as quickly as possible.

Example:

PHI - ECS 1 Valve FAIL

QRH Completed

Heading to LSGG

Crew may also use the aircraft Wi-Fi system, in an abnormal or urgent situation, to use the iPhone to send emails or messages which may be quicker and easier than using the aircraft. Crew must remember to include the aircraft tail so that messages can be sent back to the aircraft.

3.1.2 Departure Contingency Procedures

Refer to OM-A, [Operating Procedures](#).

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3.1.3 Diversion in Case of Serious Technical Failure

Once the ATC has been informed and the memory items completed, the Commander shall determine, through the use of RSIC and DORDAR, whether a diversion is required, or whether the aeroplane can continue to destination.

As per OM-A, [Operating Procedures](#), advice may be available from the hotline. Normal runway factoring may no longer be required, but emergency factoring for performance considerations may be required by the emergency checklist.

If overflying mountainous terrain, crews should consider drift down profiles and limitations, which vary according to a number of factors (see AFM and Flight Profiles).

Technical failures involving configuration will further limit the altitude and fuel burn (and therefore range and endurance). Once a Mayday has been declared and the diversion initiated the runway belongs to the emergency aeroplane. This means that retaining alternate fuel, etc. is no longer a requirement as the situation may require landing with the minimum safe fuel for flight. Passengers should be briefed accordingly and also informed of external factors such as emergency services on landing.

3.1.3.1 Choice of Airfields

Land as Soon as Possible

An emergency will be declared. A landing should be accomplished at the nearest suitable airfield considering the severity of the emergency, weather conditions, field facilities, ambient lighting, aeroplane gross weight and ATC guidance. Depending on the situation criticality and considering the occupants safety, a landing off the runway may have to be attempted.

To consider landing under these circumstances the emergency should constitute an immediate threat to the safety of the aeroplane and occupants.

The decision to land on a runway with less than the AIR OPS landing distance required should only be considered if it is safer than continuing to a runway which meets the AIR OPS requirements.

Land as Soon as Practical

The emergency conditions are less urgent and although the flight is to be terminated, the degree of the emergency is such that an immediate landing at the nearest suitable airfield may not be necessary. The next suitable airfield is an airfield where in case of abnormal situations a safe landing can be performed considering the landing performance margins, weather minima, facilities, such as rescue services, customs and ground support. The selected airfield may not necessarily be the closest one.

3.2 Abnormal / Emergency Checklist Procedures

3.2.1 Checklist Methodology

Refer to OM-A, [Operating Procedures](#).

The system of managing an unexpected situation must promote the correct priority of always having one pilot dedicated to flying the aeroplane and therefore communication between crew must be clear and concise to identify any issue and share it within the crew as soon as possible.

The method for ensuring that all crew members arrive at a common understanding of any problem is as follows:

3.2.1.1 CAS Management

Any unexpected CAS messages shall be announced by the first crew member to notice them. They shall be announced stating the Warning, Caution, or Advisory alert and the message as it is displayed. The other crew member shall confirm they have understood the event by repeating the message, or where applicable, initiating an immediate action (such as RTO).

Example

Initial Call: 'Warning ENG 1 FIRE.'

Reply: 'Warning ENG 1 FIRE confirmed.'

PM cancels the Warning.

PF: 'Memory items for ENG 1 FIRE.'

Memory items are actioned by PM.

PM: 'Memory items ENG 1 FIRE complete.'

This is just an example of the initial call outs; following this, the nature and type of the situation will dictate the next response or actions.

At a convenient time, considering the CAS message, level of urgency and cockpit workload, the PF shall initiate the associated checklist, action or discussion.

NOTE

Crew may use The Electronic Checklist for Abnormal Events and Emergencies. The ECL must be checked and it shall be displayed on the MFD and items shall be ticked off as completed. The checklist shall be reset once the procedure is complete.

3.2.2 Emergency Evacuation

Refer to [CHAPTER 11 - Emergency Evacuation Procedures](#), QRH and SEP Manual.

3.2.3 Memory Items

The following scenarios require memory items:

- Rejected Take-off;
- Take off with Engine Failure at or Above V1;
- Engine Fire, Severe Damage or Separation;
- Engine Fire;
- Dual Engine Failure;

- Inadvertent Pusher Actuation;
- Electric Transfer Fail;
- Smoke / Fire / Fumes;
- Smoke Evacuation;
- Emergency Descent;
- Emergency Evacuation;
- Abnormal Start;
- Cabin Alt high;
- Electrical Emergency;
- Gear Lever cannot be moved up;
- Landing Gear WOW System Failure.

Following the completion of the memory items, when the checklist is called for ensure the memory items are read out and checked as complete before continuing with any subsequent actions.

Other situations requiring memory items without a Checklist include:

- Terrain Warning;
- TCAS RA;
- Windshear Escape Manoeuvre;
- Upset Recovery;
- Stall Recovery.

Critical Items / Actions

See OM-A, [Operating Procedures](#).

The emergency and non-normal checklists may require operation of critical items, which must be 'confirmed' by PF, or RHS pilot if the switch is only accessible to the LHS pilot, before movement of the switch or button. The following items are considered as critical items:

- Engine Thrust Levers;
- Start / Stop Knobs;
- Shut-off Buttons;
- Bottle Switch;
- Elec Emer button;
- Generator switches.

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3.2.4 Bomb Threat In Flight

Refer to OM-A, [Security](#).

There is no Least Risk Bomb location determined by the manufacturer, therefore, unless it is considered vital, it is best not to move or touch any suspect device but to try and move passengers and equipment away from the immediate area.

AEROPLANE SEARCH CHECKLIST EMBRAER PHENOM 300	
AREAS OF SEARCH	
1.	Cockpit and Cockpit Compartments
1.	Check floor including space ahead of rudder pedals, using a mirror if required.
2.	Check side pockets and oxygen mask stowage on both sides, and check smoke goggles pouches.
3.	Check behind and under the seats.
4.	Check binder stowage behind LHS.
5.	Remove the portable fire extinguisher bottle and check surrounding area.
2.	Galleys
1.	Check all the compartments in the galley area.
2.	Check behind and around the coffee machine.
3.	Check inside the ice container.
4.	Check and then remove the waste container and check the compartment.
5.	Check inside the roller door cabinet and behind the Jepps and manuals.
6.	Check inside the toys box and blanket / pillow storage bag.
7.	Check all catering and packaging.
3.	Passenger Cabin
1.	Check ashtrays / cup holders.
2.	Check surrounding area of each seat.
3.	Check all passenger seat drawers.
4.	Check seat lining and pouches.
5.	Check behind the tables and table compartment.
6.	Check life jackets and their compartments.
7.	Check the carpet (recently removed or loose?).
8.	Check all the emergency equipment compartments / areas.
9.	Check all side pouches and seat side compartments.
4.	Lavatory

AEROPLANE SEARCH CHECKLIST EMBRAER PHENOM 300	
AREAS OF SEARCH	
1.	Check inside toilet pan with the help of a torch.
2.	Open and check inside all miscellaneous compartments.
3.	Check cupboard area including FAK behind the manuals and life raft if fitted.
4.	Check waste bin and remove to check drawer.
5.	Check fire extinguisher next to toilet door.
5.	Aeroplane Exterior
1.	A full exterior inspection shall be carried out checking all possible areas that a bomb could be placed. Particular attention should be paid to the following.
2.	Check all panels and hatches not limited to but including refuel panel, battery no.2 panel, hyd panel, toilet waste panel.
3.	Check all exhausts and inlets.
6.	Baggage Compartments
1.	A full check of the rear and front baggage bays must be completed. Ensure only known baggage is present.
2.	The front baggage bay must be thoroughly checked for unknown bags or items.
3.	Check inside all covers bags, pins bags and oil cubby hole in rear baggage bay.
7.	Landing Gear
1.	During the check special attention must be given to the main gear landing bays as they are open and it would be easy to place something in the hatch. Look all round both sides for any items.
2.	Nose gear - check all the way up inside the nose gear bay and around the struts at the top.

3.2.5 Reserved

3.3 System Failures

3.3.1 Engine Failure On the Ground

Refer to OM-A, [Operating Procedures](#), and AFM 4-01.

3.3.2 Engine Failure In Flight

Refer to QRH or AFM 4-01.

3.3.3 Rejected Take-off

Procedure for Rejecting a Take-off

1. Either pilot has the authority to call for the rejection of a take-off.
2. The pilot deciding to reject the take-off shall call 'STOP STOP' in a loud, clear voice.
3. Take-offs shall be rejected for any system failure or fault annunciation below 60 KIAS.
4. Between 60 Knots and prior to V1, a rejected take-off should only be initiated for a situation involving:
 - ANY SIGN OF FIRE;
 - LOSS OF THRUST OR ENGINE FAILURE;
 - LOSS OF DIRECTIONAL CONTROL;
 - ANY RED CAS WARNING MESSAGE;
 - ANY OTHER SITUATION WHERE IT IS DEEMED SAFER TO REMAIN ON THE GROUND.
5. Once the decision has been made to reject the take-off, it is mandatory that the reject procedure is strictly adhered to without delay.
6. At or after V1, all take-offs shall continue and any problems treated as an in-flight emergency.
7. In the event of a tyre burst or symptoms of windshear above 60 knots the crew should continue the take-off as the stopping distance is greatly increased in either event. After take-off, the landing gear should initially NOT be retracted. Distance taken to stop the aeroplane is greatly increased and accelerate-stop distance is not assured in either event. The Phenom windshear alerting system does not activate until 50'. Therefore, crew need to remain vigilant for conditions around the airfield that may cause windshear. Additionally, crews should use ATC and other pilot reports to obtain information about any reported windshear events. When any suspected windshear conditions are experienced crews should take prompt action.

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3.3.3.1 Procedure for Stopping

PF	PM
'STOP STOP'	
• Thrust Levers..... IDLE • Brakes..... APPLY MAXIMUM • Directional Control..... MAINTAIN	• Confirm Spoilers have automatically deployed. • Call ATC 'Fraction xxx stopping'.

NOTE

1. After the aircraft is stopped and the parking brake has been applied the LHS pilot will assume control and call for Memory Items and / or appropriate checklists.
2. In accordance with the AFM, the brake cooling time for an aborted take-off is 20 minutes.

3.3.4 Cabin Pressurisation

Refer to AFM and Emergency Checklists.

During flight at high altitudes, all flight crew members must be prepared for an explosive decompression of the cabin. In case of such an event, an emergency descent must be initiated immediately to place the aeroplane within the oxygen prolonged usage envelope as soon as possible.

Be aware that an emergency descent as such exposes the aeroplane, its occupants and other aeroplanes in the area to further hazards. Ensure that there is no serious damage to the aeroplane structure before commencing a high-speed descent.

Do not make a steeper descent than the situation warrants.

When the aeroplane has levelled off, the Commander should address the passengers advising that a rapid descent has been made, the reason for the rapid descent, as well as any other pertinent information.

3.3.5 Power-up Anomalies

Some systems may report as failed at the end of the power-up sequence.

The Phenom has several systems including the TAWS and Garmin that conduct Built in Tests on power-up.

Before considering a system genuinely failed, a power-up anomaly should be suspected. These types of failures occur more often in adverse environments (very cold, very warm and / or humid).

Other power-up anomalies are typical of an avionic software version and the appropriate manufacturer published information should be used when dealing with these anomalies. These are Fleet Operational Letters (FOLs) and are published on Comply365.

In some cases, and when on the ground, individual system reset may solve the problem, while on others, a complete aeroplane 'power down power up' cycle may be necessary.

The Phenom conducts a test of databases and software during the power-up sequence and displays the expiry dates on the MFD.

During aeroplane power-up the GCU (Generator Control Unit) performs a BIT (built in test) which is an electrical system integrity check. A spurious failure of this test can occur which will only be evident when crews try to start the engine. If the engine fails to start on the ground after accomplishing the Engine Start procedure, the only way to clear this and to try to allow engine start is to power down all electrical power sources and wait for 2 minutes before restoring electrical power.

3.3.6 Loss of Position Information

See also: Phenom Fleet Handbook, QRH, Garmin Manual, FOLs.

GPS is the only positioning input to the Garmin 3000 FMS. If both GPS signals are lost, the aircraft does not have a conventional backup for the FMS to have positional information. The VOR and DME can be used for conventional navigation but cannot be linked to the Garmin FMS.

The AHRS takes a signal feed from the GPS, Magnetometer, and Air Data and if any of these are interrupted, an AHRS FAULT CAS is seen by the crew which means the AHRS is working in a degraded mode of operation.

Some or all of the following annunciations or CAS messages will be seen on the Garmin 3000 in the event of a loss of GPS signal.

- **GPS LOI** (PFD)
- **BOTH ON GPS 1/2** (PFD)
- **TAWS N/A** (SVS NOT AVAILABLE) (PFD)
- **AHRS 1/2 FAULT** (CAS)
- **WSHEAR NOT AVAIL** (CAS)
- **ADS-B NOT AVAIL** (CAS)
- **NO FMS POSITION** (MFD)
- **GPS NAV LOST** (GTC)
- **AHRS 1/2 GPS** (GTC)
- **DR** (HSI)

NOTE

Dead Reckoning Mode (DR) only functions in Enroute (ENR) or Oceanic (OCN) phase of flight. In DR Mode, the system uses its last-known position combined with continuously updated airspeed and heading data (when available) to calculate and display the aircraft's current estimated position.

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In all other phases, an invalid GPS solution produces a 'NO FMS POSITION' annunciation on the map and the system stops using GPS.

If any of the above messages are presented and GPS integrity is suspected, the following QRH checklists should be followed:

- GPS NAV SYSTEM FAILURE
- AHRS 1/2 FAULT

This checklist must be followed to slow the aircraft down to prevent a possible AHRS miscompare limit being exceeded which could lead to an AHRS FAIL situation.

NOTE

Ventral rudder availability has been improved to depend on AHRS yaw rate values only. Previously it also required use of roll parameters. In practical terms, lateral oscillations will not be occurring when AHRS are operating in alternate mode.

Additional Actions

1. Check the GPS Status Page (Utilities / GPS Status) on the GTC and the detailed information shown on the PFD when this page is selected on the GTC.
2. Identify the failed sensor, cross-checking aircraft position using VOR / DME.
3. Monitor aircraft position and navigation performance.
4. Inform ATC when navigational accuracy is degraded significantly, or this is suspected.

3.3.6.1 GPS Jamming

GPS jamming may be expected in areas of political instability, near military objects and / or heads of state. The Phenom has suffered GPS jamming for over two years when going into Cairo. The following messages are the first indicators of GPS jamming along with the ones shown in [Loss of Position Information](#).

- BOTH ON GPS (1/2)
- GPS LOI
- GPS NAV LOST (on the GTC)

When any of the above messages are presented:

1. Check the GPS Status Page (Utilities / GPS Status) on the GTC and the detailed information shown on the PFD when this page is selected on the GTC.
2. Use VOR / DME information to continue navigating the aircraft.
3. Monitor aircraft position and navigation performance.
4. Inform ATC when navigational accuracy is degraded significantly, or this is suspected.

GPS will recover automatically when clear of the area of jamming.

3.3.6.2 GPS Spoofing

GPS spoofing is an attempt to deceive a GPS receiver by broadcasting incorrect GPS signals. These signals may cause a complete failure of GPS similar to GPS jamming or cause the GPS position to shift from the actual position. As GPS is the primary position sensor, the FMS position will shift with the GPS position. Depending on the rate of change of GPS position, this shift may not be detected automatically.

The following are potential indicators of GPS spoofing:

- GPS LOI
- NO FMS POSITION
- LOSS OF GPS NAVIGATION (GTC)
- Map shift on PFD / MFD
- Excessive deviation between FMS position and conventional navigation sources.

Inform ATC when navigational accuracy is degraded significantly, or this is suspected.

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3.4 Fire and Smoke

3.4.1 Engine Fire On Ground

Refer to QRH or POH 2-10 Engine Fire on the Ground.

An Engine Fire indication on the ground requires the aeroplane to be brought to an immediate halt and, with the parking brake on, carry out the Engine Fire Drill memory Items. An Emergency Evacuation or Rapid Disembarkation may be ordered by the PIC if deemed necessary and appropriate.

NOTE

The Evacuation Drill does not include discharging the Fire Extinguisher and so the fire drill must be carried out first to ensure the fire bottle is discharged into the correct engine.

3.4.2 Hot Start / Tail Pipe Fire

Tail Pipe Fire

If an engine start-up is not successfully performed, some fuel fumes can accumulate in the combustion chamber and exhaust area. An immediate reattempt to start the engine may cause this fuel to ignite, which causes flames to come out of the tail-pipe. In order to avoid this situation it is highly recommended to perform dry motoring before attempting a further start.

It should be noted that a tailpipe fire will most likely NOT result in a fire warning on the flight deck.

The Phenom does not have a specific checklist for a tail pipe fire, therefore, if notified of an engine fire during start without any indications in the cockpit, the recommended actions are:

- START / STOP Knob - STOP (to shut down the engine and remove the fuel feed);
- DRY MOTOR the engine to remove any excess fuel;
- Eng Ignition Switch - OFF;
- START / STOP Knob - START then RUN;
- Wait 30 Seconds;
- START / STOP Knob - STOP;
- Eng Ignition Switch - AUTO;
- Do not use the ENG FIRE pushbutton unless you have an E1/2 Fire CAS Message.

Since the fire is burning within the turbine casing and exhaust nozzle, applying engine fire procedures and discharging the extinguisher to the space between casings and cowls, and therefore outside of the engine core, will be ineffective (and may also make it impossible to dry motor the engine, which is the quickest way of extinguishing most tail pipe fires).

If a suspected fire is associated with an E1/2 FIRE CAS message, then the above is to be disregarded and the normal E1 (2) FIRE memory and QRH actions carried out.

Hot Starts

If the engine experiences more than the usual difficulty accelerating (due to such problems as early starter cut-out, fuel mis-scheduling, or tailwinds), the engine may spend a considerable time at very low RPM (sub-idle). Normal engine cooling flows will not be effective during sub-idle operation, and turbine temperatures may appear relatively high. This is known as a hot start (or, if the engine completely stops accelerating toward idle, a hung start).

The AFM contains all temperature limits for ITT during a start and the FADEC will abort the start if the ITT exceeds the steady state limit by a value equal to half the difference between ITT transient and steady state limit (around 720°C). However, crews should not wait for the FADEC to shut the engine down as the FADEC works very close to the limits of the engine and if a hot start is seen or suspected, then crews should abort the start manually to avoid any risk of limits being exceeded.

The FADEC will also abort the start if there is no light up indication (ITT has not raised significantly after 10 seconds of fuel flow indication) and if a hung start is detected when the time to reach idle has exceeded 60 seconds. However, as detailed above for a hot start, it is best for crews not to wait for the FADEC to close the engine down, because waiting for 60 seconds for the engine to reach idle is a long time when idle is normally achieved in around 20 seconds.

The Phenom can suffer problems with engine starting in even light tail winds, which is more apparent during battery starts. The advice given in the POH is that a tail wind of more than 10 kts can lengthen the start and a higher ITT may be observed. It should be noted that if any tail wind is sufficient to turn the blades of the engine the 'wrong way', the possibility exists that the engine may not start properly. Therefore, crews should, whenever possible, carry out engine starts into a headwind or crosswind.

See QRH - Engine Abnormal Start.

3.4.3 Engine Fire In Flight

Refer to QRH.

3.4.4 APU Fire

Not applicable.

3.4.5 Aeroplane Interior Smoke and Fire

Refer to QRH SMOKE Tab.

A smoke / fire event in flight is one of the most serious situations that can be encountered on board an aeroplane.

The basic principles to cope with this situation are:

1. PROTECT the aeroplane occupants (consider any smoke as potentially toxic).
2. EVACUATE the smoke.
3. IDENTIFY and ISOLATE the source / FIGHT THE FIRE.
4. LAND AS SOON AS POSSIBLE.

Finding the exact source of smoke can be very difficult, the safest course of action is to land as soon as possible.

The completion of any prescribed procedures and extended checklists should not delay the commencement of recovery to a suitable airfield.

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3.5 Crew Incapacitation

Refer to SEP Manual.

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3.6 Emergency Landing and Ditching

3.6.1 Emergency Landing

Refer to SEP Manual, AFM 4-01 and QRH (NAP).

The AFM refers to two types of advice for emergency situations which are:

Land As Soon As Possible - A landing should be accomplished at the nearest suitable airfield considering the severity of the emergency, weather conditions, performance considerations, aeroplane weight and ATC guidance. Depending on the situation criticality and nature of the emergency a landing anywhere might be considered.

Land As Soon As Practical - The emergency conditions are less urgent and although the flight is to be terminated, the degree of the emergency is such that an immediate landing at the nearest adequate airfield may not be necessary.

3.6.2 Ditching

Refer to SEP Manual, AFM and QRH (NAP).

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3.7 Exceeding Structural Limits

3.7.1 Overweight Landing

Normally, the maximum certified landing weight shall not be exceeded. Overweight landings, however, may be necessary in abnormal or emergency situations where, in the Commander's judgment, this course of action provides a higher degree of safety than remaining airborne until aeroplane weight reaches its landing limit.

Situations such as serious illness of crew or passengers that require immediate medical attention may justify overweight landings. In such case, deviations from prescribed procedures to the extent required are permissible in the interest of safety.

NetJets Transportes Aéreos considers that all the following circumstances warrant overweight landings:

- Engine or airframe fire;
- Engine failure;
- Total electrical failure;
- Flight Control jam or disconnect;
- Pilot incapacitation;
- Smoke in the cabin or flight deck.

Preparation

In preparation for an overweight landing, due consideration should be given to the following requirements (these may be disregarded, if deemed necessary, during an emergency):

- System malfunctions, which affect controllability or braking capability of the aeroplane such as:
 - Hydraulic failure;
 - Landing gear or tyre failure;
 - Flight controls failure.
- Performance requirements, such as:
 - Landing field length;
 - Approach and landing climb;
 - Flap limiting speeds;
 - Tyre fuse plug capacity;
 - Go-around capability.
- Meteorological conditions, such as:
 - Ceiling and visibility;
 - Crosswind / tailwind component;
 - Contamination of runways;

- Windshear conditions;
- Sink rate.

Normal sink rate at touchdown averages 200 ft/min. Transport category aeroplanes are normally certified with a sink rate of 360 ft/min at the structurally limited TOW, and with 600 ft/min at the maximum landing weight. Therefore, structural problems should not arise, provided sink rates at touchdown do not exceed 360 ft/min.

If landing at a weight in excess of maximum landing weight is unavoidable:

- Plan to land on the longest available runway;
- Minimise sink rate and crab at touchdown;
- Report overweight landing for appropriate maintenance inspections.

CAUTION

Tyre speed limitations will be exceeded if touchdown is made in excess of 182 knots ground speed.

Any overweight landing must be recorded in the Aircraft Technical Log, and proper maintenance release obtained before further flight. An overweight landing inspection is mandatory.

3.7.2 Hard Landing

A hard landing occurs when the aeroplane impacts the ground with a greater vertical speed and force than in a normal landing. The average vertical speed in a landing is around 120 ft/min at maximum take-off weight. The definition of a hard landing is in excess of 360 ft/min at maximum take-off weight or in excess of 600 ft/min at maximum landing weight. Hard landings can be caused by weather conditions, mechanical problems, overweight aeroplane, pilot decision or pilot error.

A hard landing requires a maintenance inspection.

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3.7.3 Exceeding Speed Limitations

Any overspeed requires positive action to correct. Reduce thrust to idle and select spoilers out, if available, to reduce speed to below limits.

In order to prevent flap and gear overspeeds, a 10kt buffer should be applied to the limiting speeds when making flap and gear selections.

3.7.3.1 Flap Overspeed

In the event of an inadvertent flap overspeed, Embraer recommends the following procedure:

Do not move the flap.

Reduce speed to below limiting speed for desired flap setting.

Select flap position and monitor CAS for any malfunction indications.

If no abnormal indications are present, the flight may be continued to destination. After landing, regardless of the degree of overspeed, the aeroplane will require a maintenance inspection before next flight.

If any CAS malfunction message is displayed, the flap lever should be returned to the previous setting, speed should be maintained below the limiting speed and aeroplane landed at a suitable location in accordance with QRH procedures for the current flap position.

3.7.3.2 Landing Gear Overspeed

In the event of an inadvertent landing gear overspeed, Embraer recommends the following procedure:

The Crew shall verify the position of the landing gear using normal procedures, i.e through the green circle 'DN' indicator.

Reduce the aeroplane speed by changing the attitude and / or reducing thrust to achieve again the operational envelope in which the aeroplane can be safely flown with landing gear extended and locked (VLE speed).

If the operational deviation occurred during the approach to the destination, consider maintaining the landing gear extended until the landing.

In case of an inadvertent landing gear overspeed in a segment of flight other than the approach, the landing gear may be retracted, under pilots' discretion, in order to reduce fuel consumption and improve the aeroplane performance. In this case the landing gear can be retracted when the indicated airspeed is equal or lower than VLO, the speed at which the landing gear can be safely extended and retracted.

After landing, regardless of the degree of overspeed, the aeroplane will need a maintenance inspection before the next flight.

3.8 TAWS / Ground Proximity Warning

Refer to OM-A, [Operating Procedures](#), and AFM 3-04.

3.8.1 Caution Alerts

Trigger: 'CAUTION TERRAIN', 'CAUTION OBSTACLE'.

TRIGGER	PF	PM
CAUTION ALERT	Take immediate positive corrective action to adjust aircraft flight path as necessary to eliminate the caution alert.	• Verify the PFs response is appropriate. • Monitor attitude and airspeed.
	Evaluate the situation, including surrounding terrain, and determine if subsequent actions are required.	

3.8.2 Warning Alerts

Trigger: **PULL-UP** 'TERRAIN, TERRAIN, PULL UP, PULL UP'
'OBSTACLE, OBSTACLE, PULL UP, PULL UP'

The AFM states that in the event of a TAWS alert, pilots must use the flight controls and thrust as necessary to correct the aeroplane attitude, flight path and configuration, according to the voice message presented, to provide terrain clearance.

TAWS is provided on the aeroplane as a situational awareness tool only and must not be used solely as the basis for terrain avoidance manoeuvring decisions.

The task allocation below provides general guidance to be followed in the event of a TAWS warning.

Actions to be taken in the Event of a TAWS Warning

EITHER PILOT WILL CALL 'TERRAIN GO'	
PF Actions	PM Actions
Call 'Flap 1 (or 2)' (if flap already extended).	Select Flap 1 (or 2) on PF command.
Disengage the Autopilot (Push TO/GA button).	Retract spoilers (if extended).
Advance thrust levers to Max.	Confirm Max thrust.
Roll wings level (see notes).	
Firmly apply back pressure to set climb attitude.	Call Positive RoC.
Call for Gear UP.	Select gear up on PF command.

EITHER PILOT WILL CALL 'TERRAIN GO'	
PF Actions	PM Actions
Allow airspeed to reduce to achieve optimum RoC. Target speed is top of the yellow band.	Monitor the flightpath and speed.
Maintain configuration until TAWS warning ceases or, if in IMC, reaching the appropriate safe altitude.	Confirm appropriate safe altitude (MSA / MOCA / MORA) and QNH. Communicate with ATC as soon as possible, but do not interrupt the safe flying of the aeroplane.
Re-establish normal flight profile, verify aeroplane location and altitude.	Call passing / achieving safe altitude. Advise ATC of intentions.

NOTE

If there is no immediate and positive reaction from the PF, the PM shall take control of the aeroplane and follow procedures.

Climbing straight ahead is the recommended response unless operating in visual conditions or the pilot determines, based on all available information, that turning in addition to climbing is the safest course of action.

3.9 Windshear

Refer to OM-A, [Operating Procedures](#).

In order to avoid dangerous windshear phenomena, it is important to know what windshear is and in which meteorological and geographical environment it can be expected. The following definition seems to be the most suitable for aviation:

Windshear is a sudden or rapid change in wind direction and / or speed along the flight path of an aeroplane.

3.9.1 Windshear Recognition

The Phenom has a computer based Windshear Alerting System and is an aid to detect windshear. However, the protection of the aeroplane and occupants is down to the recognition of the event and response by the crew.

The following conditions should be considered as indications of windshear:

- Indicated airspeed variations in excess of 15 kts;
- Ground speed variations;
- Variations in wind direction and speed;
- Vertical speed excursions of 500 ft/min;
- Pitch attitude excursions of 5 degrees;
- Glideslope deviation of 1 dot;
- Heading variations of 10 degrees;
- Unusual throttle positions.

3.9.2 Standard Response / Recovery Technique

The call 'WINDSHEAR GO' should be made to differentiate between this and a normal go-around.

Carry out a maximum performance go-around and DO NOT change configuration.

Standard response of the flight crew should be:

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Either Pilot calls 'WINDSHEAR GO'

PF	PM
Disengage the Autopilot (Push TO / GA button). Advance the Throttles to Max. Roll Wings level. Smoothly apply back pressure to raise the nose initially to 15 degrees nose up attitude.	Confirm Max Thrust. Confirm Spoilers retracted.
Do not change configuration except spoilers	
Adjust pitch attitude smoothly in order to achieve the maximum rate of climb and to minimise any unwanted rates of descent due to the windshear. Unusual stick forces may be required to maintain pitch attitude during airspeed variations away from normal target airspeed. Low airspeed must be accepted. Flying at airspeeds below normal reference speeds may be required in order to utilise the full performance capability of the aeroplane.	Monitor flight path. Maintain running callouts of vertical speed and speed trend until out of windshear. Visual lookout. Communicate with ATC as soon as possible but do not interrupt the safe flying of the aeroplane.
When clear of windshear, carry out normal Go-around actions.	

NOTE

1. The aim is to attain the maximum possible performance from the aeroplane in order to avoid ground contact. To do this, we are attempting to hold the aeroplane at maximum Angle of Attack (AoA). Whilst operating in this regime, control inputs must be as smooth as possible. Any additional G loading caused by rough handling may be enough to exceed max AOA.
2. Climbing straight ahead is the recommended response unless operating in visual conditions or the pilot determines, based on all available information, that turning in addition to the climbing is the safest course of action.

3.10 TCAS

Refer to OM-A, [Operating Procedures](#).

3.10.1 Traffic Advisory (TA)

Trigger: Upon a Traffic Advisory, the PF and PM procedures are:

TRAFFIC ADVISORY	
PF	PM
• Monitor TCAS display (look inside). • Direct PM where to look. • Be prepared to fly manually in the event of a RA.	• Look outside. • Attempt to acquire visual contact. • Report intruder's position and progress to PF.

3.10.2 Resolution Advisory (RA)

Trigger: Upon a Resolution Advisory, the PF and PM procedures are:

RESOLUTION ADVISORY	
PF	PM
• AUTOPILOT OFF; • Follow TCAS command.	• Monitor the manoeuvre (look inside). • Inform ATC by calling. 'Fraction XXX, TCAS RA'

3.10.3 Clear of Conflict

Trigger: When Clear of Conflict, the PF and PM procedures are as follows:

CLEAR OF CONFLICT	
PF	PM
• Return to intended flight path.	• Monitor the manoeuvre (look inside). • Inform ATC by calling. 'Fraction XXX, CLEAR OF CONFLICT, RETURNING TO (assigned clearance).' or 'Fraction XXX, CLEAR OF CONFLICT, (assigned clearance) RESUMED.'

3.11 Lightning Strikes

Refer to OM-A, [Operating Procedures](#).

If a lightning strike is confirmed or is suspected, it shall be recorded in the Aircraft Technical Log and Maintenance shall be informed. A maintenance lightning inspection is mandatory, and a proper maintenance release shall be obtained before further flight.

A lightning strike that causes damage to the aeroplane or the loss of any systems also requires an MOR to be submitted.

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3.12 Exceeding Cosmic Radiation Limits

Refer to OM-A, [Operating Procedures](#).

Cosmic Radiation detection equipment is not required as the aeroplane ceiling is FL450.

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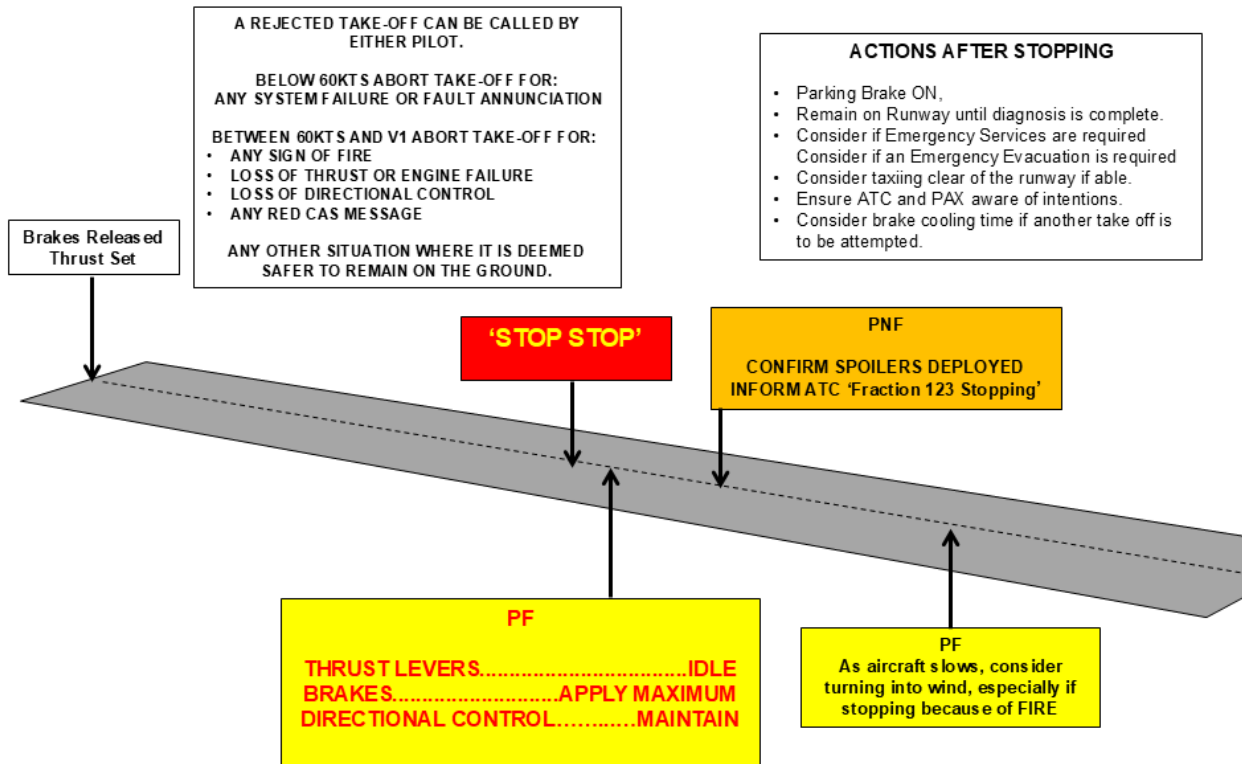
3.13 Profile Diagrams

3.13.1 Rejected Take-off

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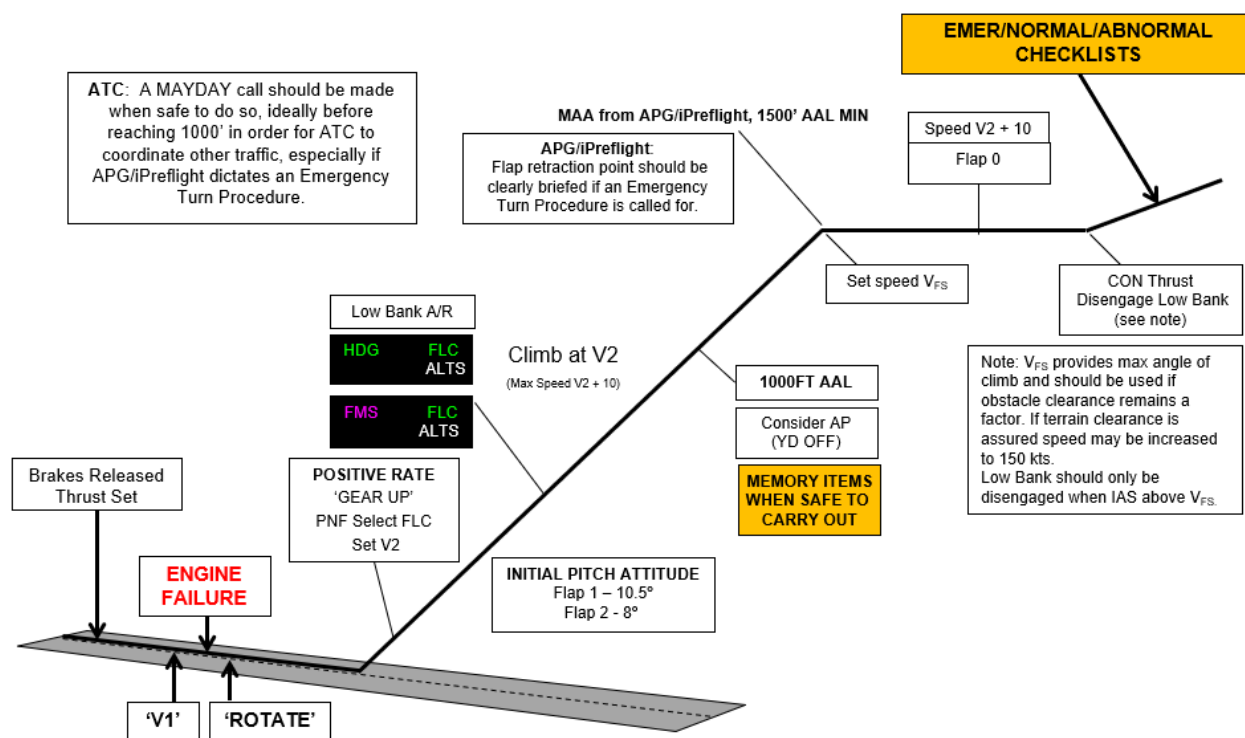
Phenom 300

3.13.1 REJECTED TAKE OFF

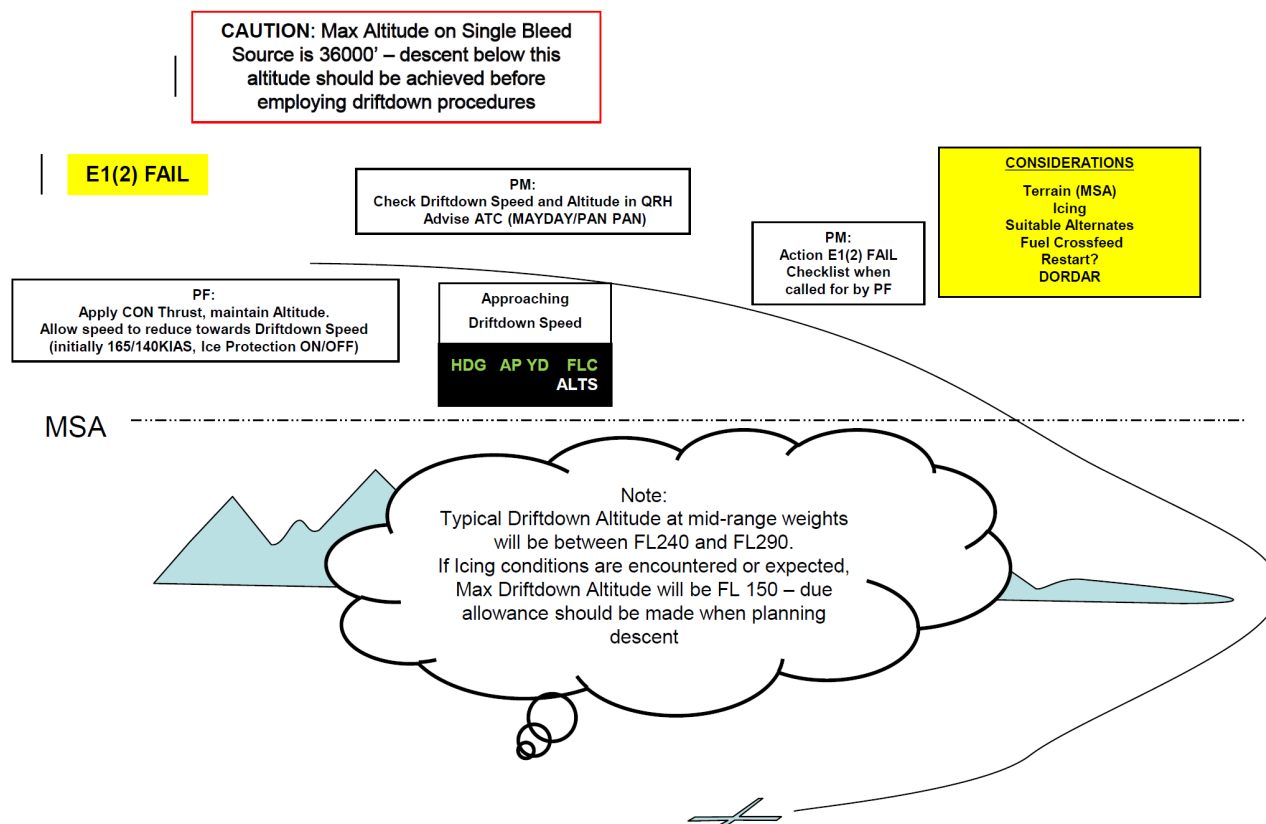


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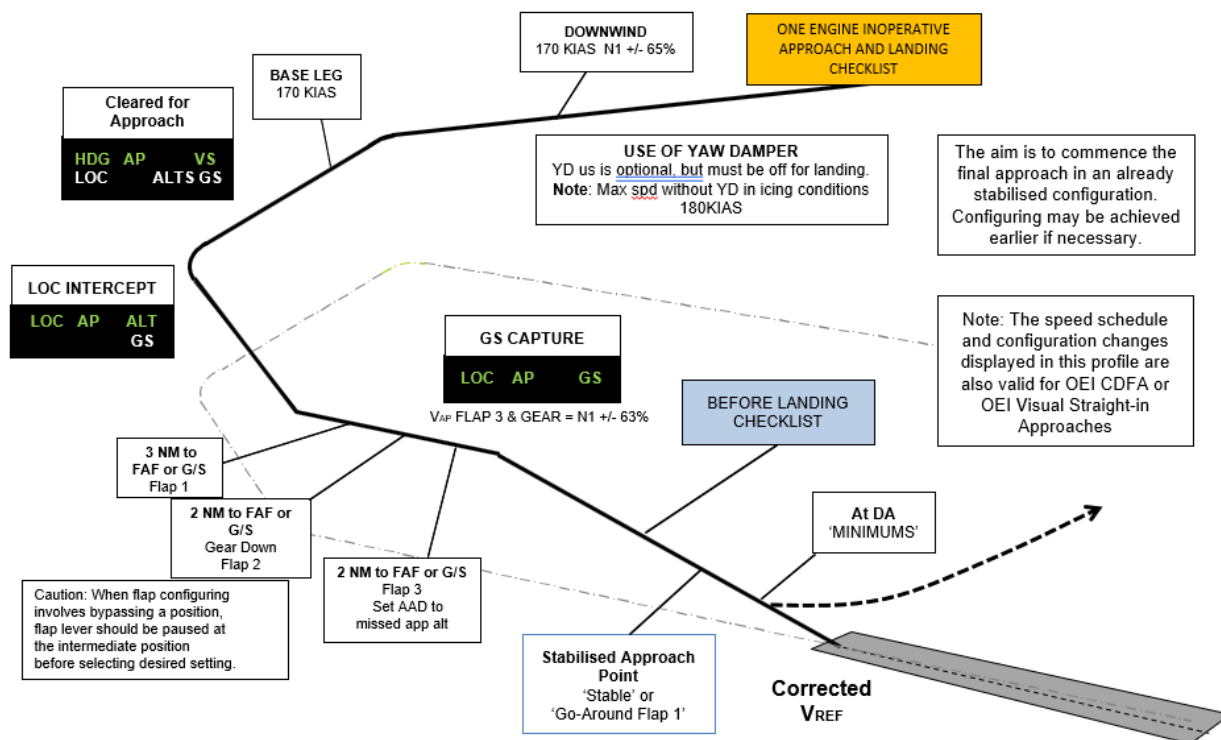
3.13.2 Engine Failure After V1



3.13.3 Engine Failure During the Cruise



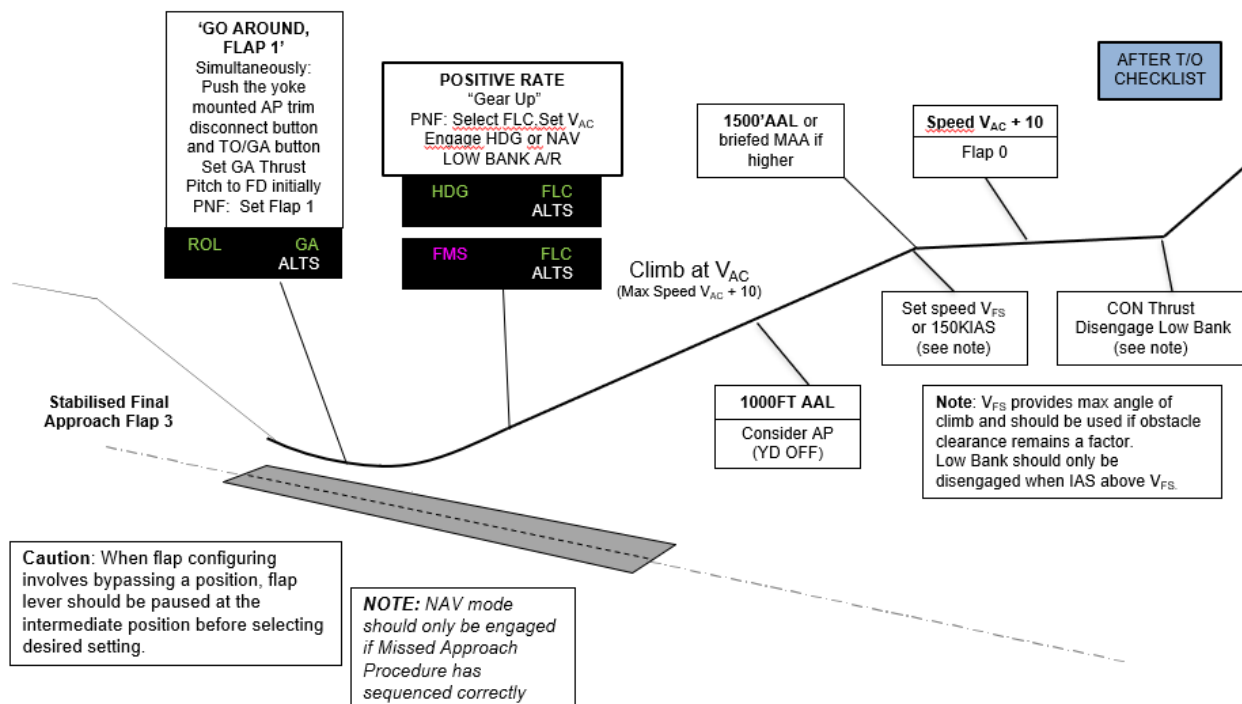
3.13.4 One Engine Inoperative Approach and Landing



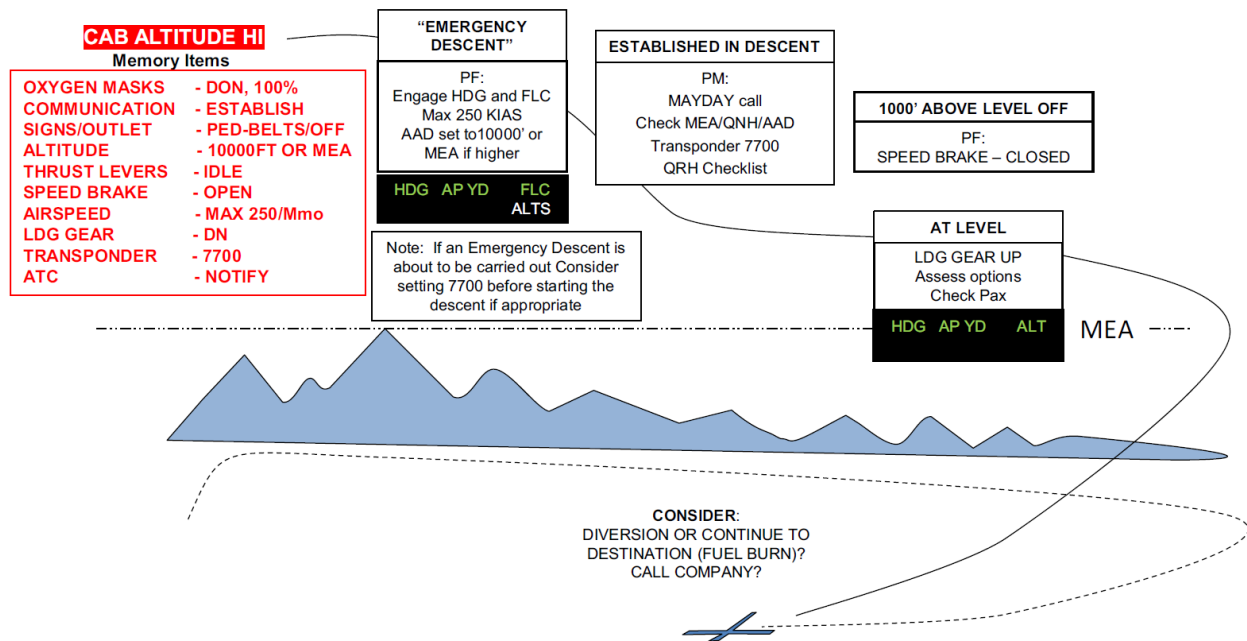
NOTE

Setting of Missed Approach Altitude. When in ALT mode prior to intercepting the final approach glidepath, PF or PM to set MAA, if not completed prior, PM shall set MAA at initiation of final descent.

3.13.5 One Engine Inoperative Go-Around



3.13.6 Emergency Descent



NOTE

Phenom 300E:

The Emergency Descent Mode (EDM) may engage during the procedure. Altitude, airspeed and heading should be manually selected as required.

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Chapter 4 - Performance

4.1 Performance Data

For Performance information and limitations, please refer to Comply 365 Document 114, Aircraft Calculations and Assumptions, OM-A 8.1.2.4.1, AFM, POH and QRH.

Performance data is available from the PDCS web interface which is used by dispatch to provide performance data in the flight release as well as the iPreFlight application on the EFB where crew can make their own selections.

4.1.1 Supplementary Performance Data

For Performance information and limitations, please refer to OM-A Technical Annex, AFM, POH and MEL.

4.1.2 Reserved

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4.2 Additional Performance Data

For Performance information and limitations, please refer to the AFM, the POH and MEL.

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Chapter 5 - Flight Planning

5.1 Pre-Flight / In-Flight Planning

Refer to OM-A, [Policy and Procedures for In-flight Fuel Management](#), [Operational Flight Plan \(OFP\)](#), [Determination of the Quantities of Fuel and Oil Carried - Fuel Policy](#), as well as the Performance Section of the AFM and POH.

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5.2 In-Flight Fuel Monitoring

Refer to OM-A, [Policy and Procedures for In-flight Fuel Management](#), [Operational Flight Plan \(OFP\)](#), [Determination of the Quantities of Fuel and Oil Carried - Fuel Policy](#), as well as the Performance Section of the AFM and POH.

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5.3 ETOPS Flight Planning

Not applicable.

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Chapter 6 - Mass and Balance

6.1 General

Refer to OM-A, AFM and POH for all instructions and limitations pertaining to Mass and Balance.

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6.2 Pre-Flight Procedure

The Garmin system is not designed to calculate aeroplane CG position and simply uses a number of mass default parameters as well as crew inputs and the 'fuel on board sync' function to determine the aeroplane's take-off and landing mass.

The 'Operating Weight' tab should have the aeroplane BEW already shown as well as the 'Pilot and Stores' figure. These should both be the same as the Basic Empty Weight and Non Variable Weight figures seen on the flight release M&B sheet.

As a check the Basic Operating Weight should be the same as the OPNLWT on the OFP.

In the '**Payload**' section of the **Weight and Fuel** app there are three boxes where the crew can input mass parameters:

- No. of Passengers
- Mass of Passengers
- Cargo

The standard passenger weights are now different for Male, Female, and Children. Therefore do not use PAX @ Each boxes but insert the AV PLD figure into the Cargo box. This will take into account everything carried and includes the standard weights for the passengers and luggage.

If additional luggage or cargo is carried by the passengers (in excess of the standard weights) then dispatch will need to be informed to update the M&B sheet and the Cargo figure can be updated.

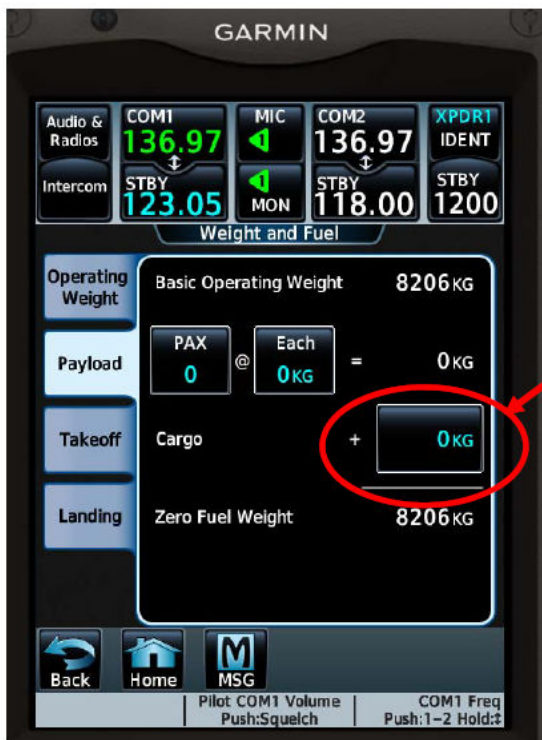
The standard masses used for AIR OPS calculations are as follows:

Male:	96 Kg / 212 Lbs (this mass includes 6 Kg / 13 lbs of hand luggage)
Female:	78 Kg / 172 Lbs (this mass includes 6 Kg / 13 lbs of hand luggage)
Child:	35 Kg / 77 Lbs (this mass includes 6 Kg / 13 lbs of hand luggage)
PAX luggage:	13 Kg / 29 Lbs

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The following action items will complete the pre-flight preparation:

Luggage	Scales	Flight Dispatch	GTC
One piece of luggage per pax / standard weight	No action	No action	No action
More than one piece of luggage per pax / standard weight	No action	Reply to the flight release with total mass of luggage in order to receive the new release with the updated payload	Update the payload mass figure in the Cargo section box
If the luggage (or part of the luggage) is non-standard mass	Weigh the nonstandard luggage using the scales onboard	Reply to the flight release with total mass of luggage in order to receive the new release with the updated payload	Update the payload mass figure in the Cargo section box



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Chapter 7 - Loading

Refer to OM-A, [Loading and Securing of Items in the Aeroplane](#), AFM and POH.

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Chapter 8 - Configuration Deviation List

8.1 Publication

The Configuration Deviation List (CDL) is contained within the Minimum Equipment List (MEL), which is an ANAC-approved document based on the Master Minimum Equipment List, AFM and AFM CDL Supplement published by the aeroplane manufacturer.

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8.2 Location

The ANAC-approved MEL (containing the CDL) is held on file in the Maintenance department of NetJets Transportes Aéreos, a copy is placed on-board each aeroplane in the fleet, and is available on Comply365.

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Chapter 9 - Minimum Equipment List

9.1 Publication

The Minimum Equipment List (MEL) is an ANAC-approved document, based on the Master Minimum Equipment List published by the aeroplane manufacturer.

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9.2 Location

The ANAC-approved MEL is controlled by the Maintenance department and held on file in the Document Control department of NetJets Transportes Aéreos, and a copy thereof is placed on board each aeroplane in the fleet, and is also available on Comply365.

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Chapter 10 - Survival and Emergency Equipment Including Oxygen

10.1 Survival and Emergency Equipment

Refer to *Safety Equipment and Procedures Manual*.

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10.2 Oxygen Requirements

Refer to OM-A, [Oxygen Equipment and Supply Requirements](#), and the AFM.

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Chapter 11 - Emergency Evacuation Procedures

11.1 Preparation for Emergency Evacuation

Once the aeroplane is stopped, the LHS pilot takes control and the RHS pilot becomes responsible for conducting a safe evacuation when ordered. Checklist items are accomplished by the LHS pilot, except for checking and opening the exits.

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11.2 Emergency Evacuation

Full Evacuation Procedures are in the SEP Manual, Chapter 3.

Memory Items are on the final page of the QRH.

In the event of an emergency evacuation being required, the LHS pilot will take control and order the evacuation after completing the required checks.

LHS	RHS
LHS - 'PREPARE FOR EVACUATION'	
• Thrust Levers - IDLE. • Emergency / Parking Brake - ON. • Start / Stop Knobs - STOP. • SHUT-OFF 1 & 2 Buttons - PUSH IN. • PRESN MODE Switch - MAN. • DUMP Button - PUSH IN. • ATC - Notify. • Emergency Evacuation - PERFORM. • BATT 1 and 2 Switches - OFF.	• Ensure Emergency / Parking Brake - ON. • Leave Seat and Enter Cabin to Prepare for Evacuation. Consider: • External Hazards / Available Exits. • Briefing Passengers. • Where to go on leaving the aeroplane.
LHS 'THIS IS AN EMERGENCY, EVACUATE, EVACUATE'	
	• Open Main and / or Emergency Exit. • Instruct Passengers ('Open Seat Belts, Come This Way, Leave Bags Behind'). • Direct Pax Upwind to a Safe Area.
• Confirm aeroplane safe and leave the aeroplane ensuring the cabin is clear of all persons.	
PIC • Take charge of PAX Care. • Coordinate with Emergency Services.	

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Chapter 12 - Aeroplane Systems

Refer to the AFM and POH.

12.1 Garmin Navigational Database Update

All guidance for updating the Garmin System is available in the Phenom Fleet Handbook.

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